Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired

A Research Presented to the Faculty of
Malinta National High School (Senior High School)

In Partial Fulfillment
of the Requirements for the Subject Capstone Research

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May 2024

RESEARCH PLAN

A. Rationale

A well packed first aid box is usually a symbol of readiness and assurance. However, for those people who are diagnosed and living with visual impairments, accessing, and utilizing these essential objects during an emergency can be a daunting and anxiety-inducing experience.

Visual impairment is a broad term that covers a range of diseases, from partial to complete blindness of a person. Individuals with visual impairments experience varying degrees of difficulty in perceiving their surroundings, which significantly affects their interaction with physical objects. In the context of medical emergency, deciphering labels, differentiating bandage sizes, and identifying medications becomes a significant challenge. The stress of fumbling through unfamiliar objects during a medical emergency, potentially worsening the situation through accidental misuse.

According to global estimates provided by the World Health Organization (WHO), 2.2 billion people live with vision impairment, placing them at heightened risk during disasters. A study published in "Prevention Web" underscores their vulnerabilities, emphasizing their frequent exclusion from disaster preparedness planning. In the Philippines, statistics provided by "UP Manila National Institutes of Health's Philippine Eye Research Institute" around two million Filipinos are visually impaired, often leaving them reliant on others during a critical moment. This lack of accessibility creates anxiety in those visually impaired individuals. This scenario underscores the need for accessible solutions that prioritize the safety of visually impaired individuals during critical moments. In coordination to that, medical companies produce the solution of significance of first-aid kit role in emergency preparedness for Visually impaired individuals during calamities. It transforms into a critical instrument of self-reliance through offering the ability to address minor injuries, and even extend aid to others with confidence.

However, the traditional first aid kits lack accessibility features, leaving visually impaired individuals dependent on sighted assistance, potentially delaying crucial treatment, and hindering self-reliance.

This gap in accessibility necessitates a solution that empowers visually impaired individuals and promotes safety during critical moments. With that, this study aims to create a first aid kit equipped with sensor-activated audio description for the visually impaired, integrating smart technology to address the limitations of the traditional first aid kit. This innovative feature empowers individuals to independently navigate the kit, fosters a sense of safety and belonging within the visually impaired community.

B. Research Questions, Hypothesis and Expected Outcomes

B. 1. Statement of the Problem

The objective of the study is to develop Vision Aid, an Arduino-based first aid kit equipped with a sensor-activated audio description for the visually impaired. This research seeks to evaluate the product's functionality by simulating a first response kit for a visually impaired individual and assessing its components.

This study will attempt to answer the following questions:

1.What is the performance of the Vision Aid component in terms of:

1.1 space testing,

1.2 timing access; and

1.3 audio efficacy?

2. What is the level of satisfaction of visually impaired individuals with Vision Aid in terms of:

2.1 usability,

2.2 functionality; and

2.3 products' efficiency?

3. How do experts from the Visual Impaired Communities perceive the product?

B. 2. Hypothesis of the Study

A hypothesis is a way for researchers to make a prediction about the outcome of an experiment or study. The following hypothesis will be tested in this study:

1. Individuals with visual impairments will express high levels of satisfaction with the usability and functionality of Vision Aid Kit.
2. The Vision Aid kit would perform incredibly well during testing.
3. The presence of sensor activated audio description technology provides extended positive performance benefits for the Vision Aid Kit and hence for individuals with a visual impairment.

B. 3. Expected Outcome

First Aid Kit equipped with sensor-activated audio description technology will be particularly useful for people with vision impairments by providing them with vital information during an emergency.

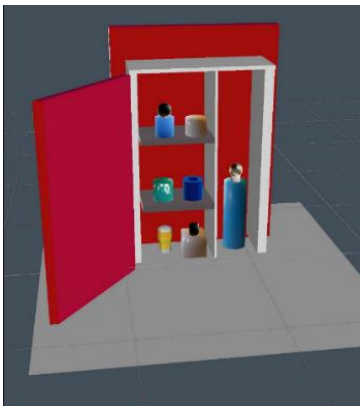
C. Description in Details of Methods of Procedures

C. 1. Procedures

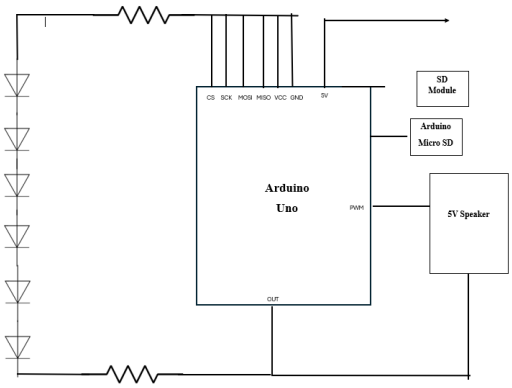
Prepare all the necessary items before building the first aid kit system for the visually impaired. Create a 3D model to help you visualize the product's overall design while you plan the

next step. Cut the plywood to the size of the first aid cabinet. Use a drill to cut at the back of the cabinet to connect wires etc. Afterward, make a schematic diagram that shows the product's inputs and outputs. Connect the infrared sensor wires to the power supply. Also, the Arduino circuit should then be connected to the power supply. Continue programming Arduino with the required code.

The prototype's design and the product's schematic diagram are presented below:



Prototype Design



Schematic Design

C. 2. Risk and Safety

- Sensor Reliability: The primary risk involves the reliability of the sensor technology in accurately detecting and identifying medicines/supplies. Failure or misidentification could lead to errors in medication administration.

- Integration Challenges: Technical challenges may arise during the integration of the sensor into the emergency kit, potentially causing delays in response during critical situations.

- Thorough Testing: Rigorous testing protocols must be implemented to ensure the sensor's functionality and accuracy, minimizing the risk of detection failures or misidentifications.

- User Training: Clear user instructions and training should be provided to ensure proper understanding and interpretation of the sensor's signals, reducing the likelihood of misuse or errors in medication administration.

C. 3. Data Analysis

To assess the effectiveness of the Vision Aid: First Response Kit for those with visual impairments, we will conduct interviews to gather qualitative feedback on users' experiences. We will also use surveys to collect quantitative data regarding satisfaction and usability. Furthermore, testing in simulated emergency situations will be used to gain further insight into overall performance and efficacy. Statistical treatments such as mean calculations and weighted averages will be utilized during analysis

D. Bibliography

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University of the Philippines Manila (2022) Highlight of visual Impairment among Filipino Children. Retrieved February 19, 2024https://www.upm.edu.ph> cpt_news Highlight on visual impairment among Filipino children [Original source: https://studycrumb.com/alphabetizer]

World Health Organization (WHO) (2023) Blindness and vision impairment. Retrieved February 19,2024https://www.who.int›...›DetailBlindnessandvisionimpairmenthttps://www.google.com/urlq=https://www.who.int/newsroom/factsheets/detail/blindnessandvisualimpairment&sa=U&ved=2ahUKEwilhOLjtreEAxUjs1YBHQjdDEEQFnoECA0QAQ&usg=AOvVaw3W74S2ZgieTe07VPIqu7Sm

CHECKLIST FOR ADULT SPONSOR (1)

Checklist for Adult Sponsor (1)

This completed form is required for ALL projects.

To be completed by the Adult Sponsor in collaboration with the student researcher(s): Kyan Ace C. Cungsod, Christopher John H. Cruz, Argie G. Jalbo, Crisallyn D. Espinar, John Francis V. Veiga, Philip James C. Carreon

Student's Name(s): Mirko R. Barcellano, Clarenz L. Dela Cruz, James I. Senilla, Julie mae G. Salimbre, Shafalie G. Dela Torre

Project Title: Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired

1. ☐ I have reviewed the ISEF Rules and Guidelines, including the science fair ethics statement.
2. ☑ I have reviewed the student's completed Student Checklist (1A) and Research Plan/Project Summary.
3. ☑ have worked with the student and we have discussed the possible risks involved in the project.
4. ☐ The project involves one or more of the following and requires prior approval by an SRC, IRB, IACUC or IBC:
   - ☐ Humans
   - ☐ Vertebrate Animals
   - ☐ Potentially Hazardous Biological Agents
     - ☐ Microorganisms ☐ rDNA ☐ Tissues
5. ☑ Items to be completed for **ALL PROJECTS**
   - ☑ Adult Sponsor Checklist (1)
   - ☑ Student Checklist (1A)
   - ☑ Research Plan/Project Summary
   - ☑ Approval Form (1B)
   - ☑ Regulated Research Institutional/Industrial Setting Form (1C) (when applicable; after completed experiment)
   - ☐ Continuation/Research Progression Form (7) (when applicable)

Additional forms required if the project includes the use of one or more of the following (check all that apply):

☑ **Humans,** including student designed inventions/prototypes. (Requires prior approval by an Institutional Review Board (IRB); see full text of the rules.)
- ☐ Human Participants Form (4) or appropriate Institutional IRB documentation
- ☐ Sample of Informed Consent Form (when applicable and/or required by the IRB)
- ☑ Qualified Scientist Form (2) (when applicable and/or required by the IRB)

☐ **Vertebrate Animals** (Requires prior approval, see full text of the rules.)
- ☐ Vertebrate Animal Form (5A)-for projects conducted in a school/home/field research site (SRC prior approval required
- ☐ Vertebrate Animal Form (5B)-for projects conducted at a Regulated Research Institution. (Institutional Animal Care and Use Committee (IACUC) approval required prior experimentation.)
- ☐ Qualified Scientist Form (2) (Required for all vertebrate animal projects at a regulated research site or when applicable)

☐ **Potentially Hazardous Biological Agents** (Requires prior approval by SRC, IACUC or IBC, see full text of the rules.)
- ☐ Potentially Hazardous Biological Agents Risk Assessment Form (6A)
- ☐ Human and Vertebrate Animal Tissue Form (6B)-to be completed in addition to Form 6A when project involves the use of fresh or frozen tissue, primary cell cultures, blood, blood products and body fluids.
- ☐ Qualified Scientist Form (2) (when applicable)
- ☐ The following are exempt from prior review but require a Risk Assessment Form 3: projects involving protists, archae and similar microorganisms, for projects using manure for composting, fuel production or other non-culturing experiments, projects using color change coliform water test kits, microbial fuel cells, and projects involving decomposing vertebrate organisms.

☐ **Hazardous Chemicals, Activities and Devices** (No SRC prior approval required, see full text of the rules.)
- ☐ Risk Assessment Form (3)
- ☐ Qualified Scientist Form (2) (required for projects involving DEA-controlled substances or when applicable)

☐ **Other**
- ☐ Risk Assessment Form (3)

☑ **I attest to the information checked above and that I have read and agree to abide by the science fair ethics statement.**

Chealand T. Dela Cruz		April 26, 2024
Adult Sponsor's Printed Name	Signature	Date of Review (mm/dd/yy)
09171778614	chealanddelacruz@gmail.com	
Phone	Email	

STUDENT CHECKLIST (1A)

Student Checklist (1A)

This form is required for ALL projects.

1. a. Student/Team Leader: Cruz, Christopher John H. Grade: 12

   Email: Cruchristopherjohnhaylar@gmail.com Phone: 09481110513

   b. Team Member: Barcellano, Mirko Z., Delacruz Clarenz T., Delina, John Francis V., Salimbre, Julie Mae G. Canicungan, Philip L. Jalbo, Angie G., Espinar, Cristelyn D. c. Team Member: Lungsod, Ryan Rice C. Delatorre, Shafaye G. Sevilla, James T.

2. Title of Project: Vision Aid: Assessing the Functionality of an Arduino - Based First Response Kit for the Visually Impaired

3. School: Malanta National High School School Phone:

   School Address: St. Jude Street, Malanta Village, Malanta, Valenzuela City

4. Adult Sponsor: Chealand T. Dela Cruz Phone/Email: CHEALANDT.DELACRUZ@gmail.com

5. Does this project need SRC/IRB/IACUC or other pre-approval? ☐ Yes ☑ No Tentative start date: _____

6. Is this a continuation/progression from a previous year? ☐ Yes ☑ No
   If Yes:
   a. Attach the previous year's ☐ Abstract and ☐ Research Plan/Project Summary
   b. Explain how this project is new and different from previous years on
   ☐ Continuation/Research Progression Form (7)

7. This year's experimentation/data collection:

   April 9, 2024 — Actual Start Date: (mm/dd/yy)

   April 10, 2024 — End Date: (mm/dd/yy)

8. Where will you conduct your experimentation? (check all that apply)
   ☐ Research Institution ☐ School ☐ Field ☑ Home ☑ Other: Vibes Massage SM City Valenzuela and Marilao and D.O.C Wellness massage @ Puregold Paso de Blas)

9. Source of Data:
   ☐ Collected self/mentor ☐ Other Describe/url: _____

10. List the name and address of all non-home and non-school work site(s), whether you worked there virtually or on-site:

Name	SM CITY VIBES MASSAGE BRANCH	D.O.C Wellness Massage - PUREGOLD SUPERMARKET
Address:	Mac Arthur Hwy, Brgy Karuhatan, Valenzuela	Corner East Service Road, Paso de Blas Rd., Valenzuela.
	Mac Arthur Hwy, Marilao, Bulacan	
Phone/ email	09328528930 / vibesmassage@yahoo.com	09338685707

11. **Complete a Research Plan/Project Summary following the Research Plan/Project Summary instructions and attach to this form.**

12. **An abstract is required for all projects after experimentation.**

APPROVAL FORM (1B)

Approval Form (1B)

A completed form is required for each student, including all team members.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and will abide by the science fair ethics statement.

Student researchers are expected to maintain the highest standards of honesty and integrity. Scientific fraud and misconduct are not condoned at any level of research or competition. Such practices include but are not limited to plagiarism, forgery, use or presentation of other researcher's work as one's own, and fabrication of data. Fraudulent projects will fail to qualify for competition in affiliated fairs and ISEF.

CRUZ, CHRISTOPHER JOHN H.		April 9, 2024
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

Crisanta H. Cruz		April 9, 2024
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Summary** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Summary** before the student begins experimentation.

SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

This project was conducted at a regulated research institution (**not home or high school, etc.**), was reviewed and approved by the proper institutional board before experimentation and complies with the ISEF Rules. **Attach (1C) and any required institutional approvals (e.g. IACUC, IRB).**

SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

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Barcellano, Mirko R.		April 01, 2024
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

Barcellano, Evelyn R.		April 01, 2024
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

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SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

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Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

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ARGIE JHUDO		April 7, 2024
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

		April 7, 2024
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

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SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

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SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

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Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

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DELACRUZ, CLARENZ, L.		4/9/2024
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

Fe delp Cruz		4/9/2024
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

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SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

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SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

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CRISALYN ESPINAR		April 9, 2024
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

ANALY MONTEBON		April 9, 2024
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

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SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

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SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

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SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

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State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

Approval Form (1B)

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James Serilla		09/09/24
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

Teodora Serilla		09/09/24
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

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SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

This project was conducted at a regulated research institution **(not home or high school, etc.)**, was reviewed and approved by the proper institutional board before experimentation and complies with the ISEF Rules. **Attach (1C) and any required institutional approvals (e.g. IACUC, IRB).**

SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name | Signature | Date of Approval (mm/dd/yy)

Approval Form (1B)

A completed form is required for each student, including all team members.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and will abide by the science fair ethics statement.

Student researchers are expected to maintain the highest standards of honesty and integrity. Scientific fraud and misconduct are not condoned at any level of research or competition. Such practices include but are not limited to plagiarism, forgery, use or presentation of other researcher's work as one's own, and fabrication of data. Fraudulent projects will fail to qualify for competition in affiliated fairs and ISEF.

Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)
Jolly Francis V. Vecina		04/04/24

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)
MARIA V. VECINA		04/04/2024

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Summary** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Summary** before the student begins experimentation.

SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

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SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

Approval Form (1B)

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a. Student Acknowledgment:

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JULIE MAE G. SATIMBRE		April 9, 2024
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

RECHELLE G. SATIMBRE		April 9, 2024
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Summary** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Summary** before the student begins experimentation.

SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

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SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

Approval Form (1B)

A completed form is required for each student, including all team members.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and will abide by the science fair ethics statement.

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RYAN ACE C. LUNGSOD		4/9/2024
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

LIEZEL P. CAJANDAB		April 9/2024
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Summary** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Summary** before the student begins experimentation.

SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

This project was conducted at a regulated research institution **(not home or high school, etc.)**, was reviewed and approved by the proper institutional board before experimentation and complies with the ISEF Rules. **Attach (1C) and any required institutional approvals (e.g. IACUC, IRB).**

SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

Approval Form (1B)

A completed form is required for each student, including all team members.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and will abide by the science fair ethics statement.

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Phillip James Lim Cariungan — Student's Printed Name

Signature

04/09/24 — Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

DOLORES LIM CARI CUNGAN — Parent/Guardian's Printed Name

Signature

04/09/24 — Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Summary** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Summary** before the student begins experimentation.

SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

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SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name — Signature — Date of Approval (mm/dd/yy)

State/National SRC Chair's Printed Name *(where applicable)* — Signature — Date of Approval (mm/dd/yy)

Approval Form (1B)

A completed form is required for each student, including all team members.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and will abide by the science fair ethics statement.

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Dela Torre, Magalic G.		04 / 09 / 24
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Summary**. I consent to my child participating in this research.

MELCHORA G. DELA TORRE		04 / 09 / 24
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB APPROVAL. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Summary** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Summary** before the student begins experimentation.

SRC/IRB Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

This project was conducted at a regulated research institution (**not home or high school, etc.**), was reviewed and approved by the proper institutional board before experimentation and complies with the ISEF Rules. **Attach (1C) and any required institutional approvals (e.g. IACUC, IRB).**

SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval(Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Summary** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

REGULATED RESEARCH INSTITUTIONAL (1C)

Regulated Research Institutional/Industrial Setting Form (1C)

This form must be completed AFTER experimentation by the adult supervising the student research either virtually or on site, conducted in a regulated research institution, industrial setting or any work site other than home, school or field.

Student's Name(s) CRUZ, CHRISTOPHER JOHN H., DELA CRUZ, CLARENZ I., VECINA, JOHN FRANCIS V., CARICUNGAN, PHILIP JAMES L. BARCELLANO, MIREE A., ESPINAR, CRISALYN O., SATINABE, JULIE MAE G., DELA TORRE, SHAFALIE G. JALDO, ARNIE G., SEVILLA, JAMES I., LUNGSOD, EYAN ACE C.,

Title of Project VISION AID: ASSESSING THE FUNCTIONALITY OF AN ARDUINO-BASED FIRST RESPONSE KIT FOR THE VISUALLY IMPAIRED

To be completed by the Supervising Adult in the Setting (NOT the Student(s)) after experimentation:
(Responses must be on the form as it is required to be displayed at student's project booth; please do not print double-sided.)

Research was supported at my work site:

1. Did you or your proxy (e.g. graduate student, postdoc, employee) mentor or provide substantial guidance to the student researcher? ☑ Yes ☐ No
   a. If no, describe your and/or your institution's role with the student researcher and his/her project (e.g. supervised use of equipment on site without ongoing mentorship and sign below.

   b. If yes, complete questions 2–5.

2. Is the student's research project a subset of your ongoing research or work? Use questions 3, 4 and 5 to detail how the student's project was similar and/or different from ongoing research or work at your site. If this project is under a grant and needs to be acknowledged, please list the grant statement here. ☐ Yes ☑ No

3. Describe the independence and creativity with which the student:
   a. developed the hypotheses or engineering goals for the research project

      NOT APPLICABLE. THIS RESEARCH PROJECT IS NOT A SUBSET OF OUR ONGOING RESEARCH OR WORK.

   b. designed the methodology for his/her research project

      NOT APPLICABLE. THIS RESEARCH PROJECT IS NOT A SUBSET OF OUR ONGOING RESEARCH OR WORK.

   c. analyzed and interpreted data

      NOT APPLICABLE. THIS RESEARCH PROJECT IS NOT A SUBSET OF OUR ONGOING RESEARCH OR WORK.

(Continued on next page)

Regulated Research Institutional/Industrial Setting Form (1C) Continued

Student's Name(s) CRUZ, CHRISTOPHER JOHN H., DELA CRUZ, CLARENZ T., DECINA, JOHN FRANCIS V., CARICUNGAN, PHILIP BARCELLANO, MIKKO R., ESPINAR, CRISALYN D., SAMMORE, JULIE MAE G., JAMES L. JALDO ARGIE G., SERILLA, JAMES I., LUNGSOD, RYAN ACE C., DELA TORRE, SHAFALIE G.

4. Detail the student's role in conducting the research (e.g. data collection, specific procedures performed). Differentiate what the student observed and what the student actually did.

NOT APPLICABLE. THIS RESEARCH PROJECT IS NOT A SUBSET OF OUR ONGOING RESEARCH OR WORK.

5. Did the student(s) work on the project as part of a group? ☑ Yes ☐ No
Were there other high school students present? If yes, please list the student names and describe how their work was related or different from the work of this project.

THE RESEARCHERS WORKED AS A GROUP FOR THIS PROJECT. HOWEVER, NO OTHER HIGH SCHOOL STUDENTS WERE INVOLVED ASIDE FROM THEM.

I attest that the student has conducted the work as indicated above and that any required review and approval by institutional regulatory board (IRB/IACUC/IBC) has been obtained. Copies are attached if applicable. I further acknowledge that the student will be presenting this work publicly in competition and I have communicated with the student research regarding any requirements for my review and/or restrictions of what is publicized.

CHEQUARD T. DAGCAZ Supervising Adult's Printed Name		Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired Title
Mariano National High School Institution		April 26, 2024 Date Signed (must be after experimentation) (mm/dd/yy)

QUALIFIED SCIENTIST FORM (2)

Qualified Scientist Form (2)

May be required for research involving human participants, vertebrate animals, potentially hazardous biological agents, and hazardous substances and devices. Must be completed and signed before the start of student experimentation.

Student's Name(s) Christopher John H. Cruz, Argie G. Jadao, Crisalyn O. Espinar, John Francis V. Vector, Philip James Cortenyo, Shaffanie Delatorre Mirko R. Barcellano, Clarenz C. DelaCruz, James I. Sevilla, Julie Mae Salimbe, Ryan Ace Longboa

Title of Project Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired

To be completed by the Qualified Scientist:

Scientist Name: ROMLYN GARINGO

Educational Background: _____ Degree(s): _____

Experience/Training as relates to the student's area of research:

Position/Institution: _____ Email/Phone: _____

1. Have you reviewed the ISEF rules relevant to this project and the science fair ethics statement relevant to this project? ☑ Yes ☐ No
2. Will any of the following be used?
   a. Human participants ☐ Yes ☑ No
   b. Vertebrate animals ☐ Yes ☑ No
   c. Potentially hazardous biological agents (microorganisms, rDNA and tissues, including blood and blood products) ☐ Yes ☑ No
   d. Hazardous substances and devices ☐ Yes ☑ No
3. Will this study be a sub-set of a larger study? ☐ Yes ☑ No
4. Will you directly supervise the student? ☑ Yes ☐ No
   a. If no, who will directly supervise and serve as the Designated Supervisor?
   b. Experience/Training of the Designated Supervisor: _____

To be completed by the Qualified Scientist:

I certify that I have reviewed and approved the Research Plan/Project Summary prior to the start of the experimentation. If the student or Designated Supervisor is not trained in the necessary procedures, I will ensure her/his training. I will provide advice and supervision during the research. I have a working knowledge of the techniques to be used by the student in the Research Plan/Project Summary. I understand that a Designated Supervisor is required when the student is not conducting experimentation under my direct supervision.

ROMLYN GARINGO
Qualified Scientist's Printed Name

Signature

April 6, 2024
Date of Approval (mm/dd/yy)

To be completed by the Designated Supervisor when the Qualified Scientist cannot directly supervise.

I certify that I have reviewed the Research Plan/Project Summary and have been trained in the techniques to be used by this student, and I will provide direct supervision.

Designated Supervisor's Printed Name

Signature

Date of Approval (mm/dd/yy)

Phone

Email

VISION AID: ASSESSING THE FUNCTIONALITY OF AN ARDUINO-BASED FIRST RESPONSE KIT WITH SENSOR-ACTIVATED AND AUDIO DESCRIPTION FOR VISUALLY IMPAIRED

Christopher John Cruz
Mirko Barcellano
Argie Jaldo
Clarenz Dela Cruz
Crisalyn Espinar
James Serilla
John Francis Vecina
Julie Mae Satimbre
Kyan Ace Lungsod
Philip James Caricungan
Shafalie Dela Torre
Proponents

Chealand T. Dela Cruz
Research Adviser

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Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired

Christopher John H. Cruz, Mirko R. Barcellano, Argie G. Jaldo, Clarenz L. Dela Cruz, Crisalyn D. Espinar, James I. Serilla, John Francis V. Vecina, Julie Mae G. Satimbre, Kyan Ace C. Lungsod, Philip James L. Caricungan, and Shafalie G. Dela Torre

Malinta National High School – St. Jude St., Malinta Village, Valenzuela City

ABSTRACT

First Aid Kit equipped with a sensor-activated audio description, created especially for people with vision impairments. Acknowledging the distinct obstacles experienced by people with vision impairments in emergency situations, Vision Aid incorporates cutting-edge technologies and haptic design characteristics to provide timely aid and ensure security. The study's purpose is to assess the functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired. The system, known as Vision Aid, automatically response to an audio description technology when the sensor detects a medical supply at 8-13 cm distances.

The outcome of the study illustrates Vision Aid functionality and product efficiency through space testing, timing access, and audio efficacy. Medical supplies play an important role in deciding time access, demonstrating the system's responsiveness across vast distances. Similarly, expert satisfaction is consistently high across functionality, usability, and product efficacy, indicating Vision Aid's success. Expert feedback, on the other hand, provides vital insights into the system's benefits, potential downsides, and enhancement suggestions, all of which contribute to the improvements of Vision Aid design and functionality over time.

Overall, this research study supports Vision Aid as a practical Arduino-based First Response Kit for the Visually Impaired. The findings indicate its ability to contribute considerably to visually impaired people in varied situations, highlighting its position as a key tool in the ongoing efforts to lessen the impact of calamities in the Philippines as well as globally.

Keywords: Vision Aid, Sensor-Activated, Audio Description, Medical Supplies, Arduino-Uno

INTRODUCTION

A well-packed first aid box provides immediate assistance in a medical emergency, containing necessary medical supplies and equipment for responding to an incident. Having a well-

stocked first aid kit can help treat minor injuries, prevent infections, and stabilize serious injuries until professional medical assistance comes. Nowadays, being prepared for unexpected situations is an important part of everyone's life, especially for visually impaired people who require immediate attention and response. Also, first aid kit can be a lifesaver, it is crucial to note that most of these kits are designed to meet different needs, often leaving out the needs of individuals with visual impairments (Emergency Saver, 2022). According to the statistics from the World Health Organization (WHO), approximately 2.2 billion people live with some forms of vision impairment globally. Being blind or visually impaired does not mean losing the independence of getting response and action from certain calamities. Additionally, the World Health Organization (WHO) stated that visual impairment occurs when the visual system's functionality declines due to an eye condition. Individuals with visual impairments encounter significant challenges during emergency situations, like natural disasters, accidents, or medical emergencies. Their limited vision can hinder their ability to assess their surroundings, locate emergency exits, obtain critical information, and communicate with emergency responders. Consequently, there is an urgent need for innovative and personalized approaches to address the unique needs of people with visual impairments in such situations. This is significant in ensuring their safety and well-being during emergencies. As stated by Abdullah-Al-Wadud (2019), individuals have greatly benefited from the rapid integration of technology into daily life, particularly those with special needs. However, there are still difficulties that visually impaired people must overcome, particularly when it comes to taking their medications. Even with technological developments, individuals with low vision frequently rely on non-visual guidance like touch or smell to recognize the medicines they are taking. These procedures are prone to mistakes, which may lead to drug accidents. People who suffer from visual impairments are also more likely to encounter limitations in their freedom, mobility, and academic performance (Welp et al., 2016). Vision, the most dominant of our senses,

is critical to every aspect and stage of our lives. We take vision for granted, but without it, we struggle to learn, walk, read, participate in school, and work. Vision impairment occurs when an eye condition affects the visual system and its vision functions (World Health Organization, 2023). Voice recognition software can be very useful for blind individuals as it enables them to recognize what they need or what to pick up in the first aid kit. This greatly facilitates communication and information access for users with low vision (Halpin, 2024). An approach that enhances safety in emergencies while empowering those with vision impairments. Considering this, the goal of this project is to develop a first aid kit with sensor-activated audio description for the blind that incorporates smart technology to overcome the drawbacks of the conventional first aid kit. According to the study of Ho et al., (2022), blindness and visual impairment are becoming more important public health concerns in the Philippines, a low-middle-income country in Southeast Asia. It is believed that 4 million Filipinos suffer eye disorders that go undetected due to socioeconomic obstacles. Many community-based health centers lack the equipment required for complete eye exams, and regional hospitals have similar budget constraints. Human resources for health have also proven challenging to organize in villages, as most ophthalmologists work in major cities. Additionally, Kuriakose et al. (2020) noted that humanitarian conditions might be considered when constructing navigation systems. The development of navigation tools for visually impaired individuals is now a major focus of assistive technology research. A study published in "Prevention Web" emphasizes their vulnerabilities, highlighting their frequent omission from disaster preparedness planning. In accordance with "UP Manila National Institutes of Health's Philippine Eye Research Institute," around two million Filipinos are vision impaired, leaving them dependent on others during critical moments. This lack of accessibility causes anxiety among visually challenged people. This scenario highlights the importance of accessible solutions that prioritize the protection of visually impaired people during critical periods. In

addition, medical companies develop a solution for the importance of first-aid kits in emergency preparedness for visually impaired people during disasters. It develops into an essential tool for self-reliance by allowing people to treat minor injuries and confidently assist others. Nevertheless, standard first aid packages lack accessibility features, making visually impaired people reliant on sighted help, potentially delaying critical care and impeding self-reliance.

The research expands the scope of assistive technologies to encompass emergency preparedness and first response. It builds upon the findings from the study "Medication identification aid for visually impaired patients" by Almuzaini and Abdullah-Al-Wadud (2019). The goal of the study is to give visually impaired people a complete solution that improves their safety, independence, and well-being in emergency circumstances by including supply identification and medication capabilities in our first response kit. In conclusion, this research helps visually impaired people's quality of life and health outcomes by bridging the gap between medication management and emergency preparedness. According to Almukainzi et al., (2020), people who are vision impaired usually encounter numerous difficulties when going about what they do every day. Also stated these individuals are disabled and can't distinguish between the names of medications and color. Thus, one of the challenges a person may encounter with a visual impairment is the failure to recognize what medication to take (Kentab et al., 2015). This study addressed the lack of accessibility and safety for visually impaired individuals during emergencies due to the inadequacy of traditional first aid kits.

This experimental study titled Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired aims to create a first aid kit equipped with sensor-activated audio description for the visually impaired. Through integrating smart technology, the limitations of the traditional first aid kit which lacks accessibility features are addressed. This study focusses only on creating a first aid kit box

equipped with sensor-activated audio description to help people with visual impairment. Other recommendations that contributed to the needs of people with visual impairment were not part of the study such as, the assistant in applying the supplies, and monthly checklist of the supplies. The researchers conducted this study in Malinta, Valenzuela City during the academic year 2023 to 2024.

The expected outcome of the study is a first aid kit equipped with sensor-activated audio description technology which is useful for people with vision impairments since it has provided them with vital information during emergencies.

MATERIALS AND METHODS

This section of the research paper acts as the study's foundation, outlining a comprehensive structure for the research methodology, including how the study is conducted, collection of the information, and analyzation of the results. These components, containing the research instrument, play a significant part in maintaining the study's strength and reliability.

Methods and Techniques of the Study

This study, titled “Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired” employs experimental research design. This method is characterized by the design of a sensor-activated audio description technology prototype for use in a first aid kit. It allows accurate use of these necessary supplies in an emergency while carefully managing extraneous factors. In the context of vision aid, this approach assists in determining the causal relationships between specific design components and the functionality of the system for more research and useful applications offering priceless insights. As a result, the researchers were able to draw precise conclusions regarding the technology's functionality. Numerous studies use experimentation and regard it as one of the most

reliable research techniques for evaluating a product's functionality. Establishing causation is one of its noteworthy advantages. The primary goal of a research study is data collection, which enables the investigator to address research questions. based on these categories and the benefits and disadvantages of applying the techniques by Taherdoost, H. (2021).

Research Instrument

The study "Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired" utilized both qualitative and quantitative research instruments, resulting in a strong and multidimensional data collection strategy. This combination effectively used each method's distinct capabilities to provide a comprehensive understanding of Vision Aid's operational performance and user experience.

Quantitative research instruments were useful in gathering objective, numerical data about Vision Aid's functionality. These instruments monitored sensor data, including infrared sensor events and response times, which provided quantitative insights into the system's operating efficacy. This data is subsequently subjected to rigorous statistical analysis, allowing for a more exact assessment of Vision Aid's performance in sensor-activated audio description for the visually impaired. Additionally, user surveys based on Likert-scale questions collected vigilant quantitative input on user satisfaction with system usability functionality and product efficiency.

In contrast, the qualitative component of the research is strengthened by interviews, which allowed researchers to delve deeper into the experience aspects of Vision Aid. Engaging with the field of visually impaired individuals gave essential insights regarding the prototype’s operation. Open-ended questions prompted participants’ thoughts, concerns, and recommendations, providing the study with an abundance of rich and contextually detailed data.

The combination of qualitative and quantitative data proved to be crucial. The findings from this dual approach formed the foundation for evidence-based suggestions and iterative development of Vision Aid's design, functionality, and user experience. This approach has considerably improved its suitability for real-world applications in a first aid kit equipped with a sensor-activated audio description, strengthening its usefulness and effectiveness.

Research Instrument 1: Space Testing

Space testing with the Arduino-based first aid kit within the constrained space ensured its operation, including sensor responsiveness, audio playback, and overall system performance. This ensures the infrared sensors are placed optimally within the space to detect the necessary medical supplies. Additionally, this examined the sensors' location to accommodate any obstacles or maximize their efficacy.

Table 1. Vision Aid component in terms of Space Testing

Name of Medical Supplies	Distance *(Functional, Not Functional)*				
	3 cm distance	5 cm distance	8 cm distance	10 cm distance	13 cm distance
Betadine					
Cotton					
Band-Aid					
Gauze					

Agua Oxigenada					
Alcohol					

Research Instrument 2: Timing Access

The time access guarantees that audio playback occurs for a certain length following each sensor activation and which sensor readings are printed to the serial monitor at a regulated rate. After identifying a medical supply with an infrared sensor, the code uses the TMRpcm library to adjust the level and play an audio file. The delay following each audio replay influences how long the system pauses before checking the next sensor value. This tested the audio playback capabilities by activating the Infrared sensors. In addition, it verified each audio file plays for the appropriate amount of time before proceeding to the next action.

Table 2. Vision Aid Component in terms of Timing Access

Name of Medical Supplies	Time (in seconds)			
	Trial 1	Trial 2	Trial 3	Average
Betadine				
Cotton				
Band-Aid				
Gauze				
Agua Oxigenada				
Alcohol				
Average Time:				

Research Instrument 3: Audio Efficacy

Conducting an audio efficacy test on the product is provided results in audio efficacy under various settings, such as differing ambient noise levels or distances between the sensors and the speakers. Therefore, this is useful in determining the resilience and reliability of audio playback in real-world circumstances.

Table 3. Vision Aid component in terms of Audio Efficacy *(4- Clearly Audible, 3- Audible, 2- Inaudible, 1- Highly Inaudible)*

Name of Medical Supplies	Number of Trials				Verbal Interpretation
	Trial 1	Trial 2	Trial 3	Average	
Betadine					
Cotton					
Band-Aid					
Gauze					
Agua Oxigenada					
Alcohol					

Research Instrument 4: Level of Satisfaction of Visually Impaired Individuals

Conducting a survey on the level of satisfaction of visually impaired individuals with a first aid kit equipped with audio-description technology is essential to assess the product usability, functionality, and efficiency. The feedback to this survey evaluated its effectiveness and identified

the areas of improvement that can help the researcher to make the product overall performance great as the expected outcome of the study.

Table 4. Survey on Visually Impaired Individuals with Vision Aid in Terms of Usability, Functionality, and Product Efficiency *(4- Highly Satisfied, 3- Satisfied, 2- Unsatisfied, 1- Highly Unsatisfied)*

Indicators	**4**	**3**	**2**	**1**
Usability				
1. The sensor activation on the Vision Aid Kit is easy to use.				
2. The audio descriptions for the first aid supplies are clear and easy to understand.				
3. The Vision Aid Kit helped me to find the supplies more easily.				
4. I am confident of finding the specific supplies I need using the Vision Aid Kit's audio instructions.				
5. The Vision Aid Kit is easy to navigate in an emergency.				
6. I am comfortable using the Vision Aid Kit independently.				
Functionality				
1. The audio descriptions in the Vision Aid Kit provide all the information I need to use the first aid supplies safely and effectively.				
2. The voice instructions are accurate and reliable in guiding me through each supply.				
3. The Vision Aid Kit features seem helpful for common first aid situations I might encounter.				
4. I believe the Vision Aid Kit allowed me to provide basic first aid assistance to myself or others in an emergency.				
5. I am confident in the Vision Aid Kit's ability to help me manage a first aid situation.				
6. The Vision Aid overall performance is in great level.				
Product Efficiency				
1. The sensor on the Vision Aid Kit responds quickly and accurately to my touch.				
2. The audio instructions play smoothly without delays or interruptions.				
3. The battery life of the Vision Aid Kit seems enough for an emergency.				

4. I feel confident troubleshooting any minor technical issues with the Vision Aid Kit.				
5. The Vision Aid Kit seems reliable and functions efficiently in emergency situations.				

Research Instrument 5: Semi-Structured Interview Questions

To gauge the perceptions of the Visually Impaired Community towards our product, we employed a semi-structured interview methodology. This approach aimed to elicit detailed insights into their perspectives on the functionality of the First Response Kit for the Visually Impaired. Through this interview process, we sought to gather expert opinions, identify areas for improvement, and ensure that our technology aligns with established health and safety standards.

Interview Questions:

1. How do you perceive the benefits of Vision Aid: First Response Kit for the Visually Impaired?

2. What potential drawbacks do you foresee with the Vision Aid: First Response Kit for the Visually Impaired?

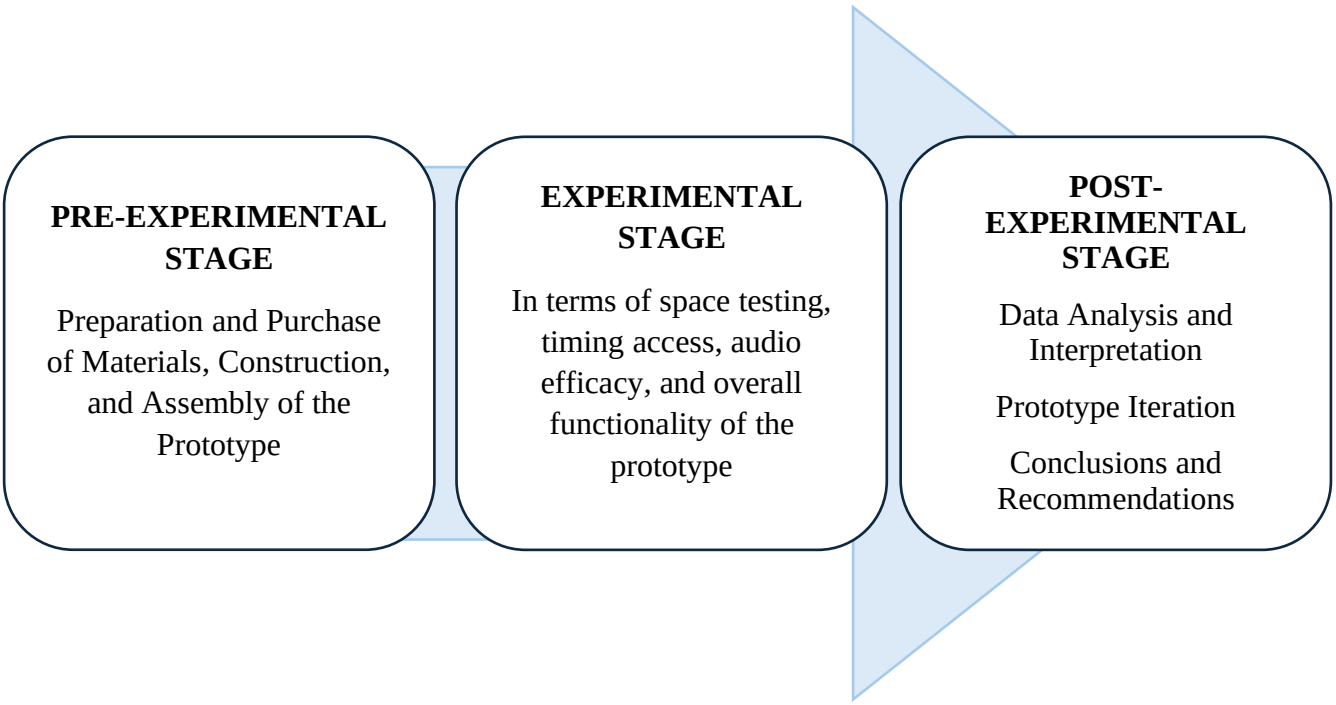
3. Do you have any recommendations to enhance the product further based on your expertise and experiences?

Data Gathering Procedures

In arranging to methodically collect and analyze relevant data for our study, an arrangement of carefully planned information gathering strategies were executed, guaranteeing a deliberate and thorough approach to getting significant insights.

Figure 1

Three Experimental Process



Pre-Experimental Stage

This research titled "Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired" consists of two crucial stages: procurement and material acquisition followed by the prototype building and assembly.

Preparation:

During the preparation and material sourcing stages, a systematic approach to planning and purchasing is necessary. The first and most critical stage of preparation is acquisition. The researchers carefully chose and determined the materials required to make the prototype. This comprises crucial components for the project's technological core, such as an Arduino, an infrared sensor, a microSD card, a SD card module, a breadboard, a 5v amplifier, a speaker, and a first aid kit box. The Arduino Uno Infrared Sensor first aid kit has been specially designed for the visually

impaired. The main processor is the Arduino Uno microcontroller, which is programmed using codes to interpret inputs from the Infrared sensors. For instance, when a person reaches for something necessary, that sensor detects it, and when a gesture is detected by the system, its integrated microprocessor translates the user's action. Also included in this piece of hardware are infrared sensors, speaker, an SD card module, and one Micro SD card making hands-free operation possible during emergencies where human dexterity may be impaired. The unit is nicely segmented into sections for different first aid supplies such as alcohol, betadine, Band-Aid, cotton, gauze, and Agua Oxigenada. An Infrared Sensor mounted on each container distinguishing hand movements enables precise and clear supply retrieval. While simultaneously serves as a critical auditory feedback channel between users with a visually impaired disability to simulate the First Aid Kit prototype as an auxiliary embodiment of their impairment interface the use of an infrared sensor in this facility is indispensable it plays a very vital role in terms of generating sound effects during interaction processes between any blind person and this first aid kit model.

Furthermore, the next critical stage in prototyping is writing and improving the codes in Arduino, which served as the prototype's main processor. The code must include PINs for the Infrared sensors and audio output device or speaker. This step ensures code clarity and simplicity of reference by initializing the defined pins for their respective uses. The Infrared sensor pins are configured as input pins to read sensor data, while the speaker pin is configured as an output pin to transmit audio signals. This function reads input from the infrared sensors and provides auditory feedback based on detected gestures or system conditions.

Assembly:

The next critical step is to build and assemble the prototype. This phase comprised the physical embodiment of the project's concept, bringing together the many components to form a

workable system. The Arduino-based first aid kit with infrared sensor interaction and audio feedback demands meticulous attention to detail when connecting sensors, audio output devices, and microcontrollers. Configuring the program to ensure smooth operation adds another level of complexity to the process. It's a painstaking process that necessitates a methodical approach to guarantee that all components work together seamlessly.

The initial step in developing the prototype is to set up the initial First Aid Kit box. The first aid kit box is divided into six compartments, with a total capacity of six items. The next step in assembling the first aid kit box is to drill a hole for the wiring to pass through the infrared sensors; space is also left to accommodate the Arduino, and wiring. It is also ensured that the screws, doors, and materials utilized in the first aid kit are durable and well-made. Furthermore, the next step is to connect the six Infrared Sensors, Micro SD Card Module, and Speaker to the programmed Arduino Uno. The six infrared sensors are placed in each of the first aid kit box's six compartments. A 5V speaker is also used to improve the device's auditory efficacy or feedback.

Initially, all necessary components were acquired to begin building the visually impaired-friendly first aid kit prototype. Before moving further with the next steps, a 3D model prototype is created to visualize the product's entire design.

The prototype's design is visually represented as Figure 2, while the product's schematic diagram is presented as Figure 3.

Figure 2

Prototype Design

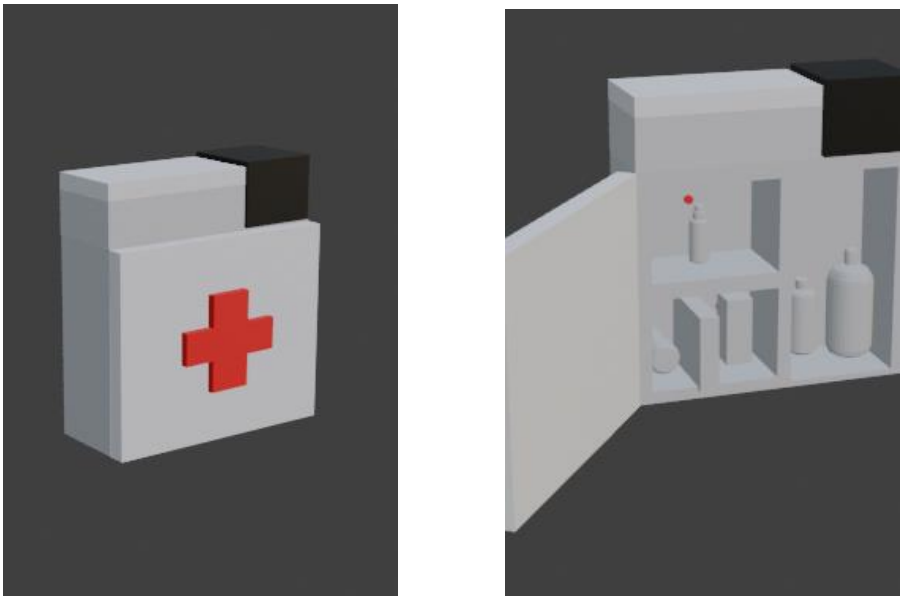
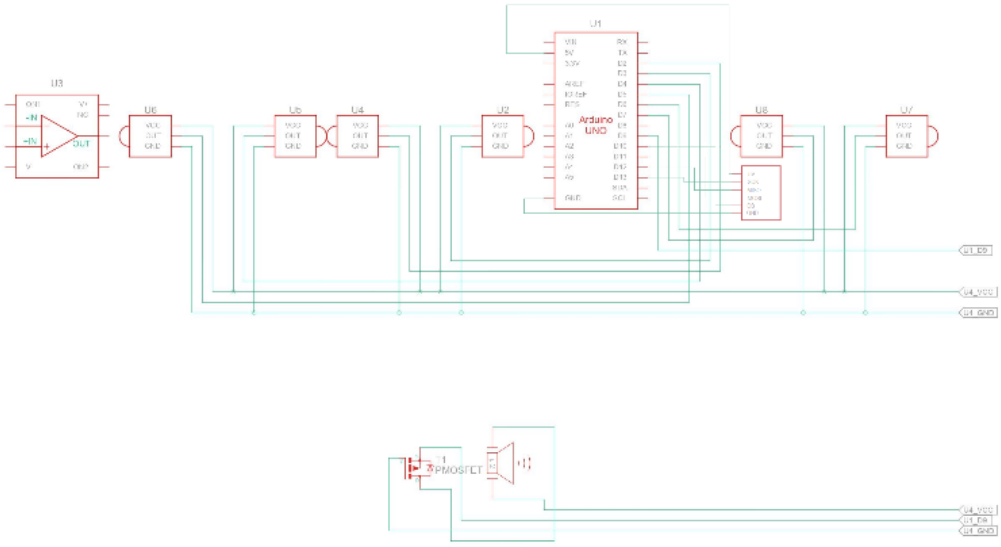


Figure 3

Schematic Diagram



Experimental Stage

The experimental stage of the study titled "Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired" is designed to make a first aid kit prototype that has a prototype of sensor-activated audio description technology. Regarding accessibility features for visually impaired people, typical first aid kits have limitations that this prototype seeks to address. Following the prototype's development, a rigorous testing phase commenced to guarantee the functionality of the sensor-activated audio description technology as intended. Test scenarios meticulously simulate real-life emergencies where individuals with visual impairments might require access to the first-aid kit. This testing encompassed the space, sensor activation timing, and audio efficacy. The researcher conducted a comprehensive "space test" to determine the optimal activation range of the infrared sensor. This involved getting the supply inside the prototype by a simulated user and measuring the distance at which the sensor triggers the audio instructions. For the sensor activation timing test, the researcher assessed the time it takes for the sensor to activate after a user enters its detection zone. This measured using high-precision timers to ensure a swift and responsive activation for timely assistance. For audio efficacy, the researcher evaluated the clarity and intelligibility of the audio instructions delivered by the first-aid kit. This involved testing in various noise environments to simulate real-world emergencies and ensure the audio is easily understood. User feedback is also incorporated to assess the comprehensiveness and helpfulness of the audio descriptions.

This process may involve addressing usability issues identified during testing, enhancing sensor precision and refining the descriptions. To assess sensor activation range, timing, and audio efficacy in emergencies, the researcher intends to test the device in real life on visually impaired people while actively soliciting user feedback.

Post Experimental Stage

The collected data from the testing is analyzed to evaluate the effectiveness of the prototype in providing timely and accurate audio to people with visual impairments when taking the emergency kit. Also, the researcher tried to check if the sensor is activated, audio efficacy and timing, and overall feedback on the experiences of the users. After data analysis, the next stage is to undertake prototype iterations, where the knowledge gathered from the tests is applied to improve the effectiveness of Vision Aid Kit. To improve the functionality and usability of the prototype, all usability concerns discovered during testing are fixed and enhanced. Lastly, the conclusions include an extensive summary of the Vision Aid Kit. The project intended to improve accessibility for individuals with visual impairments by developing and testing a prototype first aid kit incorporating sensor-activated audio description technology. Data analysis demonstrated its efficacy in actual emergencies. Concurrently, suggestions are formulated according to the study's results, offering significant direction for the advancement and possible modifications of Vision Aid Kit. These suggestions should aid in the ongoing development of the Vision Aid Kit, which benefits those who are blind or visually impaired.

Data Processing & Statistical Treatment

In the research study titled "Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit with Sensor-Activated and Audio Description for Visually Impaired", a comprehensive data processing and statistical treatment approach implemented to analyze the collected data. Qualitative feedback from interviews is thematically analyzed to identify common themes and insights regarding the usability and functionality of the Vision Aid kit box. Surveys are conducted to collect quantitative data on satisfaction and usability levels among visually impaired individuals. Testing results from simulated emergency situations underwent statistical

analysis to evaluate the performance and efficacy of the Vision Aid kit, particularly focusing on the sensor-activated audio description technology. Additionally, statistical treatments were applied to assess the reliability and accuracy of the sensor technology integrated into the kit. Overall, the integration of data processing and statistical treatment in this study provided a comprehensive analysis of the Vision Aid kit's effectiveness, user satisfaction, and impact on emergency preparedness for visually impaired individuals.

RESULTS

This section provides a thorough analysis of the data gathered, providing valuable insights on the functionality and performance of Vision Aid as a reliable helper for those with visual impairments during medical emergencies.

In addition, this study determined the functionality of Vision Aid as an Arduino-based first aid kit equipped with a sensor-activated audio description for the visually impaired. As evidence to its functionality the following data were gathered:

Table 1. Vision Aid component in terms of Space Testing

Name of Medical Supplies	Distance *(Functional, Not Functional)*				
	3 cm distance	5 cm distance	8 cm distance	10 cm distance	13 cm distance
Betadine	Not Functional	Not Functional	Functional	Functional	Functional
Cotton	Not Functional	Functional	Functional	Functional	Functional

Band-Aid	Not Functional	Not Functional	Functional	Functional	Functional
Gauze	Not Functional	Not Functional	Functional	Functional	Functional
Agua Oxigenada	Not Functional	Not Functional	Functional	Functional	Functional
Alcohol	Not Functional	Functional	Functional	Functional	Functional

Table 1 presents an in-depth examination of Infrared Sensor performance in terms of system responsiveness at various distances. The data includes five exact distances—3 cm, 5 cm, 8 cm, 10 cm, and 13 cm—from the medical supplies. The descriptive terms of medical supplies such as betadine, cotton, band-aids, gauze, agua oxigenada, and alcohol were evaluated for their ability to detect functionality at various distances, providing insights into the system's responsiveness, specifically functional and non-functional interpretation across different proximity levels.

At 3 cm, the detection of medical supplies continuously non-functional. Moving to 5 cm, the detection of the functionality of medical supplies shows modest variability, with two functional supplies, namely cotton and alcohol, and the sensor response of other medical supplies is inactive. The remaining three distances are 8 cm, 10 cm, and 13 cm. The detection of its functionality in medical supplies yielded functional and reliable results.

When the data analyzed collectively, the medical supplies produced a consistent rapid response at distances as far as 8 cm and 13 cm, with functional interpretations for sensor responsiveness, audio playback, and overall system performance. However, as the distance

decreased to 3 cm and 5 cm, there is an evident increase in the detection of medical supplies without audio playback.

Table 2. Vision Aid Component in terms of Timing Access

Name of Medical Supplies	Time (in seconds)			
	Trial 1	Trial 2	Trial 3	Average
Betadine	0.83	0.81	0.92	0.85
Cotton	1.17	1.35	1.39	1.30
Band-Aid	0.88	0.89	0.92	0.90
Gauze	1.20	1.17	1.34	1.24
Agua Oxigenada	1.91	1.79	1.69	1.80
Alcohol	0.55	0.96	1.20	0.90
Average Time: 1.12 seconds				

In the Vision Aid access timing evaluation, the duration it took to detect an infrared sensor is assessed in seconds across a series of trials. Betadine, specific timings were documented for Trials 1, 2, and 3, providing values of 0.83, 0.81, and 0.92 seconds, respectively. The sum of these trials yielded an average time of 0.85 seconds for this infrared sensor. This tedious process is repeated for each of the six infrared sensors. Cotton's average detection time is 1.30 seconds, calculated from individual trial times of 1.17, 1.35, and 1.39 seconds. Similarly, Band-Aid trial times were 0.88, 0.89, and 0.92 seconds, with an average detection time of 0.90 seconds. Gauze had trials of 1.20, 1.17, and 1.34 seconds, with an average detection time of 1.24 seconds. Agua Oxigenada, on the other hand, showed trial timings of 1.91, 1.79, and 1.69 seconds, for an average detection time of 1.80 seconds. The final medical supply, alcohol, had an average time of 0.90

seconds, with the first trial taking 0.55 seconds, the second taking 0.96 seconds, and the third taking 1.20 seconds.

The aggregate average time across all trials is 1.12 seconds. This average time provides a summary measure of the Infrared sensors' detection performance in seconds.

Table 3. Vision Aid component in terms of Audio Efficacy *(4- Clearly Audible, 3- Audible, 2- Inaudible, 1- Highly Inaudible)*

Name of Medical Supplies	Number of Trials				Verbal Interpretation
	Trial 1	Trial 2	Trial 3	Average	
Betadine	4	4	4	4	Clearly Audible
Cotton	3	3	4	3	Audible
Band-Aid	4	4	4	4	Clearly Audible
Gauze	4	4	4	4	Clearly Audible
Agua Oxigenada	3	3	4	3	Audible
Alcohol	3	4	4	4	Clearly Audible

Table 3 presented data on Vision Aid's audio efficacy in terms of evaluating the resilience and reliability of audio performing in real-world circumstances. The medical supplies were analyzed, and three exact trials were conducted to detect the medical supplies using infrared sensors for their auditory efficacy in a speaker. Additionally, it provides a verbal interpretation indicating that 4 is clearly audible, 3 is audible, 2 is inaudible, and 1 is highly inaudible. For the initial medical supplies, Betadine specific audio documented for Trials 1, 2, and 3, yielding values of all four. The total of these trials resulted in a verbal interpretation of clearly audible for

this specific medicinal supply, with an average of 4. This meticulous procedure was repeated for each of the six medicinal supplies. Cotton, on the other hand, displayed a verbal interpretation of audible based on individual trial times of 3, 3, and 4, with an average of 3. Similarly, Band-Aid and Gauze had trial times of 4 for all trials, yielding an average of 4 and a verbal interpretation of Clearly Audible. Agua Oxigenada's trial times were 3, 3, and 4, resulting in a verbal interpretation of Audible with an average of 3. Alcohol, the final medical supply, had an average of four, with three for the first trial and four for the second and third trials. Hence, the verbal interpretation of alcohol is clearly audible.

Overall, this indicates that the audio efficacy of medical supplies in detection is proportional to the distance of the infrared sensors, as most medical supplies are clearly audible. Cotton and Agua Oxigenada, on the other hand, are audible to the real-world circumstances.

Table 4. Survey on Visually Impaired Individuals with Vision Aid in Terms of Usability, Functionality, and Product Efficiency

Indicators	General Weighted Mean	Verbal Interpretation
Usability	3.63	Highly Satisfied
Functionality	3.67	Highly Satisfied
Product Efficiency	3.62	Highly Satisfied

(3.25-4.00 - Highly Satisfied, 2.50- 3.24 - Satisfied, 1.75-2.49 - Unsatisfied, 1.00-1.74 - Highly Unsatisfied)

The data presented on table 4, derived from a survey on visually impaired individuals that assess their level of satisfaction in terms of usability, functionality, and product efficiency of the Vision Aid or Arduino-based first aid kit product equipped with audio-description technology for people with visual impairments. Accompanied by corresponding verbal interpretations, the general weighted mean served as a quantitative gauge of visually impaired individuals' level of satisfaction with the product.

The product's technology impressively achieved a general weighted mean of 3.63 in terms of "Usability", making it in a "Highly Satisfied" range of verbal interpretation. This implied that visually impaired individuals perceived Vision Aid as effective assistance, very accessible, fitted to the purpose of the product, and very user-friendly. The product's technology in functionality, with a general weighted mean of 3.67, is likely marked as "Highly Satisfied" interpretation. The participants express that Vision Aid Kit exceeds performance expectations, is informative in terms of clear instructions, has very helpful features, and empowers them to handle first aid situations effectively. The product efficiency, marked as 3.62 general weighted mean, positioning it within the "Highly Satisfied" interpretation. The responses indicate that the Vision Aid kit operates quickly, delivers instructions clearly, and is user-friendly even with minor technicalities, making it a dependable tool for emergencies.

The consistent verbal interpretation of "Highly Satisfied" across the three indicators provide strong evidence that the Vision Aid is successful as an Arduino-based first aid kit product equipped with audio-description technology for people with visual impairment, meeting and surpassing the product expectations. In addition to confirming the effectiveness of the technology, this survey data offers valuable insights for potential improvements, with the goal of further enhancing the user experience and optimizing the overall performance of Vision Aid.

Table 5. Experts Perception on Vision Aid Based on Its Advantages and Disadvantages.

QUESTIONS	PARTICIPANT	RESPONSES	CODES	THEME
Question Number 1:	FE 1	Napaka dali lang gamitin at makuha ang mga supply na kailangan ng mga bulag.	Ease of Use	

How do you perceive the benefits of Vision Aid: First Response Kit for the Visually Impaired?	FE 2	May nagbibigay ng sounds kung ano ang kukunin namin.	Auditory Feedback	Accessibility for Visually Impaired Individuals
	FE 3	Isa ito sa mga problemang kinakaharap nila kapag may kailangan sila at wala silang matatawag para matulungan sila.	Increase Independence	
Question Number 2: What potential drawbacks do you foresee with the Vision Aid: First Response Kit for the Visually Impaired?	FE 1	Kailangan niya ng malakas na sounds kapag ito ay kinukunan ng supply sa loob ng First Aid Kit, kasi masyado siyang mahina para saakin.	Auditory Feedback	Auditory Feedback and Volume
	FE 2	Try niyong mas lakasan pa ang tunog nito kapag ito ay nagsasalita na.		
	FE 3	Ang kakulangan na aming nakita ay hindi sapat ang espasyo na mayroon ang isang supply.	Space and Organization	Space and Organization
Question Number 3: Do you have any recommendations to enhance the product further based on your expertise and experiences?	FE 1	Kailangan talaga na mas malakas pa ang sound na maririnig ng bulag, kasi kailangan nila yun.	Auditory Feedback and Volume	Enhanced User-Centered Design
	FE 2	Siguro wag niyong limitahan ang mga ilalagay, at mas lakasan pa ang talks nito. At mas maganda kung may mga gamot ito, kasi malaking tulong to para sa mga bulag.	Product Inclusivity and Variety	
	FE 3	Bigyan niyo ng sapat na espasyo ang bawat supply kasi malalaki ang mga kamay ng bulag.	Space and Organization	

Table 6 consolidated participant's perspectives on the perceived advantages, drawbacks, and improvement suggestions for Vision Aid—an Arduino-based first response kit for the visually impaired. The responses to three specific questions were organized into codes and overarching themes.

Experts have provided valuable feedback in response to inquiries about the perceived benefits of Vision Aid. The fact that the product is user-friendly and easy to use for those who are blind or visually impaired has been emphasized in these replies, highlighting how clear sound feedback and simple operation make this kit more accessible. A significant advantage is the ability to assist in emergency situations through spoken instructions and easy access to necessary supplies.

However, the potential drawbacks associated with Vision Aid have also been identified by experts. For example, Expert 1 pointed out the need for louder feedback when retrieving the supplies from the kit since the current sound was considered too soft. The volume of the spoken instructions should also be increased, as suggested by Expert 2. The importance of clear and differentiated auditory signals for visually impaired people is underlined by these concerns. In addition, Expert 3 highlighted a lack of adequate space between supplies, which can lead to inadvertent distortion of other items, particularly for users with low visual sensitivity.

Moreover, recommendations for improvement of the product were presented by experts. To accommodate visual impairments, these suggestions emphasized the need for more powerful audio signals and a greater degree of separation between supply channels. To better guide users, Expert 1 suggested a stronger sound system, while Expert 2 suggested increasing the variety of supplies available. In addition, to be more comfortable during an emergency, Expert 3 recommended battery-powered operations.

Finally, both strengths and areas of improvement in Vision Aid have been highlighted by experts' comments. Its user-friendly design and sound feedback are also clear benefits. At the same time, the need for louder sounds and more organized storage of supplies is identified as a limitation. These findings are useful guidance for refining the kit to make it more effective as a visual aid and improving its total effectiveness.

DISCUSSIONS

The analysis of the data gathered from the study on Vision Aid indicated that the functionality and performance of the kit were assessed, including its sensor-activated audio description for the visually impaired during medical emergencies. The study found that the distance from the medicinal supplies affects when they become functional in Vision Aid.

Space testing demonstrates that the distance may affect the infrared sensors and the audio playback influences the system's responsiveness. When the distance is between 3 cm and 5 cm, the system shows less responsiveness and late audio feedback, but when it is between 8 cm, 10 cm, and 13 cm, the responsiveness of the system improves. Gines B. et al. (2002) indicated that distance may affect the infrared sensor's responsiveness. Measuring the distance between 1 cm and 5cm is expected to have errors in the infrared sensors' distance. Subsequently, Infrareds are small and cheap sensors that provide an operational range between 10 and 70 centimeters. (Gonzales et al. 2006)

Every medical supply has a varied timing of detections for the infrared sensor evaluation, and each medical supply has a different audio playback speed. With a detection time of 0.85 seconds, betadine is the most rapid, while agua is the slowest, taking 1.80 seconds. This implies that the infrared sensors' responsiveness and the quality of the audio playback may be impacted by the distance between the sensors and the supply. The ability to detect medical supplies consistently nonfunctional at 3 cm. When the medical supplies were moved to 5, 8, 10, and 13 centimeters away, their functionality varied; some were dormant, and others were functional.

According to the data analysis, medical supplies consistently provided quick responses at 8 and 13-cm distances, with functional interpretations for audio playback, sensor responsiveness, and overall system performance. Nonetheless, there is a rise in the identification of medical supplies without audio playback at closer ranges of 3 and 5 cm. Therefore, the distance from the

medicinal supplies affects when they become functional in Vision Aid. A study conducted by Han & Lin (2002) states that in the traditional periodic real-time task model, tasks are scheduled to be executed once during each of their periods. A task's completion in one period is dependent upon its completion in any subsequent period.

In audio efficacy of the Vision Aid Kit in detecting different medical supplies. Using the sensors Betadine, Band-Aid, Gauze, and Alcohol consistently received a verbal interpretation of clearly audible, while Cotton and Agua Oxigenada were rated as audible. Hence, audio efficacy has a vital role for every visually impaired individual. Michael A. N. et al. (2013) stated that technologies like text-to-speech and other forms of audio information technology have been a huge help to those with visual impairments.

This study on Vision Aid revealed that the distance from the medical supplies impacts their functionality within the kit. Experts provided feedback suggesting improvements such as stronger audio signals and better supply channel organization. The survey results showed high satisfaction levels among visually impaired individuals regarding the usability, functionality, and product efficiency of Vision Aid. The timing evaluation indicated varying detection times for different medical supplies, with betadine being the fastest at 0.85 seconds and agua oxigenada the slowest at 1.80 seconds. Joshi et al. (2020) emphasized the importance of sensor-based technologies and computer vision in creating efficient assistive devices for visually impaired individuals.

The results of the survey indicated a high level of expert satisfaction with the vision aid first response kit. The consistent "Highly Satisfied" classification across all indicators underscored the overall success of the vision aid kit in meeting or surpassing authorities' expectations. This aligns with the findings of a study by Jashanbadi & Bonab (2023), the correct performance of the

services attracts the expert's satisfaction with the product, whose good performance is not only influenced by the latest technology.

The study findings suggest that the distance between the sensors and medical supplies significantly impacts the responsiveness and audio playback of the Vision Aid system. Noel (2018) stated that in comparison to far space, near space has lower temporal acuity. Additionally, response time to audiovisual targets presented asynchronously indicated a window of time for quick detection of a variety of stimulus asynchronies that were more pronounced in near space than in far space. However, a study conducted by Abe et al. (2014) mentions that they opposed changes in the sound's frequency response, the perception of distance is heavily influenced by the sound pressure level, at least in semi-anechoic spaces. As a result, they used the limitations approach to determine which sound pressure level most closely matched the virtual sound source's visually perceived distance.

CONCLUSIONS

The discussion on vision aids highlights the significant impact of spatial proximity on the operational effectiveness and performance of aids for visually impaired individuals. According to the study, the proximity of the infrared sensors to the medical supplies in the Vision Aid system has a significant effect on the sensors' responsiveness and audio playback quality. This emphasizes the importance of careful spatial planning and precise sensor calibration to maximize the effectiveness of the system, as demonstrated by previous studies on the responsiveness of sensor technology.

Furthermore, the experiment revealed varying time intervals for detecting various medical supplies, with variations in response times seen across different goods. For example, betadine demonstrated the shortest detection time, whereas agua oxigenada showed a slower reaction. These findings highlight the complex connection between distance, sensor precision, and auditory

interpretation in the Vision Aid system, consistent with prior research that emphasizes the importance of audio technologies in assisting people with visual impairments.

The subject's feedback on prospective improvements for the Vision Aid system, such as enhancing audio signals and organizing supplies, highlights the significance of continuous refinement and user-centric design in creating efficient assistive technology. This supports the claim that sensor-based technologies play a crucial role in developing effective solutions for people with visual impairments.

It emphasizes the importance of giving priority to user feedback and expert input to improve the performance of the system and ensure user happiness. Visually impaired individuals have validated the effectiveness of the Vision Aid system by reporting high levels of satisfaction with its usability, usefulness, and efficiency. The consistent positive feedback highlights the effectiveness and importance of the visual assistance solution in achieving its intended goal, reflecting the research findings on the relationship between service performance and expert satisfaction.

To summarize, the study's findings highlight the complex relationship between proximity, sensor responsiveness, auditory effectiveness, and user satisfaction when creating and assessing assistive technology for visually impaired individuals. Future generations of the Vision Aid system can improve functionality, usability, and overall impact on those with visual impairments by focusing on spatial dynamics and technological precision. The study's scholarly insights and empirical evidence establish a strong basis for furthering research and innovation in the field of assistive technologies. This emphasizes the significance of user-centered design and ongoing improvement to develop inclusive and efficient solutions for various user populations.

RECOMMENDATIONS

Based on the results and conclusions, the following recommendations were made: To ensure that everything in the kit supplies is organized and understandable, adjust the audio pace and volume level. To make the audio easier to comprehend, it can also be translated into Filipino. In addition, add an additional number of items in the kit, such as medications, as it might be more challenging for the blind or visually impaired to recognize prescription labels. Furthermore, utilize an Arduino Uno with two 5V outlets to eliminate the need for two plugs. Add an extra feature can be made to ensure that the supplies that are going to be included in the kit are not interchangeable with one another. To guarantee that the device keeps working even when there are electricity outages, use additional sources of energy like batteries powered by solar technology.

ACKNOWLEDGEMENT

This study was made possible by a dedicated group of individuals who provided unflinching support, allowing the researchers to persevere in the face of difficulties and disappointments. The researchers would like to express their profound gratitude to everyone who helped make this study possible.

To their **parents**, for their undying love and persistent moral and financial support throughout the most difficult stages of their journey.

To the research adviser, **Mrs. Chealand T. Dela Cruz**, for her unwavering support and guidance. Her knowledge guided the researchers toward a better grasp of various misconceptions observed during the investigation.

To the **employees and authorities of Vibes Massage and D.O.C Wellness Massage**, for their cooperation and informative feedback during the survey and interview, which greatly contributed to the study's overall success.

To **Mr. Ej De Castro**, for his relentless patience in imparting knowledge and guiding the researchers through the fundamentals of Arduino Uno operation and application.

To **Mr. Romlyn O. Garingo,** for providing ideas on the use of amplifier components, which substantially enhanced the research.

To **Ms. Cindy De Guzman Damasco**, for supporting and donating the first aid kit box to the researchers for research completion.

To the school's assistant principal and school head, **Mr. Silverio M. Agustin and Mr. Edwin L. Valera**, for their unwavering support, encouragement, and assistance to the researchers.

And above all, to the **Almighty Father** for supplying us with inner strength, courage, and determination. This research would not have been possible without his direction and grace.

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APPENDICES

APPENDIX A:

REQUEST FOR SUPPORT DURING VISION AID SIMULATION IMPLEMENTATION

April 11, 2024

Danabelle Jamaleto
DANABELLE MANAGER / Reception 1

VIBES Massage
SM City - Valenzuela Branch

Greetings!

In partial requirement in the subject, Capstone Research, we the Group 4 of 12 STEM - Maxwell from Malinta National High School - Senior High, are presently completing the research titled ***"Vision Aid: First Response Kit for the Visually Impaired"***.

In connection with this, we are asking for your assistance in testing the efficacy of our product, Vision Aid. We would like to seek some advice, suggestions, and any help you could offer that will be helpful in improving the quality and functionality of our Arduino-based first response kit for the visually impaired. Rest assured that all the data gathered will be kept confidential to secure the privacy of the school and participants.

We extend our gratitude in advance, hoping that you'll consider our request.

Thank you and God Bless!

Sincerely,

Christopher John H. Cruz
Team Leader

Noted,

Chealand T. Dela Cruz
Research Adviser

April 11, 2024

D.O.C Wellness Massage
Puregold – Paso de Blas

LUZVIMINDA VENTURA

Greetings!

In partial requirement in the subject, Capstone Research, we the Group 4 of 12 STEM - Maxwell from Malinta National High School - Senior High, are presently completing the research titled ***"Vision Aid: First Response Kit for the Visually Impaired"***.

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Thank you and God Bless!

Sincerely,

Christopher John H. Cruz
Team Leader

Noted:

Chealand T. Dela Cruz
Research Adviser

April 11, 2024

VIBES Massage
SM City – Marilao Branch

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We extend our gratitude in advance, hoping that you'll consider our request.

Thank you and God Bless!

Sincerely,

Christopher John H. Cruz
Team Leader

Noted:

Chealand T. Dela Cruz
Research Adviser

APPENDIX B:

REQUEST FOR PARTICIPATION IN SURVEY

April 11, 2024

Janelle Manalo
DANABERE MATAMOTO / RECEPTIONIST

VIBES Massage
SM City - Valenzuela Branch

Greetings!

In partial requirement in the subject, Capstone Research, we the Group 4 of 12 STEM - Maxwell from Malinta National High School - Senior High, are presently completing the research titled ***"Vision Aid: First Response Kit for the Visually Impaired"***.

In connection with this, we would like to request your permission to conduct our survey among some of the Visually Impaired Individuals of VIBES Massage at SM City Valenzuela. This is part of our initiative to gather feedback from them in evaluation of our product's usability, functionality, and efficiency. Rest assured that all the data gathered will be kept confidential to secure the privacy of the school and participants.

We extend our gratitude in advance, hoping that you'll consider our request.

Thank you and God Bless!

Sincerely,

Christopher John H. Cruz
Team Leader

Noted:

Chealand T. Dela Cruz
Research Adviser

April 11, 2024

D.O.C Wellness Massage
Puregold – Paso de Blas

LUZHIMINDA VENTURA

Greetings!

In partial requirement in the subject, Capstone Research, we the Group 4 of 12 STEM - Maxwell from Malinta National High School - Senior High, are presently completing the research titled *"**Vision Aid: First Response Kit for the Visually Impaired**"*.

In connection with this, we would like to request your permission to conduct our survey among some of the Visually Impaired Individuals of D.O.C Wellness Massage at Puregold Paso de Blas. This is part of our initiative to gather feedback from them in evaluation of our product's usability, functionality, and efficiency. Rest assured that all the data gathered will be kept confidential to secure the privacy of the school and participants.

We extend our gratitude in advance, hoping that you'll consider our request.

Thank you and God Bless!

Sincerely,

Christopher John H. Cruz
Team Leader

Noted:

Chealand T. Dela Cruz
Research Adviser

April 11, 2024

VIBES Massage
SM City - Marilao Branch

odamasen
Ornalyn Damasen - Service Aid

Greetings!

In partial requirement in the subject, Capstone Research, we the Group 4 of 12 STEM - Maxwell from Malinta National High School - Senior High, are presently completing the research titled ***"Vision Aid: First Response Kit for the Visually Impaired"***.

In connection with this, we would like to request your permission to conduct our survey among some of the Visually Impaired Individuals of VIBES Massage at SM City Marilao. This is part of our initiative to gather feedback from them in evaluation of our product's usability, functionality, and efficiency. Rest assured that all the data gathered will be kept confidential to secure the privacy of the school and participants.

We extend our gratitude in advance, hoping that you'll consider our request.

Thank you and God Bless!

Sincerely,

Christopher John H. Cruz
Team Leader

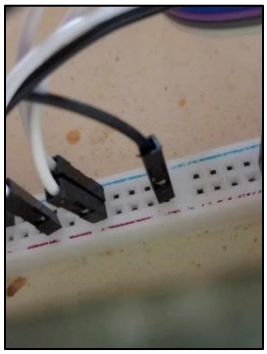
Noted:

Cheafald T. Dela Cruz
Research Adviser

APPENDIX C

INVENTORY OF MATERIALS UTILIZED IN PROTOTYPE CONSTRUCTION AND THEIR RESPECTIVE FUNCTIONS

The following materials were used by the researchers in creating the Arduino based first aid kit equipped with a sensor-activated audio description:

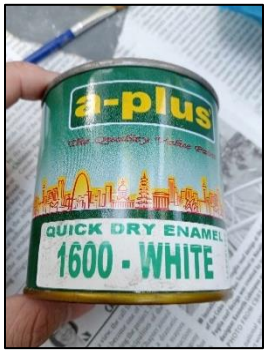


Breadboard

Sandpaper

Paint Brush

Masking Tape



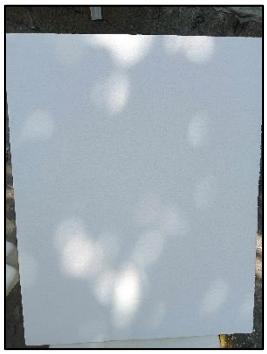
White Paint

Paint Roller

Glue Sticks

Triangle Ruler

Wooden Ruler



Styrofoam

Betadine

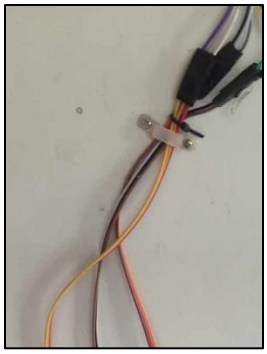
Cotton

Band Aid

Gauze

Agua

Alcohol



M to M Jumper Wires

Memory Card

Arduino Uno

Micro SD Card

Arduino Cable

Infrared Sensor

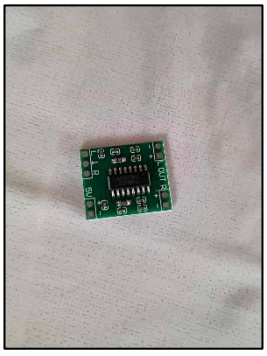
Chisel

Wood Glue

Grinder

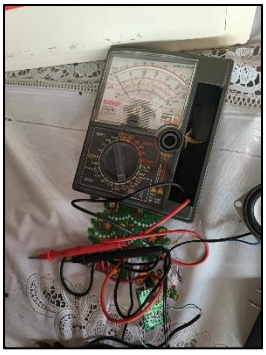
Cabinet Knob

Scissor

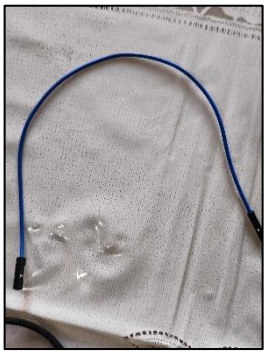


Amplifiers

Speaker



Electric Tester



F to F Jumper Wires

Wire Cutter

Soldering Iron

Charger Block

USB Cable

Side Cutting Pliers

Steel Wire

Circular Saw

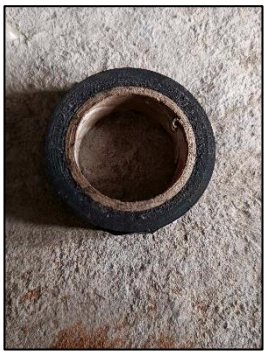
Saw

AC DC Adaptor

Plywood

Cable

Electric Drill



Electrical Tape

Wire Insulation

1. Breadboard: A breadboard is a standardized platform used for designing and testing circuit designs without the requirement of soldering. The breadboard is composed of a plastic board with a grid of holes that are internally connected by metal strips. This design allows for the convenient insertion and interconnection of components.

2. Liha: Also referred to as sandpaper, is utilized as an abrasive substance in woodworking and finishing tasks. Liha paper is utilized to refine surfaces, eliminate flaws, and ready them for painting, staining, or varnishing by establishing a more consistent texture.

3. Paint Brush: Tool consisting of bristles affixed to a handle, is employed to administer paint or coatings onto our first aid kit box. The paintbrushes we utilize possess a variety of shapes, sizes, and bristle kinds to accommodate specific applications and desired finishes.

4. Masking tape: A pressure-sensitive adhesive tape that is used to temporarily cover and protect surfaces while painting and handicraft. It is specifically designed for the purpose of furnishing the First Aid Kit Box. Masking tape offers precise delineation when applied to edges, eliminating paint seepage and guaranteeing a polished outcome.

5. White Paint: Applying white paint to first aid kit boxes can enhance contrast and visibility for those with visual impairments, facilitating their ability to locate the kit in emergencies. The pristine and unblemished look of white paint additionally improves the general ease of use and accessibility of the first aid kit for those with varying visual capabilities.

6. Paint Roller: A cylindrical equipment that utilizes a fabric or foam cover to apply paint evenly and efficiently to expansive surface areas. Paint rollers are utilized to paint the cover of the first aid kit because of their efficiency and ability to provide consistent coverage.

7. Glue Sticks: Solid adhesives contained in twist-up tubes that undergo melting when applied, creating a temporary bind between materials upon cooling. In order to affix the speaker and other cables from above the box, we utilize adhesive sticks.

8. Triangle Rulers: Utilized in geometric building and technical drawing to establish and measure accurate angles.

9. Wooden Rulers: Measuring instruments constructed from wood or alternative substances, which display length indicators in standardized units such as inches, centimeters, or millimeters. Wooden rulers are vital instruments for accurate measuring and arrangement, functioning as representations of linear dimensions.

10. Styrofoam: To maintain a uniform and systematic arrangement, we secure the contents in the first aid bag using Styrofoam. The Styrofoam inserts facilitate the establishment of a consistent gap between each supply, preventing their movement and enhancing the sensors' ability to detect them.

11. Betadine: This material provides a specific role on sensor 1 that detects the product; once it is removed from the sensor, it will trigger and produce sounds.

12. Cotton: This material is used on sensor 2 that detects the product; once it is removed from the sensor, it will trigger and produce sounds.

13. Band Aid: This material provides a use for sensor 3. that detects the product; once it is removed from the sensor, it will trigger and produce sounds.

14. Gauze: This material provides a role for sensor 4. that detects the product; once it is removed from the sensor, it will trigger and produce sounds.

15. Agua: This material provides a use for sensor 5. that detects the product; once it is removed from the sensor, it will trigger and produce sounds.

16. Alcohol: This material provides a use for sensor 6. that detects the product; once it is removed from the sensor, it will trigger and produce sounds.

17. Male to Male Jumper Wires: The Male to Male Jumper Wires is attached to the female plug. A power cord is inserted into the outlet to establish an electrical connection with the power source, typically the outlet.

18. Memory Card: This has been used for programming that has the audio that is attached to the SD card.

19. Arduino Uno: Serves as the heat-responsive suppression system's brain. It is in charge of directing and coordinating the different parts and operations of the system. Combined with sensors to identify high temperatures, the Arduino With the data it has gathered, Uno is able to make observations in real time.

20. Micro Arduino SD Card: Provides an easy and effective way to add SD card capability to Arduino programs. Users can put audio files on an SD card, which the Arduino can read and play back with only one simple input.

21. Arduino Cable: Provides a simple to use and useful method to power the Arduino board. With a USB power adapter, users may easily energize their Arduino projects via the USB Cable to it.

22. Infrared Sensor: The Infrared Sensor is a vital tool use to helps people who are visually impared to recognize the materials in our product.

23. Chisel: A chisel is a sharp-bladed piece of equipment used to carve wood to cover our product, improving both their look and simplicity.

24. Stikwel Wood Glue: Stikwel wood glue is use particularly for connecting wood surfaces, giving woodworking projects a strong and permanent adhesive bond. Its composition guarantees consistent adherence and durable outcomes for wood bonding and construction.

25. Grinder: A grinder is an electric equipment with a circular rough blade that is frequently used to shape and smooth wood surfaces for our product's covers. It effectively slices through wood, enabling precise shape.

26. Wood Cabinet Knob: A wood cabinet knob is a strong, physical handle that is designed specifically for our product's door, making it easier for those with vision impairments to open.

27. Scissor: It used to cut variety of materials such us wires, tape and even other materials.

28. Amplifiers: Use to increase the amplitude of a signal, making it louder or stronger.

29. Speaker: Use to let blind people know the items taken from the first aid box.

30. Electric Tester: Use to check the presence of electrical voltage in a circuit or electrical device.

31. Female To Female Jumper Wires: Use to connect, extend, or connect components on a breadboard.

32. Wire Cutter: Use to cut some wires and to make the cuts clean and precise.

33. Soldering Iron: Use to connect the wire to the amplifier and the sensors.

34. Charger Block: Use to plug in the power outlet to provide the necessary power for the amplifier.

35. Usb Cable: Use to connect to the charger block to travel the power from the charger block into the amplifier.

36. Side Cutting Pliers: To cut wire by indenting and wedging a cable apart.

37. Stainless Steel Wire: The steel wire is a metallic, cylindrical, and flexible strand that is used to conduct electricity.

38. Circular Saw: The portable electric circular saw, with the blade attached to a motor shaft, is probably the most used saw. With the proper blade it can cut almost any material—wood, metals, plastics, fibreglass, cement block, slate, and brick. On wood it can rip, crosscut, and make angle cuts.

39. Saw: Tool for cutting solid materials to prescribed lengths or shapes.

40. AC DC Adaptor: A device that converts alternating current (AC) from a wall outlet into direct current (DC).

41. Plywood: Plywood provides robust structural support and is commonly used for sheathing, roofing, and flooring.

42. Cable Ties: Cable ties are the simplest and most common tool for cable management. They are usually made of plastic or nylon, and have a locking mechanism that allows you to secure and tighten the cables together.

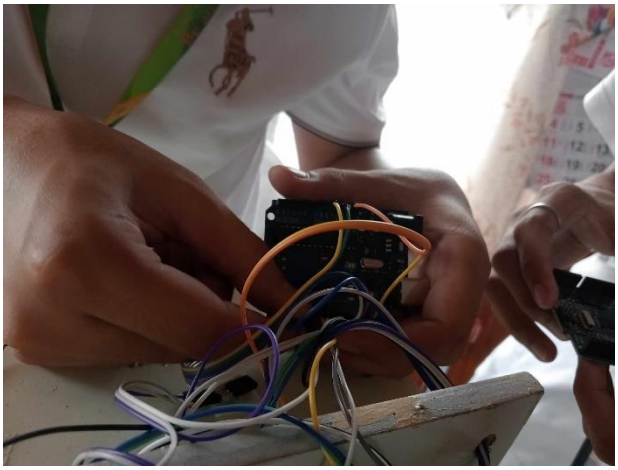
43. Portable Electric Drill: Its an tool for drilling holes in wood and other soft materials.

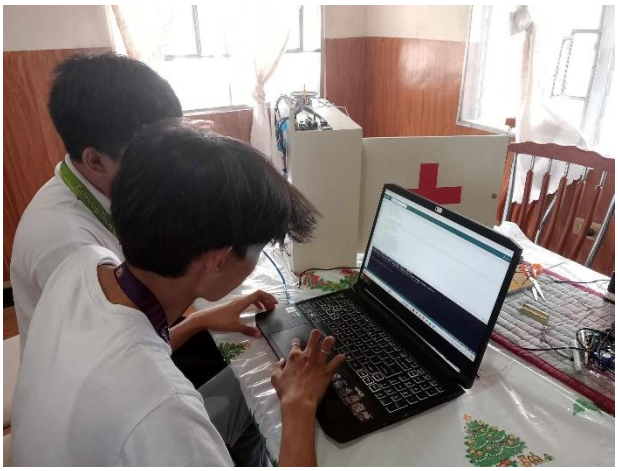
44. Electrical Tape: Electrical tape is a safety tape for wires, used to cover and insulate a broad range of cables, wires and other materials that conduct electricity.

45. Wire Insulation: Covers and holds wires together. It keeps the electrical current contained.

APPENDIX D

METHODS AND OBSERVATION



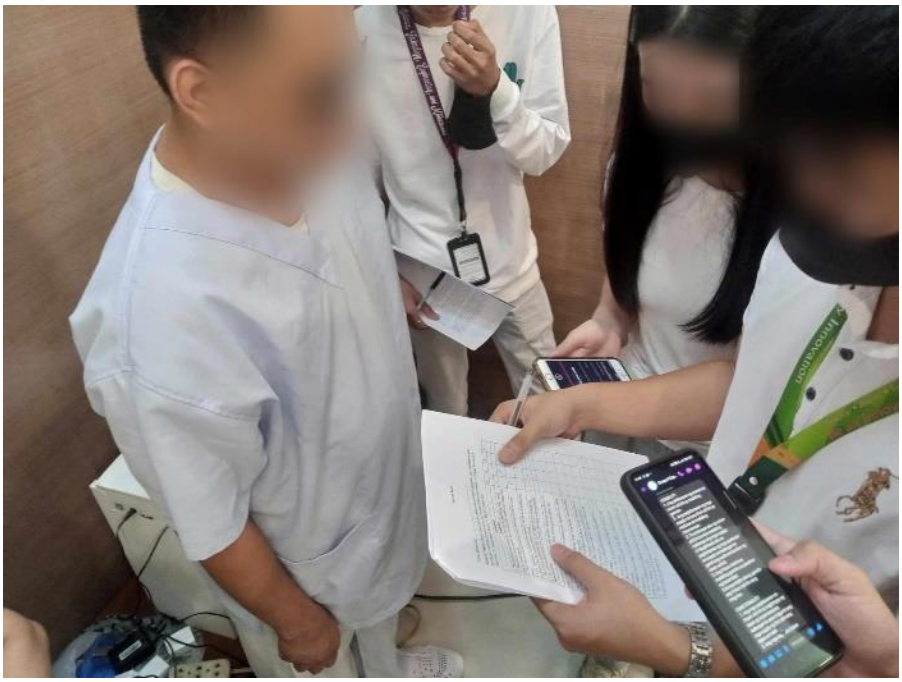


Construction of the Prototype

Preparation of the Vision Aid Simulation

Actual Product Testing

Conduct Semi Interview to Vibes Massage And D.O.C Wellness Massage Officials



Actual Data Gathering

APPENDIX E

INTERVIEW TRANSCRIPTION

1. How do you perceive the benefits of Vision Aid: First Response Kit for the Visually Impaired?

Researcher: Paano mo nakikita ang mga benepisyo ng Vision Aid: First Response Kit para sa mga May Kapansanan sa Paningin?

Field Expert 1: Ang product na ito ay napaka dali lang gamitin at makuha ang mga supply na kailangan ng mga bulag, kapag emergency kasi may nag sasalita kaya alam na alam nila kung ano ang mga kukunin nila sa loob ng first aid kit.

Field Expert 2: Malaking tulong ito sa mga bulag kasi isa ito sa mga problemang kinakaharap nila kapag may kailangan sila at wala silang matatawag para matulungan sila, ang pagbibigay ng sounds kung ano ang kinuha ay malaking tulong para sa kanila.

Fireld Expert 3: Sa totoo lang po ay madali po siyang gamitin kasi nga po may nagbibigay ng sounds kung ano ang kukunin namin, at ito ay napakadaling gamitin para sa mga bulag.

2. What potential drawbacks do you foresee with the Vision Aid: First Response Kit for the Visually Impaired?

Researcher: Anong mga potensyal na kakulangan ang nakikita mo sa VisionAid sa: First Response Kit para sa May Kapansanan sa Paningin?

Field Expert 1: Sa aking nakita at obserbahan ay kailangan niya ng malakas na sounds kapag ito ay kinukunan ng supply sa loob ng First Aid Kit, kasi masyado siyang mahina para saakin. Ayun kasi ang mahalag para sa mga bulag kasi wala silang nakikita, kaya kinakailangna ng sapat na sounds para sa kanilang kaalamn sa pag kuha ng supply sa loon ng First Aid Kit.

Field Expert 2: Una po ay try niyong maglagy ng mga gamut tapus mas lakasan pa ang tunog nito kapag ito ay nagsasalita na.

Field Expert 3: Ang kakulangan na aming nakita ay hindi sapat ang espasyo na mayroon ang isang supply, kagaya nalang pag kunaha ang Agua oxinada na katabi ng alcohol, pag kinuha ang Agua oxinada dahil sa hindi sapat ang space na mayroon silang dalawa ay nagagalaw din ang alcohol.

3. Do you have any recommendations to enhance the product further based on your expertise and experiences?

Researcher: Mayroon ka bang anumang mga rekomendasyon upang mapahusay pa ang produkto batay sa iyong kadalubhasaan at mga karanasan?

Field Expert 1: Ang aking rekomendasyon na ma-ibibigay ay kailangan talaga na mas malakas pa ang sound na maririnig ng bulag, kasi kailangan nila yun kasi hindi sila nakakita ang kanilang kukunin.

Field Expert 2: Siguro wag niyong limitahan ang mga ilalagay, at mas lakasan pa ang talks nito. At mas maganda kung may mga gamot ito, kasi malaking tulong to para sa mga bulag.

Field Expert 3: Bigyan niyo ng sapat na espasyo ang bawat supply kasi malalaki ang mga kamay ng bulag, kasi nag ma-massage sila at mas gaganda pa sana kung ang ilalagay niyo ay yung hindi gaanong basic sa bulag kagaya ng alcohol, betadine at iba pa, ang pwede niyong ilagay na mas mahirap kapain ng bulag ay pera, kasi yun yung mas nahihirapan kami kung magkano ba ang kinuha naming at ang pang huli ay mas maganda sana kung gagamitan niyo ito ng battery kasi nakasaksak lang siya, kasi kung may battery mas mapapakinabangan sa oras ng kalamidad.

APPENDIX F

ARDUINO BASED FIRST AID KIT EQUIPPED WITH A SENSOR-ACTIVATED AUDIO CODES

```
#include "SD.h"
#define SD_ChipSelectPin 10
#include "TMRpcm.h"
#include "SPI.h"

TMRpcm tmrpcm;
const int IR_sensor = 2; // Define IR sensor pin
const int IR_sensor2 = 3;
const int IR_sensor3 = 4;
const int IR_sensor4 = 5;
const int IR_sensor5 = 6;
const int IR_sensor6 = 7;
void setup()
{
pinMode(IR_sensor, INPUT); // Set IR sensor pin as input
pinMode(IR_sensor2, INPUT);
pinMode(IR_sensor3, INPUT);
pinMode(IR_sensor4, INPUT);
pinMode(IR_sensor5, INPUT);
pinMode(IR_sensor6, INPUT);
tmrpcm.speakerPin=9;
Serial.begin(9600);
if(!SD.begin(SD_ChipSelectPin))
{
  Serial.println("SD fail");
```

```
  return;
}



}

void loop() {


 int sensorValue = digitalRead(IR_sensor); // Read IR sensor value
 int sensorValue2 = digitalRead(IR_sensor2); // Read IR sensor value
 int sensorValue3 = digitalRead(IR_sensor3); // Read IR sensor value
 int sensorValue4 = digitalRead(IR_sensor4); // Read IR sensor value
 int sensorValue5 = digitalRead(IR_sensor5); // Read IR sensor value
 int sensorValue6 = digitalRead(IR_sensor6); // Read IR sensor value




  // put your main code here, to run repeatedly:
  if(sensorValue>=1){
tmrpcm.setVolume(5);
tmrpcm.play("beta.wav");
 Serial.println("BETADINE");
 Serial.println(sensorValue);
delay(2000);
  }
    if(sensorValue2>=1){
tmrpcm.setVolume(5);
```

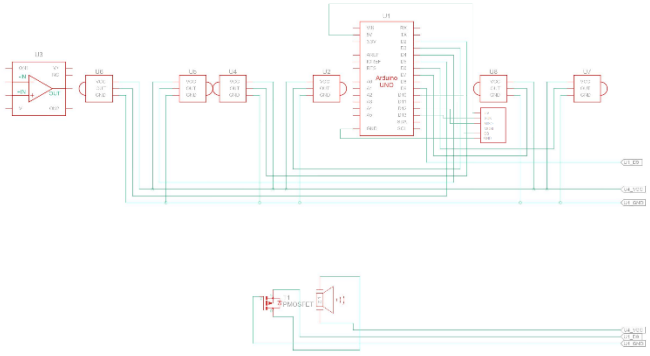
```
tmrpcm.play("cotton_1.wav");
 Serial.println("COTTON");
 Serial.println(sensorValue2);
delay(2000);
  }
    if(sensorValue3>=1){
tmrpcm.setVolume(5);
tmrpcm.play("band.wav");
 Serial.println("BANDAID");
 Serial.println(sensorValue3);
delay(2000);
  }

      if(sensorValue4>=1){
tmrpcm.setVolume(5);
tmrpcm.play("guaze_1.wav");
 Serial.println("GAUZE");
 Serial.println(sensorValue4);
delay(2000);
  }
        if(sensorValue5>=1){
tmrpcm.setVolume(5);
tmrpcm.play("agua_1.wav");
 Serial.println("AGUA");
 Serial.println(sensorValue5);
delay(2000);
  }
          if(sensorValue6>=1){
tmrpcm.setVolume(5);
tmrpcm.play("alco.wav");
```

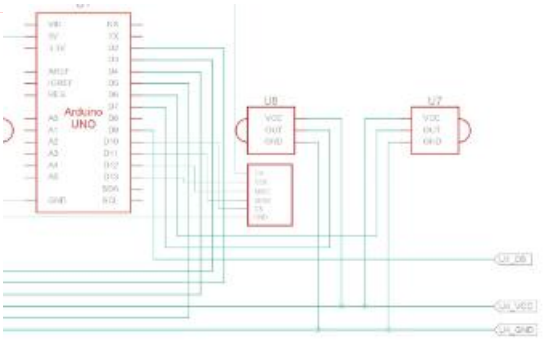
```
 Serial.println("ALCOHOL");
 Serial.println(sensorValue6);
delay(5000);
  }
   Serial.println(sensorValue);
    Serial.println(sensorValue2);
     Serial.println(sensorValue3);
      Serial.println(sensorValue4);
       Serial.println(sensorValue5);
        Serial.println(sensorValue6);
// Print IR sensor value to serial monitor
 delay(100); // Delay 100ms
```

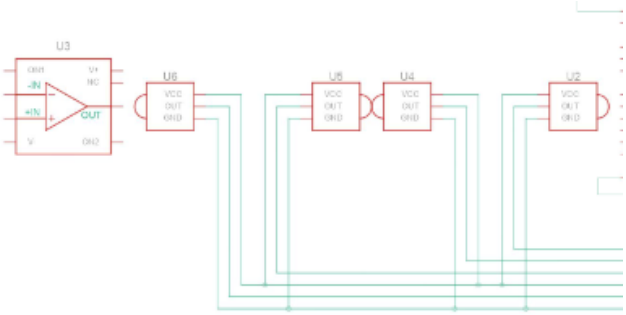
}

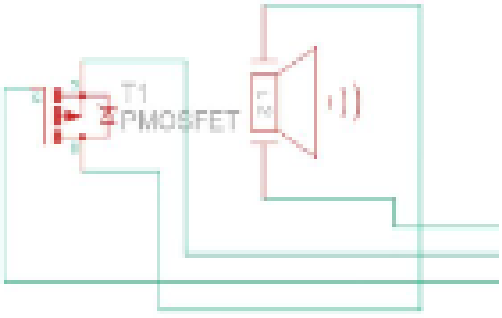
APPENDIX G

SCHEMATIC DIAGRAM



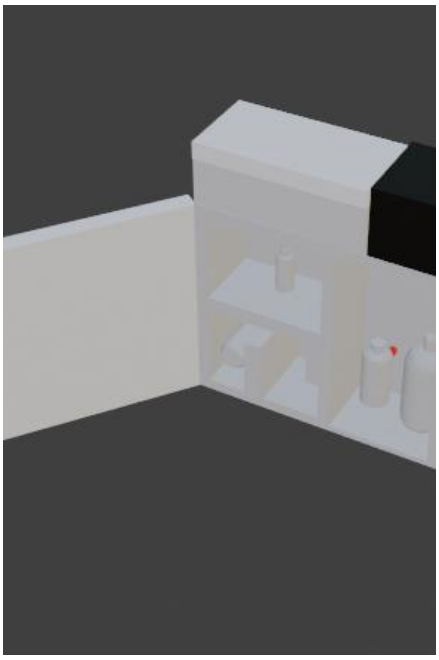


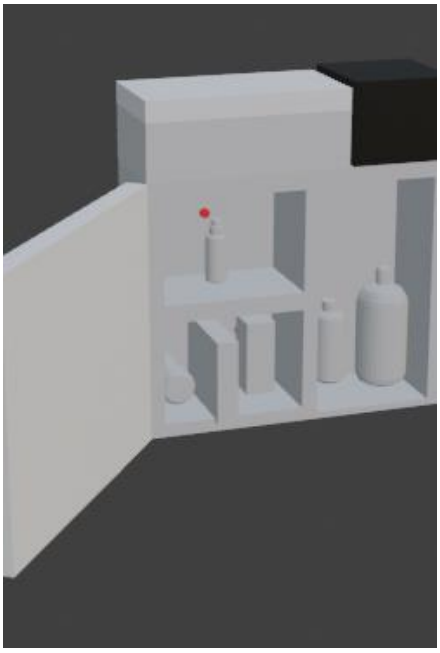


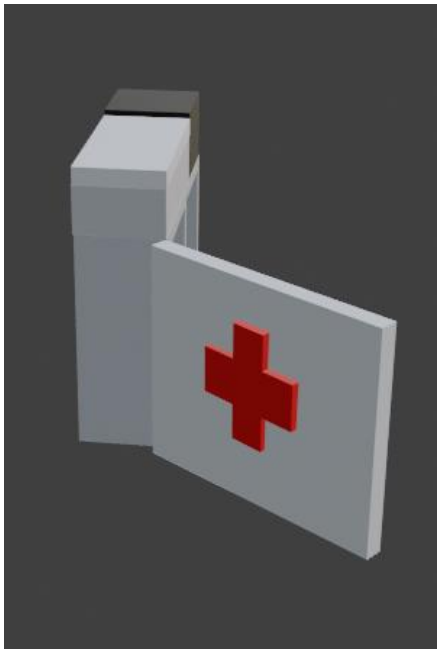


APPENDIX H

PROTOTYPE DESIGN







APPENDIX I

STATISTICAL COMPUTATION

Indicators	4	3	2	1	W.M	V.I
Usability						
1. The sensor activation on the Vision Aid Kit is easy to use.	8	4	0	0	3.67	Highly Satisfied
2. The audio descriptions for the first aid supplies are clear and easy to understand.	9	2	1	0	3.67	Highly Satisfied
3. The Vision Aid Kit helped me to find the supplies more easily.	9	2	1	0	3.67	Highly Satisfied
4. I am confident of finding the specific supplies I need using the Vision Aid Kit's audio instructions.	10	1	1	0	3.75	Highly Satisfied
5. The Vision Aid Kit is easy to navigate in an emergency.	6	5	1	0	3.42	Highly Satisfied
6. I am comfortable using the Vision Aid Kit independently.	7	5	0	0	3.58	Highly Satisfied
General Weighted Mean	**3.63**					**Highly Satisfied**
Functionality						
1. The audio descriptions in the Vision Aid Kit provide all the information I need to use the first aid supplies safely and effectively.	7	5	0	0	3.58	Highly Satisfied
2. The voice instructions are accurate and reliable in guiding me through each supply.	7	4	1	0	3.42	Highly Satisfied
3. The Vision Aid Kit features seem helpful for common first aid situations I might encounter.	10	2	0	0	3.83	Highly Satisfied
4. I believe the Vision Aid Kit allowed me to provide basic first aid assistance to myself or others in an emergency.	10	2	0	0	3.83	Highly Satisfied
5. I am confident in the Vision Aid Kit's ability to help me manage a first aid situation.	9	3	0	0	3.75	Highly Satisfied
6. The Vision Aid overall performance is in great level.	7	5	0	0	3.58	Highly Satisfied
General Weighted Mean	**3.67**					**Highly Satisfied**
Product Efficiency						
1. The sensor on the Vision Aid Kit responds quickly and accurately to my touch.	8	4	0	0	3.67	Highly Satisfied
2. The audio instructions play smoothly without delays or interruptions.	9	3	0	0	3.75	Highly Satisfied

3. The battery life of the Vision Aid Kit seems enough for an emergency.	7	4	1	0	3.42	Highly Satisfied
4. I feel confident troubleshooting any minor technical issues with the Vision Aid Kit.	7	4	1	0	3.42	Highly Satisfied
5. The Vision Aid Kit seems reliable and functions efficiently in emergency situations.	10	2	0	0	3.83	Highly Satisfied
General Weighted Mean	**3.62**					**Highly Satisfied**

Interpretation: (according to Rosario & Reyes, 2021)

3.25-4.00 – Highly Satisfied

2.50-3.24 – Satisfied

1.75-2.49 – Unsatisfied

1.00-1.74 – Highly Unsatisfied

APPENDIX J

CERTICATE OF STATISTICIAN

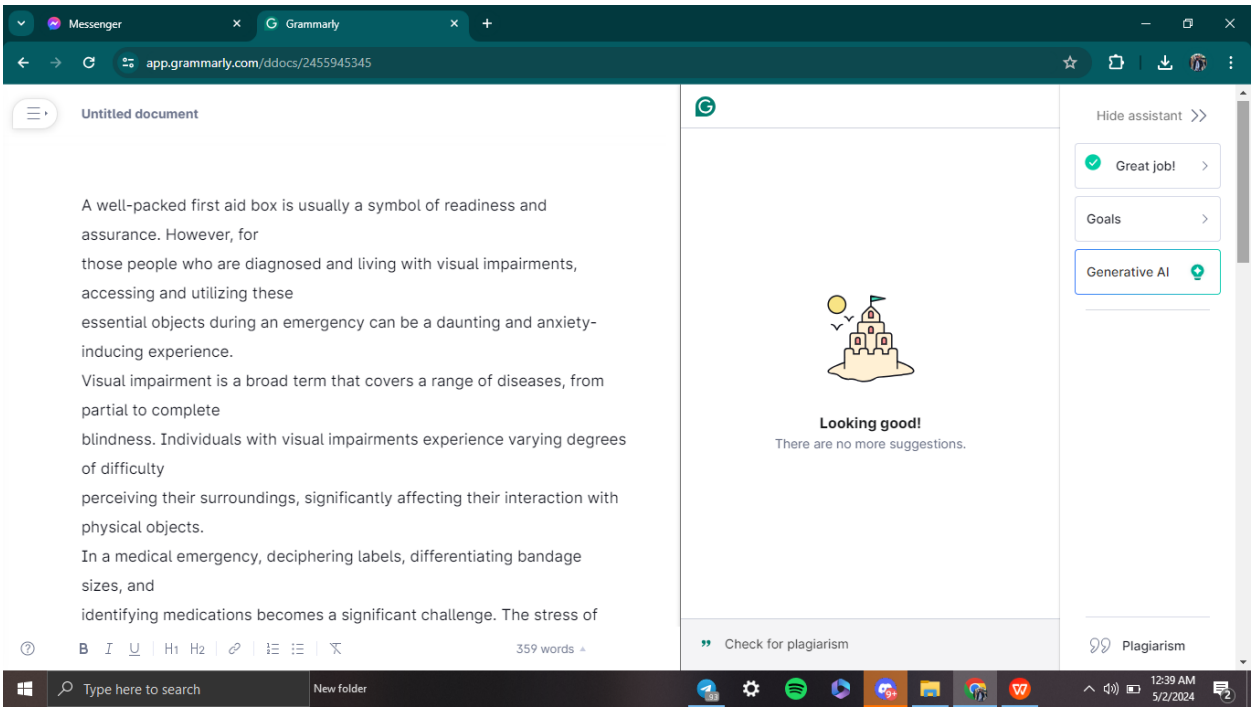
CERTIFICATION OF STATISTICIAN

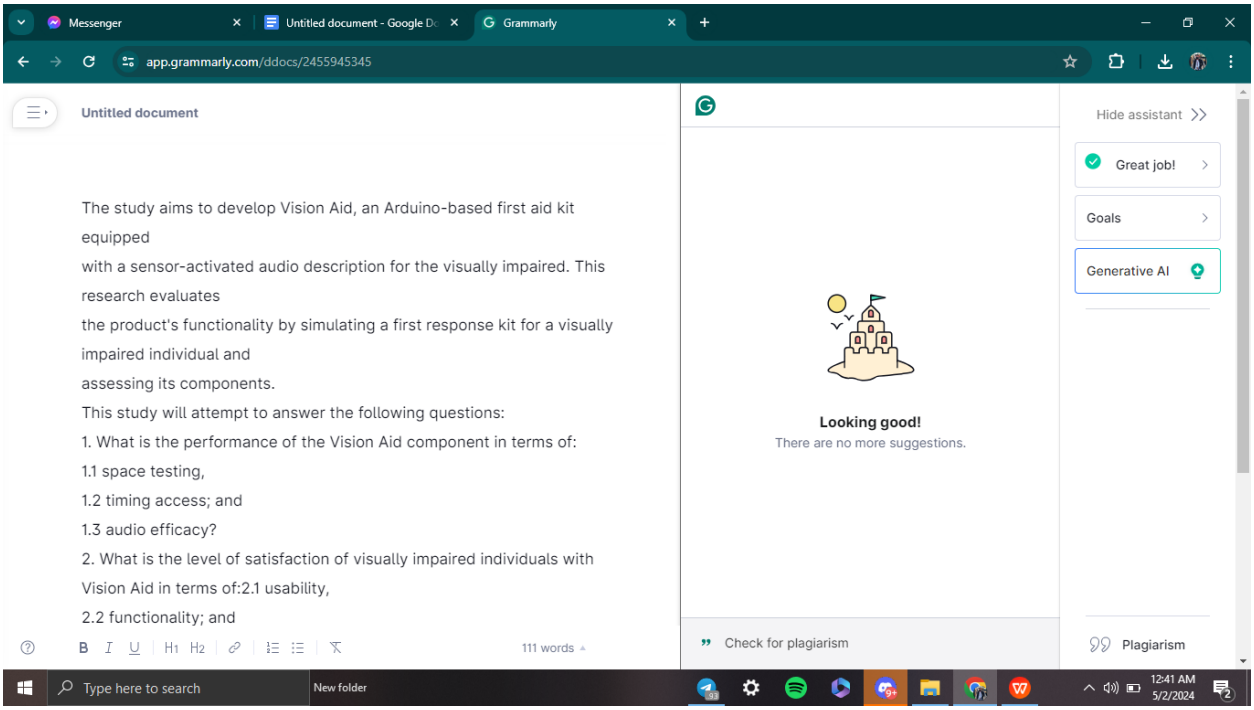
This certifies that the research paper with the title **"Vision Aid: First Response Kit for Visually Impaired"** which was done by Christopher John H. Cruz and Mirko R. Barcellano, Phillip James L. Caricungan, Clarenz L. Delacruz, Shafalie G. Dela Torre, Crisalyn D. Espinar, Argie G. Jaldo, Kyan Ace C. Lungsod, Julie Mae G. Satimbre, James l. Serilla, John Francis V. Vecina of Malinta National High School was statistically reviewed and corrected by the undersigned using the initial data set that the student-researcher provided.

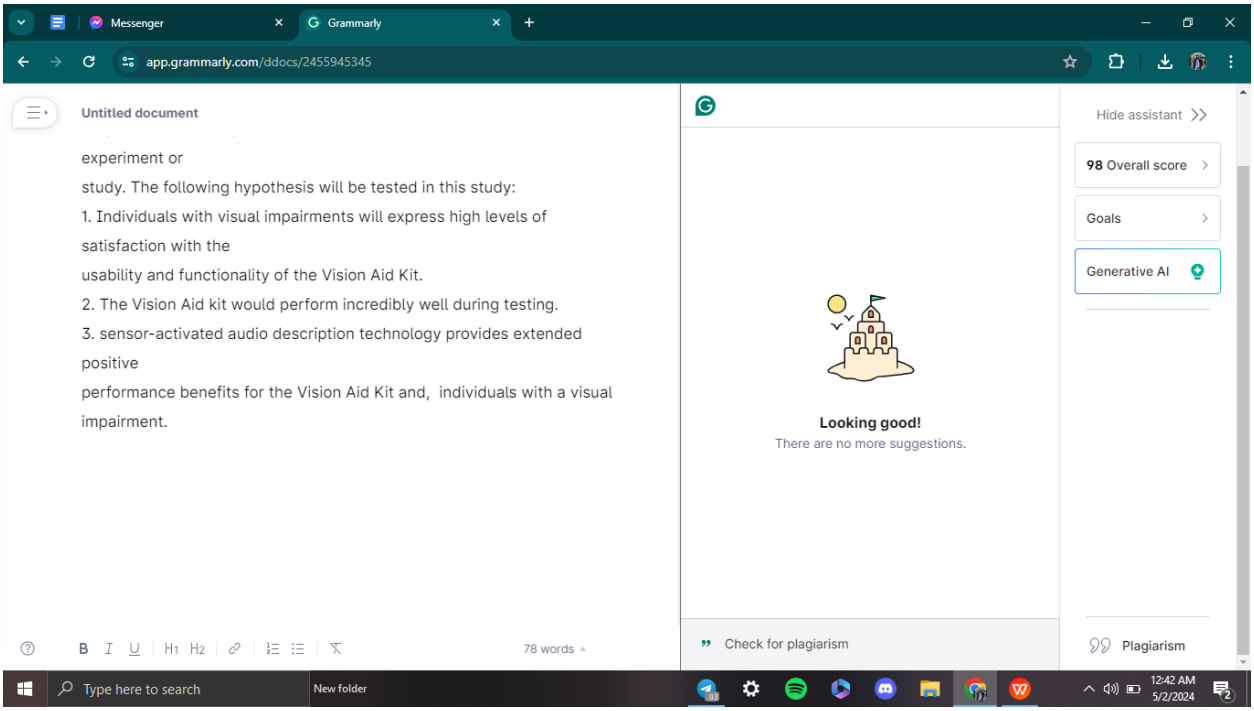
Ponzi Anthony C. Ray Del Val
Statistician
Master Teacher 1

APPENDIX K

RESULTS OF GRAMMAR TEST



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experiment or
study. The following hypothesis will be tested in this study:
1. Individuals with visual impairments will express high levels of satisfaction with the
usability and functionality of the Vision Aid Kit.
2. The Vision Aid kit would perform incredibly well during testing.
3. sensor-activated audio description technology provides extended positive
performance benefits for the Vision Aid Kit and, individuals with a visual impairment.
78 words
Hide assistant
98 Overall score
Goals
Generative AI
Looking good!
There are no more suggestions.
Check for plagiarism
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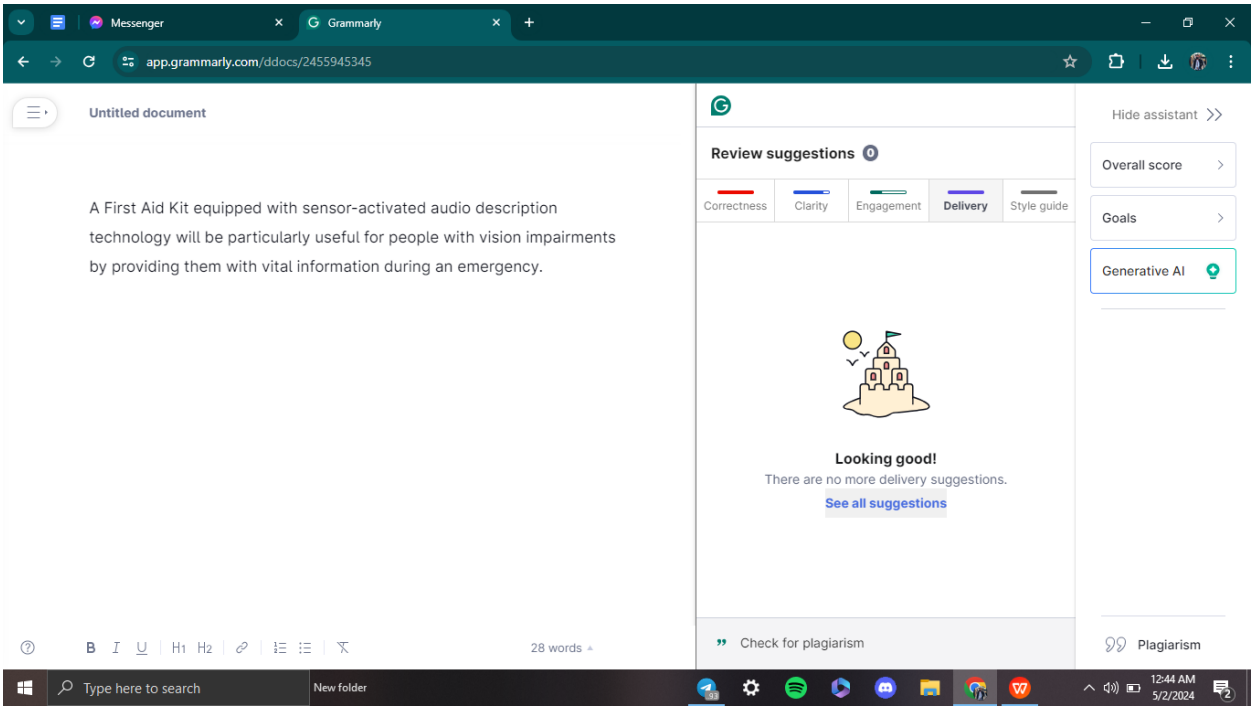
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A First Aid Kit equipped with sensor-activated audio description technology will be particularly useful for people with vision impairments by providing them with vital information during an emergency.
28 words
Review suggestions
0
Correctness
Clarity
Engagement
Delivery
Style guide
Hide assistant
Overall score
Goals
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Looking good!
There are no more delivery suggestions.
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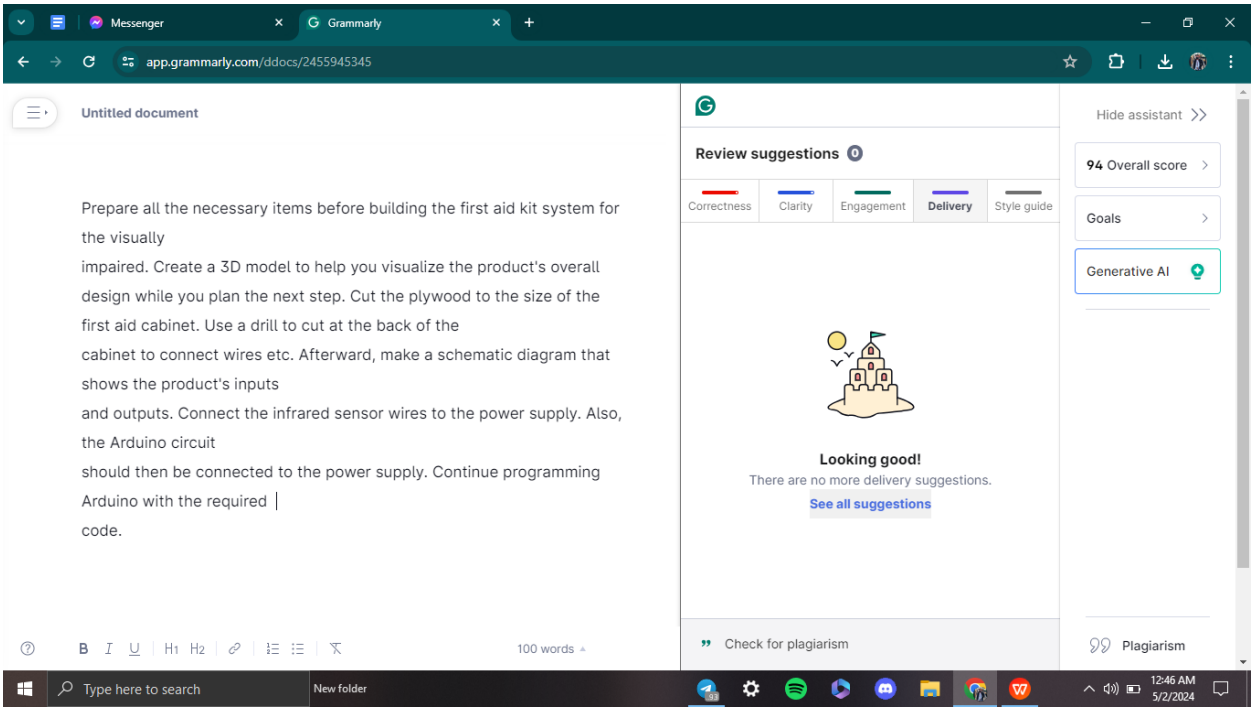
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Prepare all the necessary items before building the first aid kit system for the visually
impaired. Create a 3D model to help you visualize the product's overall design while you plan the next step. Cut the plywood to the size of the first aid cabinet. Use a drill to cut at the back of the
cabinet to connect wires etc. Afterward, make a schematic diagram that shows the product's inputs
and outputs. Connect the infrared sensor wires to the power supply. Also, the Arduino circuit
should then be connected to the power supply. Continue programming Arduino with the required
code.
100 words
Review suggestions 0
Correctness
Clarity
Engagement
Delivery
Style guide
Looking good!
There are no more delivery suggestions.
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Check for plagiarism
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94 Overall score
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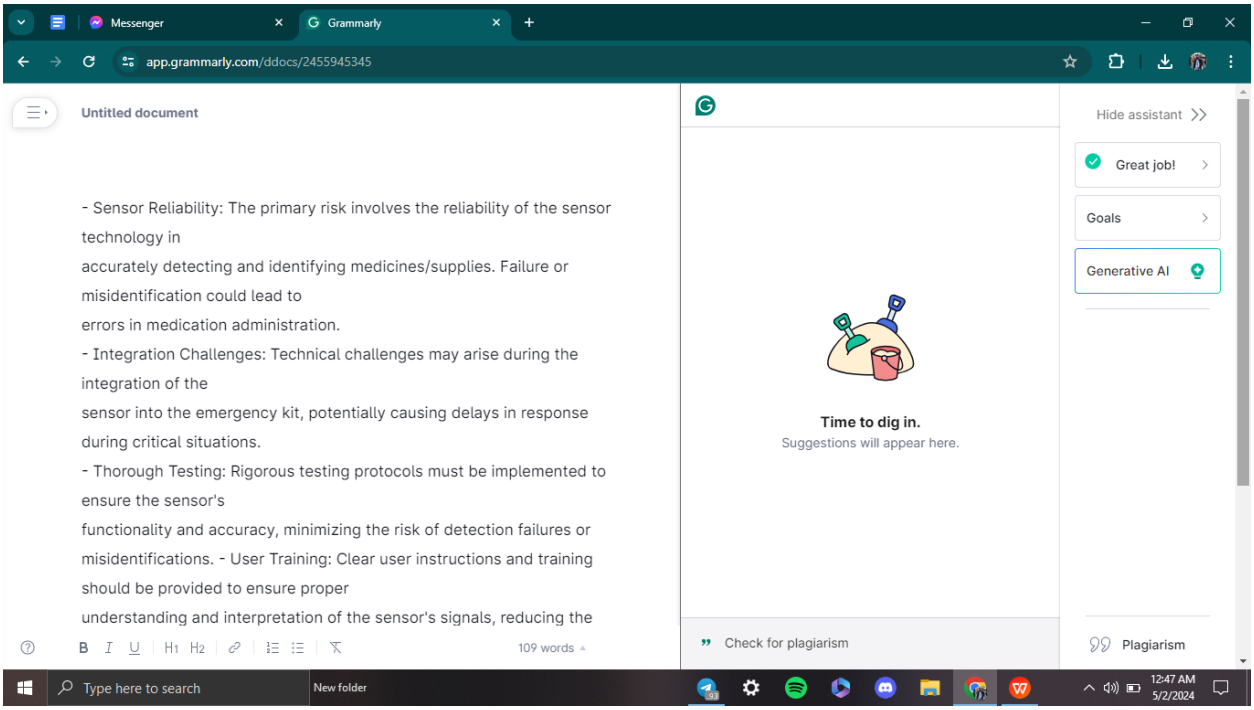
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- Sensor Reliability: The primary risk involves the reliability of the sensor technology in
accurately detecting and identifying medicines/supplies. Failure or misidentification could lead to
errors in medication administration.
- Integration Challenges: Technical challenges may arise during the integration of the
sensor into the emergency kit, potentially causing delays in response during critical situations.
- Thorough Testing: Rigorous testing protocols must be implemented to ensure the sensor's
functionality and accuracy, minimizing the risk of detection failures or misidentifications. - User Training: Clear user instructions and training should be provided to ensure proper
understanding and interpretation of the sensor's signals, reducing the
109 words
Time to dig in.
Suggestions will appear here.
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Great job!
Goals
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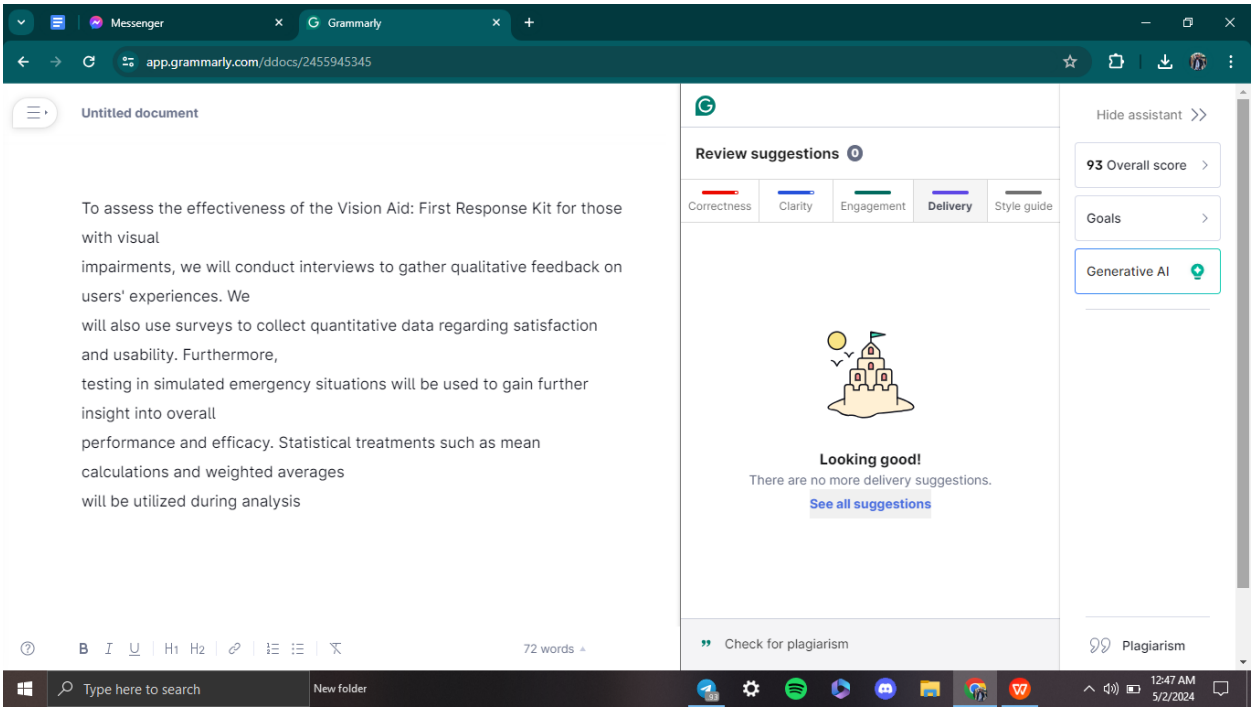
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To assess the effectiveness of the Vision Aid: First Response Kit for those with visual
impairments, we will conduct interviews to gather qualitative feedback on users' experiences. We
will also use surveys to collect quantitative data regarding satisfaction and usability. Furthermore,
testing in simulated emergency situations will be used to gain further insight into overall
performance and efficacy. Statistical treatments such as mean calculations and weighted averages
will be utilized during analysis
72 words
Review suggestions 0
Correctness
Clarity
Engagement
Delivery
Style guide
Looking good!
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See all suggestions
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Hide assistant
93 Overall score
Goals
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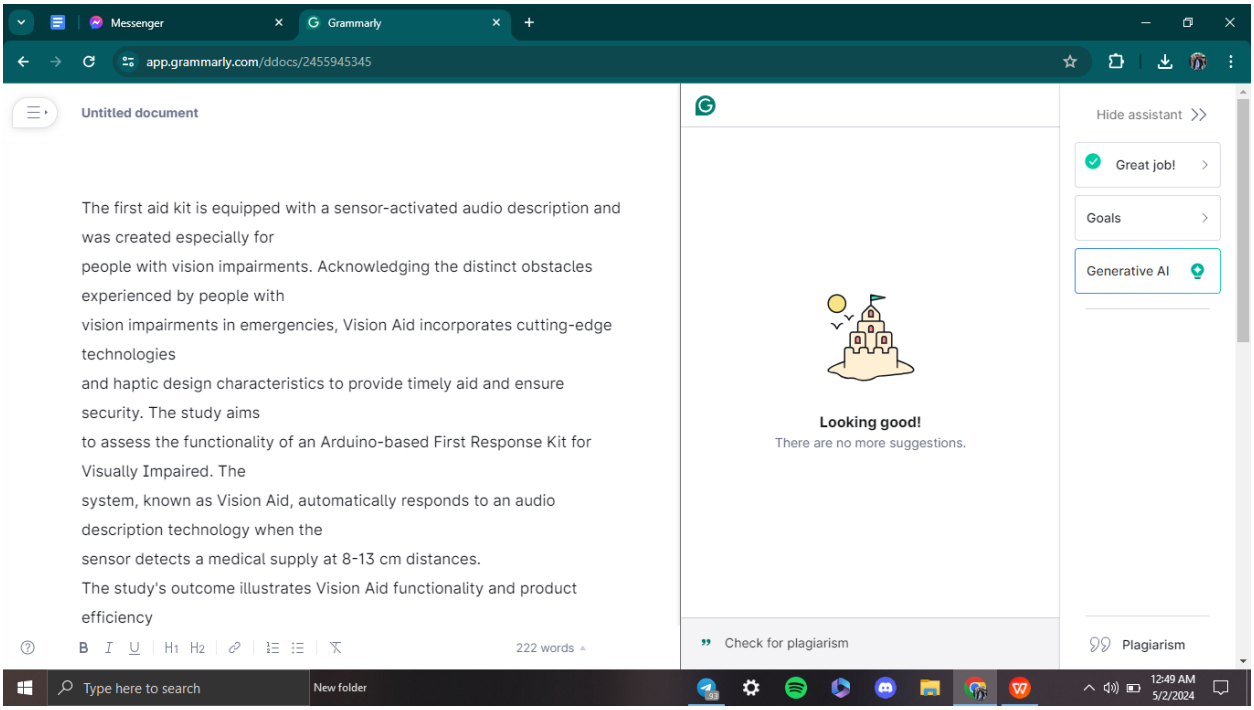
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Untitled document
The first aid kit is equipped with a sensor-activated audio description and was created especially for
people with vision impairments. Acknowledging the distinct obstacles experienced by people with
vision impairments in emergencies, Vision Aid incorporates cutting-edge technologies
and haptic design characteristics to provide timely aid and ensure security. The study aims
to assess the functionality of an Arduino-based First Response Kit for Visually Impaired. The
system, known as Vision Aid, automatically responds to an audio description technology when the
sensor detects a medical supply at 8-13 cm distances.
The study's outcome illustrates Vision Aid functionality and product efficiency
222 words
Looking good!
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Great job!
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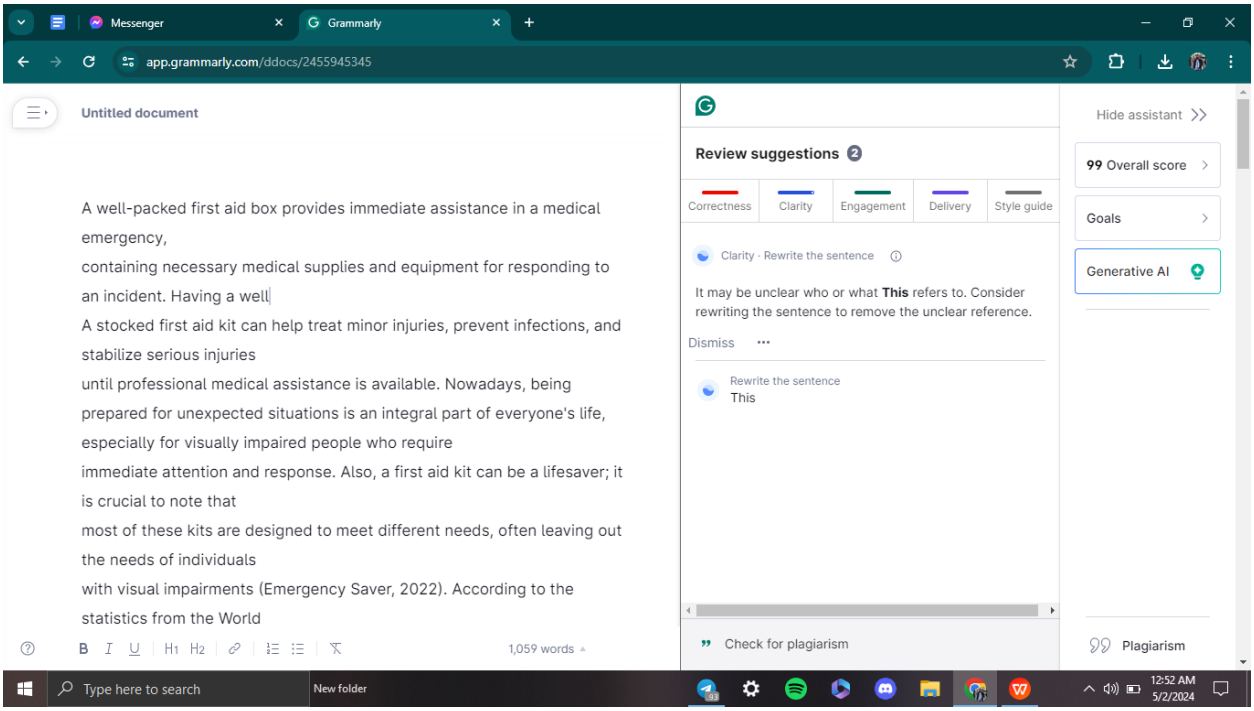
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A well-packed first aid box provides immediate assistance in a medical emergency,
containing necessary medical supplies and equipment for responding to an incident. Having a well
A stocked first aid kit can help treat minor injuries, prevent infections, and stabilize serious injuries
until professional medical assistance is available. Nowadays, being prepared for unexpected situations is an integral part of everyone's life, especially for visually impaired people who require
immediate attention and response. Also, a first aid kit can be a lifesaver; it is crucial to note that
most of these kits are designed to meet different needs, often leaving out the needs of individuals
with visual impairments (Emergency Saver, 2022). According to the statistics from the World
1,059 words
Review suggestions 2
Correctness
Clarity
Engagement
Delivery
Style guide
Clarity · Rewrite the sentence
It may be unclear who or what This refers to. Consider rewriting the sentence to remove the unclear reference.
Dismiss
Rewrite the sentence
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99 Overall score
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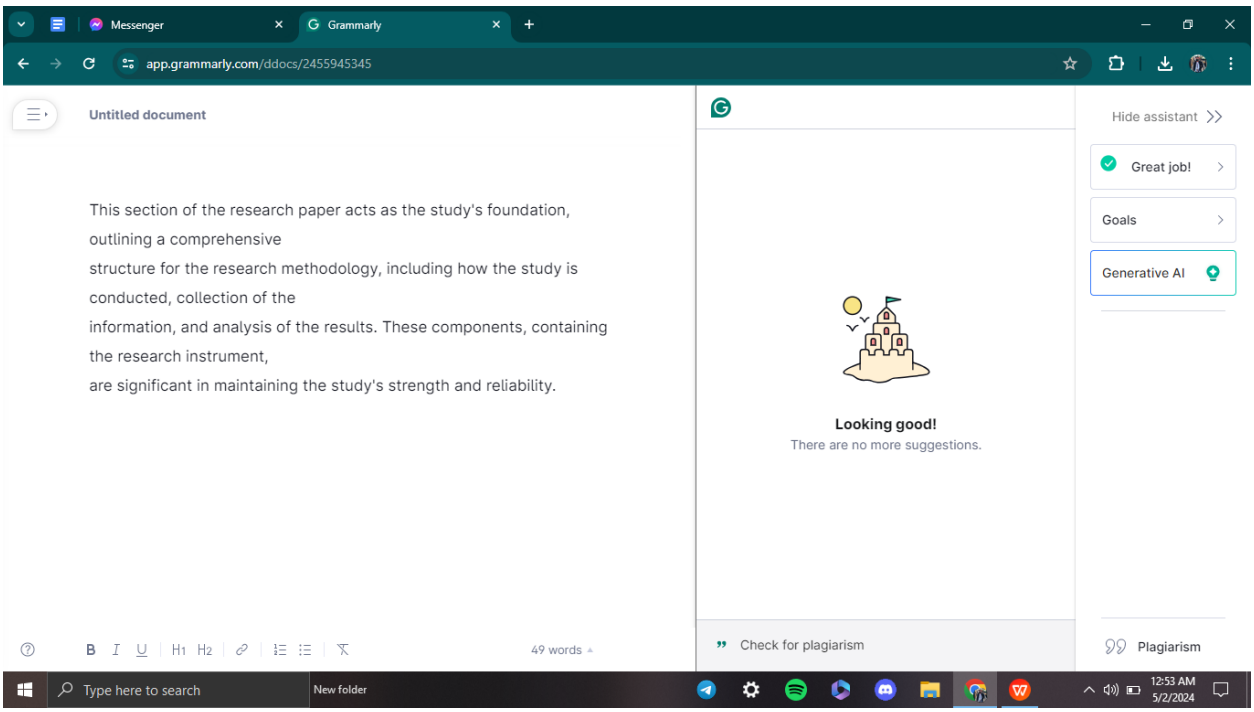
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This section of the research paper acts as the study's foundation, outlining a comprehensive
structure for the research methodology, including how the study is conducted, collection of the
information, and analysis of the results. These components, containing the research instrument,
are significant in maintaining the study's strength and reliability.
49 words
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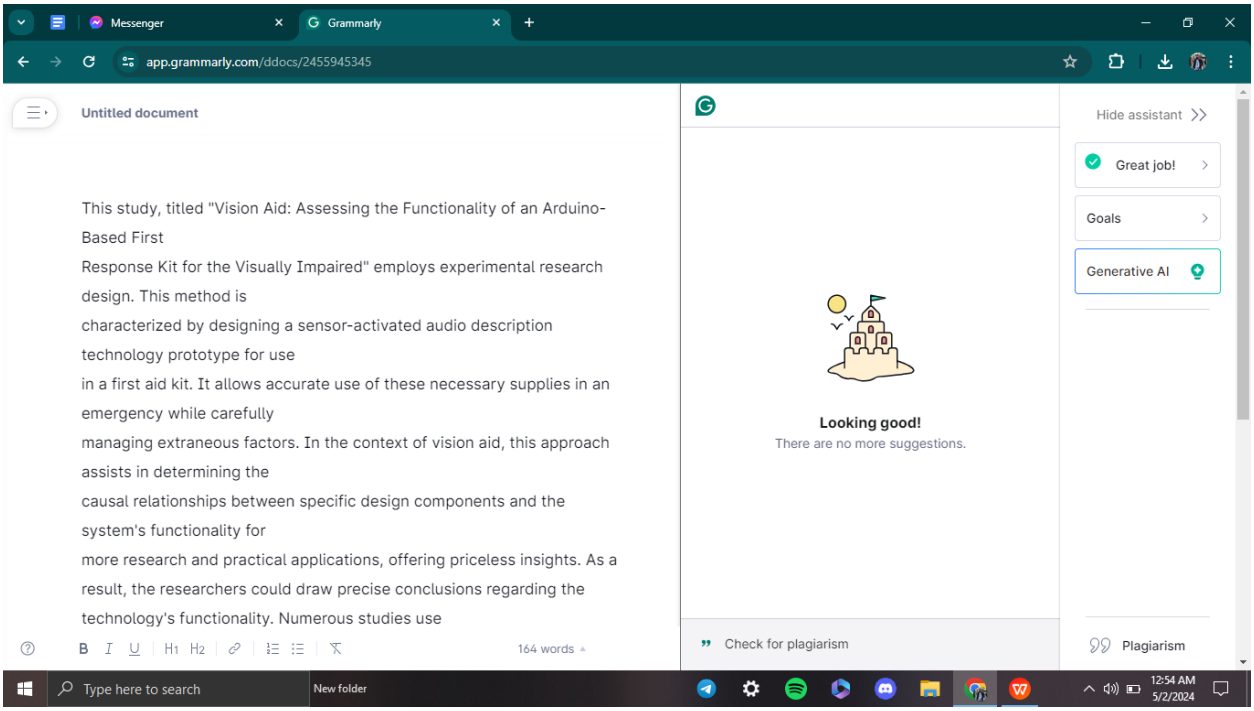
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This study, titled "Vision Aid: Assessing the Functionality of an Arduino-Based First
Response Kit for the Visually Impaired" employs experimental research design. This method is
characterized by designing a sensor-activated audio description technology prototype for use
in a first aid kit. It allows accurate use of these necessary supplies in an emergency while carefully
managing extraneous factors. In the context of vision aid, this approach assists in determining the
causal relationships between specific design components and the system's functionality for
more research and practical applications, offering priceless insights. As a result, the researchers could draw precise conclusions regarding the technology's functionality. Numerous studies use
164 words
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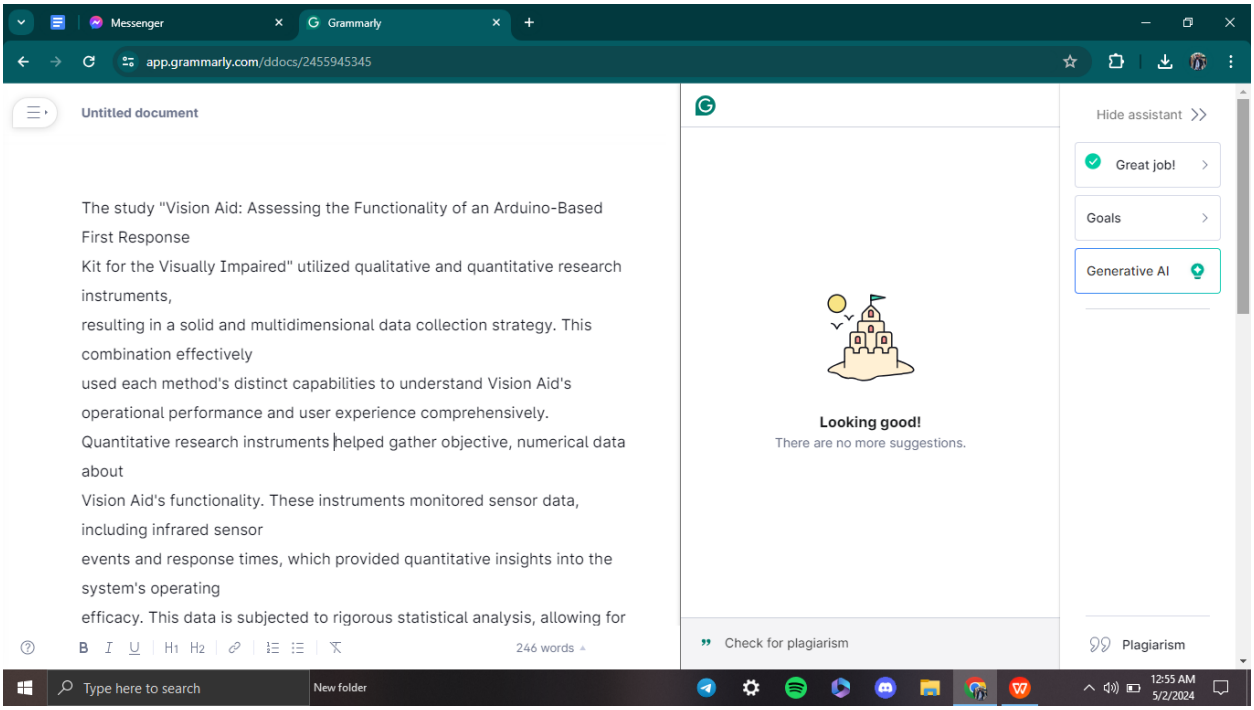
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The study "Vision Aid: Assessing the Functionality of an Arduino-Based First Response
Kit for the Visually Impaired" utilized qualitative and quantitative research instruments,
resulting in a solid and multidimensional data collection strategy. This combination effectively
used each method's distinct capabilities to understand Vision Aid's operational performance and user experience comprehensively.
Quantitative research instruments helped gather objective, numerical data about
Vision Aid's functionality. These instruments monitored sensor data, including infrared sensor
events and response times, which provided quantitative insights into the system's operating
efficacy. This data is subjected to rigorous statistical analysis, allowing for
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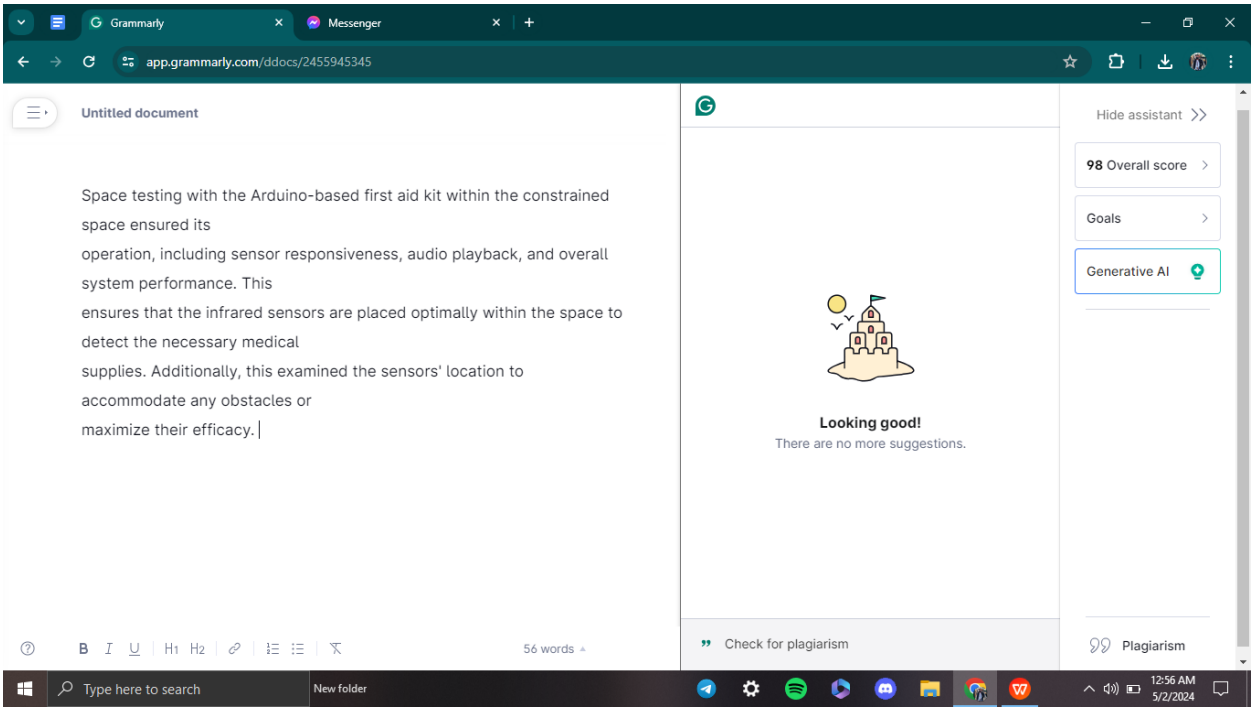
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Space testing with the Arduino-based first aid kit within the constrained space ensured its
operation, including sensor responsiveness, audio playback, and overall system performance. This
ensures that the infrared sensors are placed optimally within the space to detect the necessary medical
supplies. Additionally, this examined the sensors' location to accommodate any obstacles or
maximize their efficacy.
56 words
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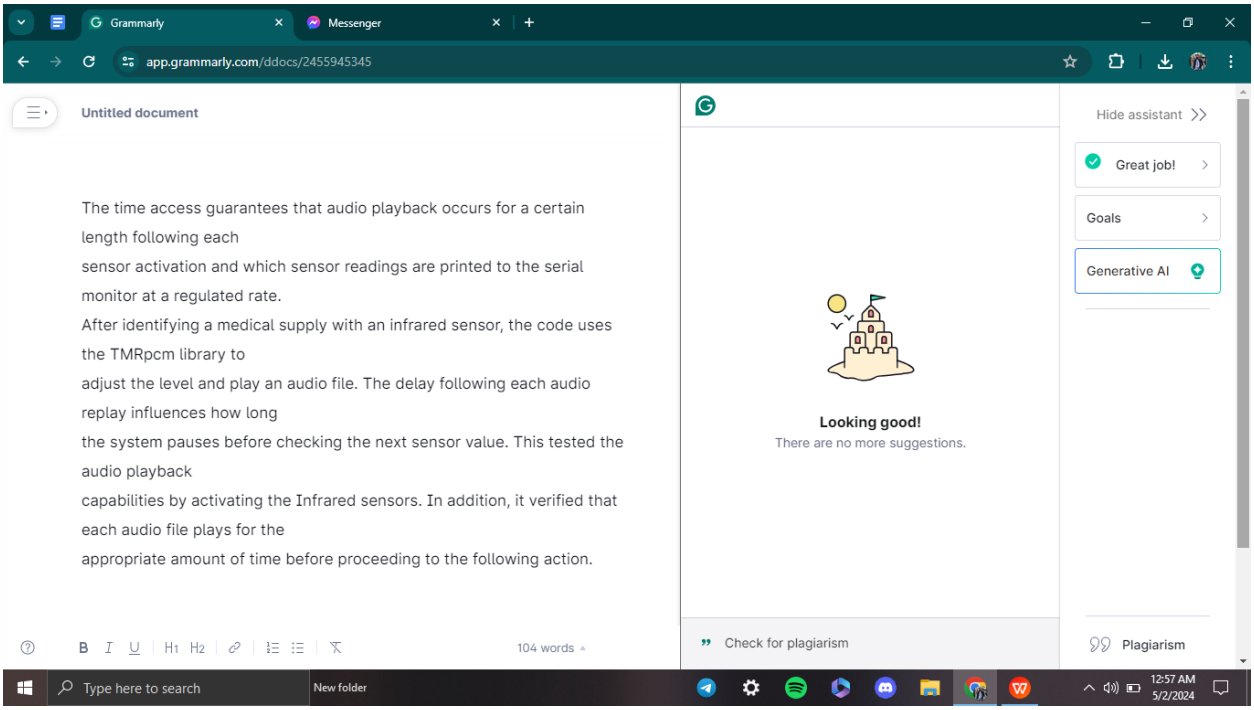
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The time access guarantees that audio playback occurs for a certain length following each
sensor activation and which sensor readings are printed to the serial monitor at a regulated rate.
After identifying a medical supply with an infrared sensor, the code uses the TMRpcm library to
adjust the level and play an audio file. The delay following each audio replay influences how long
the system pauses before checking the next sensor value. This tested the audio playback
capabilities by activating the Infrared sensors. In addition, it verified that each audio file plays for the
appropriate amount of time before proceeding to the following action.
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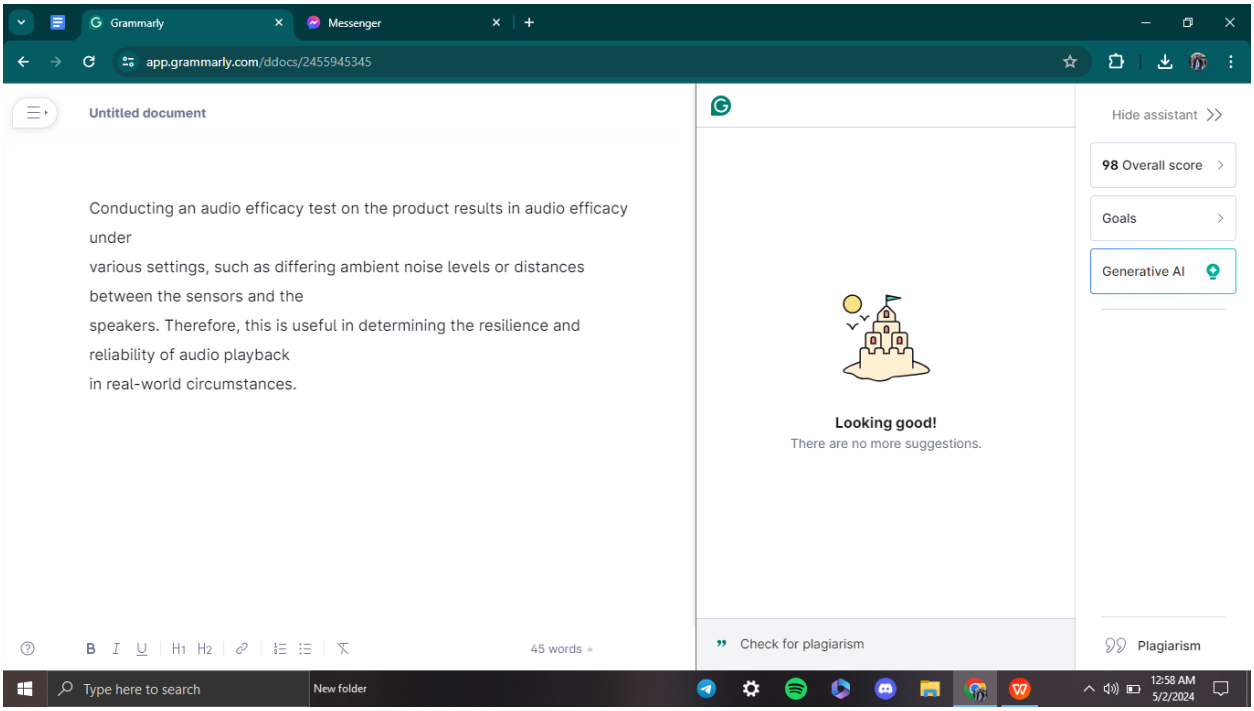
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Conducting an audio efficacy test on the product results in audio efficacy under
various settings, such as differing ambient noise levels or distances between the sensors and the
speakers. Therefore, this is useful in determining the resilience and reliability of audio playback
in real-world circumstances.
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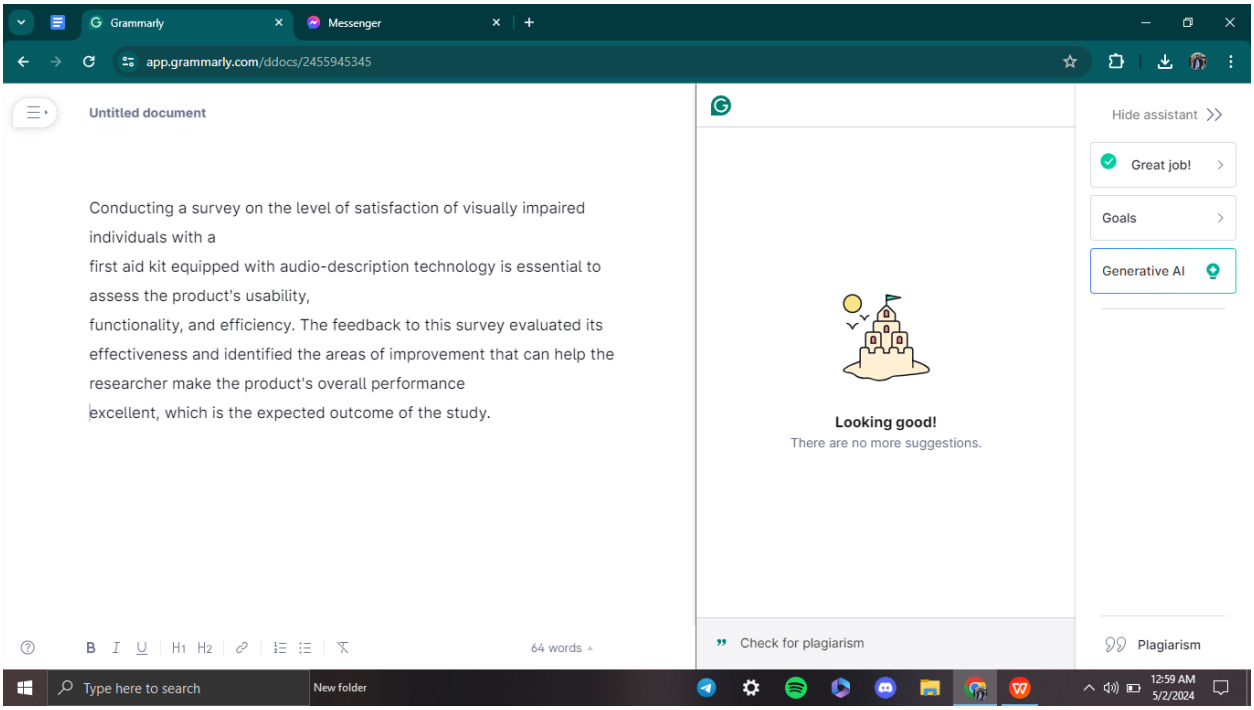
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Conducting a survey on the level of satisfaction of visually impaired individuals with a
first aid kit equipped with audio-description technology is essential to assess the product's usability,
functionality, and efficiency. The feedback to this survey evaluated its effectiveness and identified the areas of improvement that can help the researcher make the product's overall performance
excellent, which is the expected outcome of the study.
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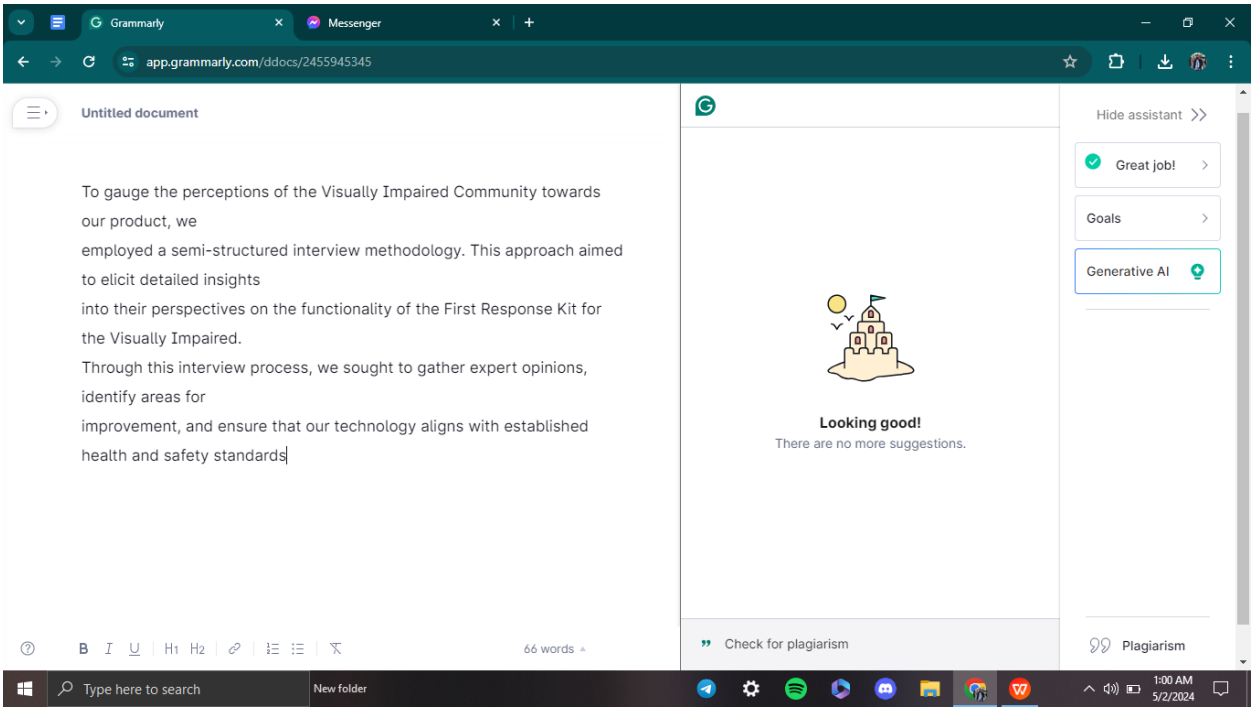
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To gauge the perceptions of the Visually Impaired Community towards our product, we
employed a semi-structured interview methodology. This approach aimed to elicit detailed insights
into their perspectives on the functionality of the First Response Kit for the Visually Impaired.
Through this interview process, we sought to gather expert opinions, identify areas for
improvement, and ensure that our technology aligns with established health and safety standards
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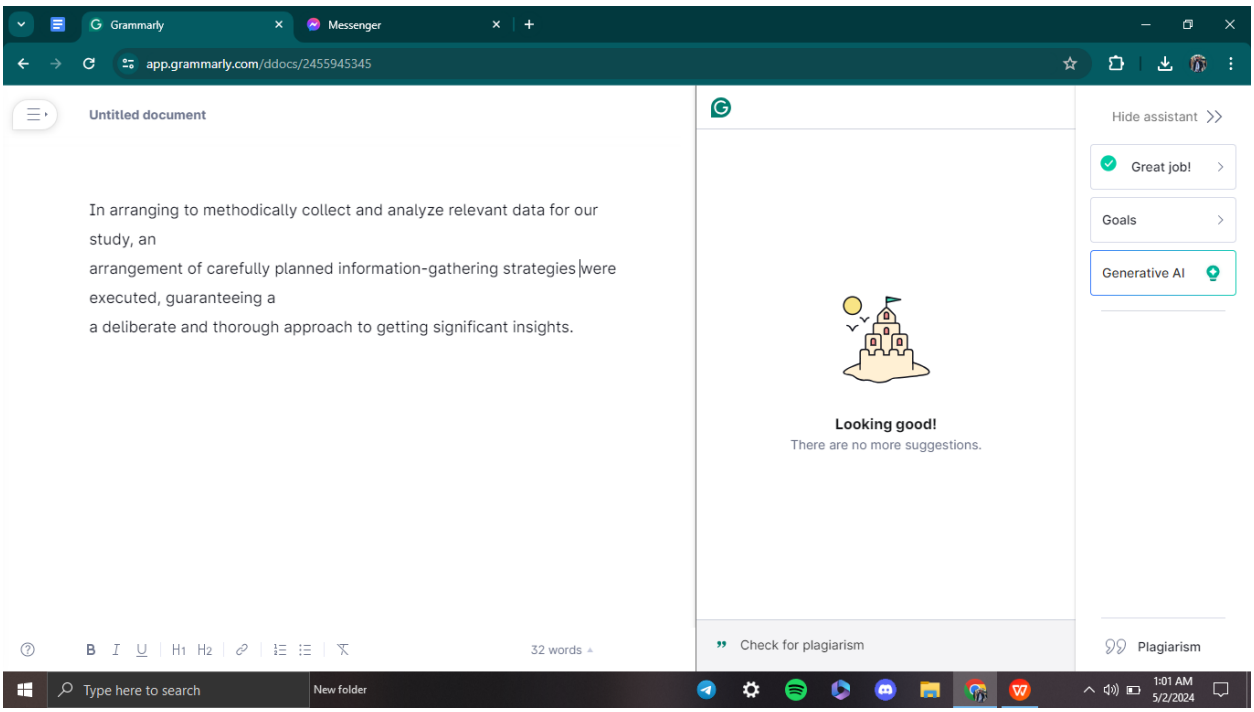
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In arranging to methodically collect and analyze relevant data for our study, an
arrangement of carefully planned information-gathering strategies were executed, guaranteeing a
a deliberate and thorough approach to getting significant insights.
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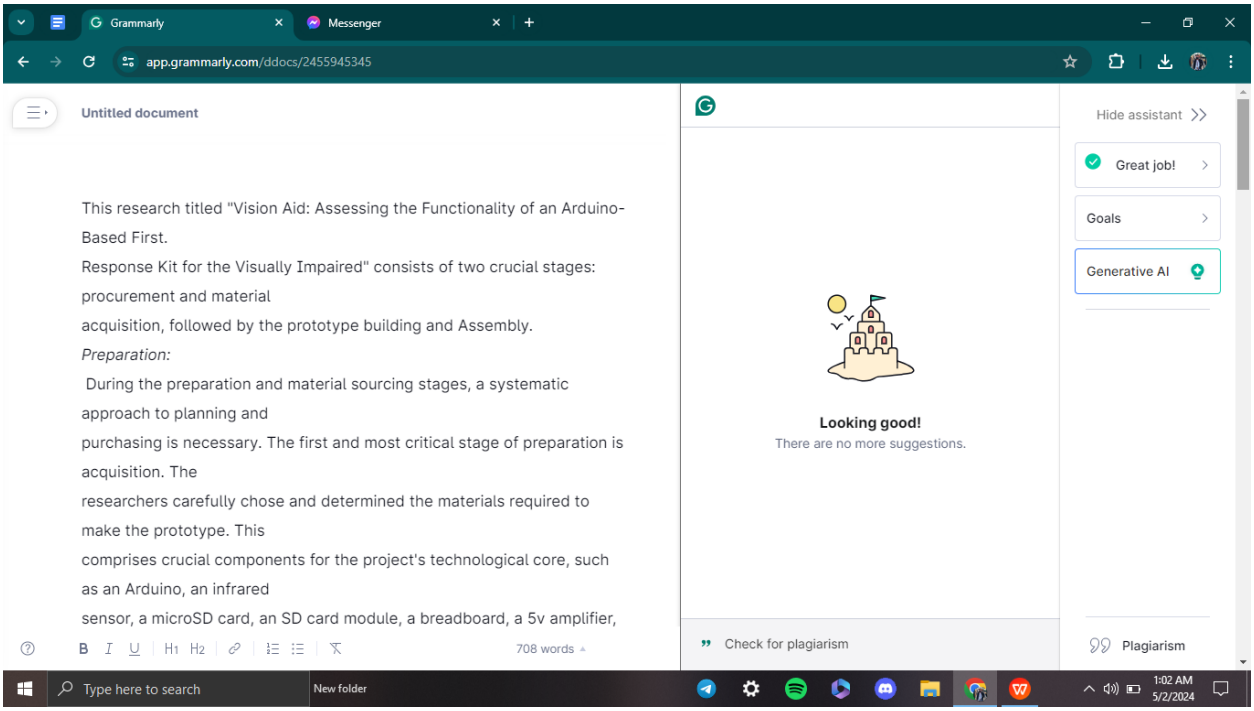
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This research titled "Vision Aid: Assessing the Functionality of an Arduino-Based First.
Response Kit for the Visually Impaired" consists of two crucial stages: procurement and material
acquisition, followed by the prototype building and Assembly.
Preparation:
During the preparation and material sourcing stages, a systematic approach to planning and
purchasing is necessary. The first and most critical stage of preparation is acquisition. The
researchers carefully chose and determined the materials required to make the prototype. This
comprises crucial components for the project's technological core, such as an Arduino, an infrared
sensor, a microSD card, an SD card module, a breadboard, a 5v amplifier,
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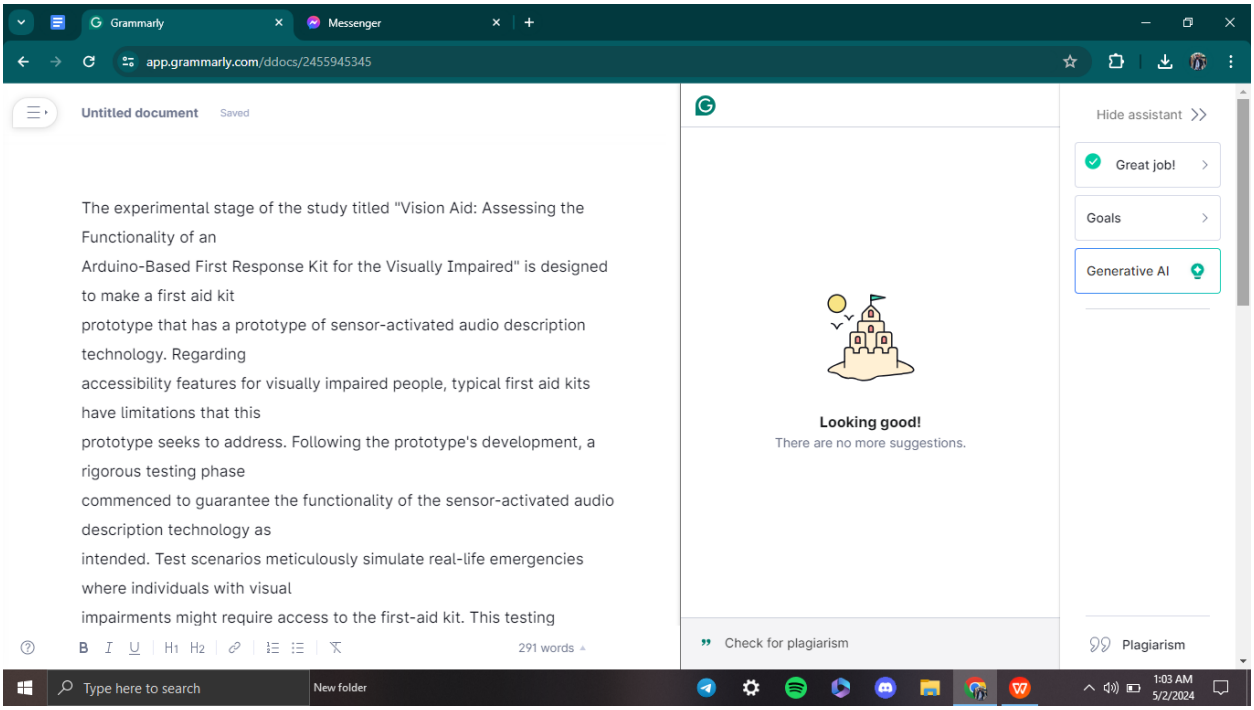
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The experimental stage of the study titled "Vision Aid: Assessing the Functionality of an
Arduino-Based First Response Kit for the Visually Impaired" is designed to make a first aid kit
prototype that has a prototype of sensor-activated audio description technology. Regarding
accessibility features for visually impaired people, typical first aid kits have limitations that this
prototype seeks to address. Following the prototype's development, a rigorous testing phase
commenced to guarantee the functionality of the sensor-activated audio description technology as
intended. Test scenarios meticulously simulate real-life emergencies where individuals with visual
impairments might require access to the first-aid kit. This testing
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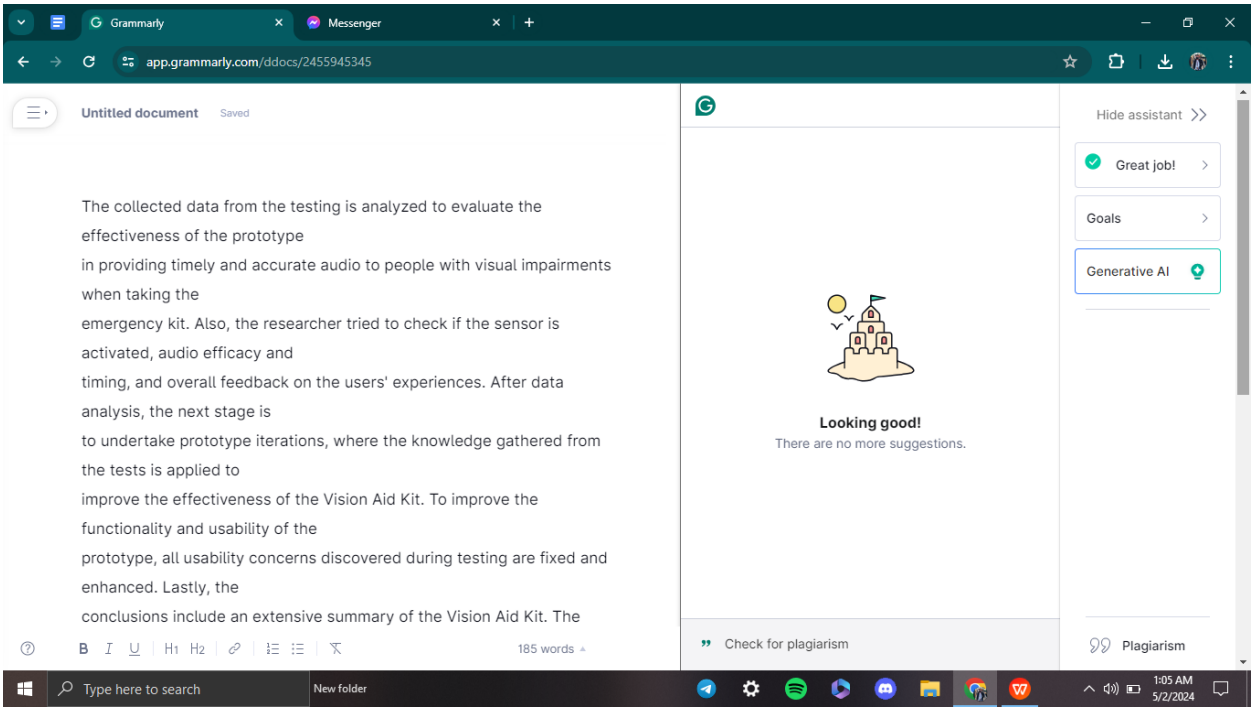
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The collected data from the testing is analyzed to evaluate the effectiveness of the prototype in providing timely and accurate audio to people with visual impairments when taking the emergency kit. Also, the researcher tried to check if the sensor is activated, audio efficacy and timing, and overall feedback on the users' experiences. After data analysis, the next stage is to undertake prototype iterations, where the knowledge gathered from the tests is applied to improve the effectiveness of the Vision Aid Kit. To improve the functionality and usability of the prototype, all usability concerns discovered during testing are fixed and enhanced. Lastly, the conclusions include an extensive summary of the Vision Aid Kit. The
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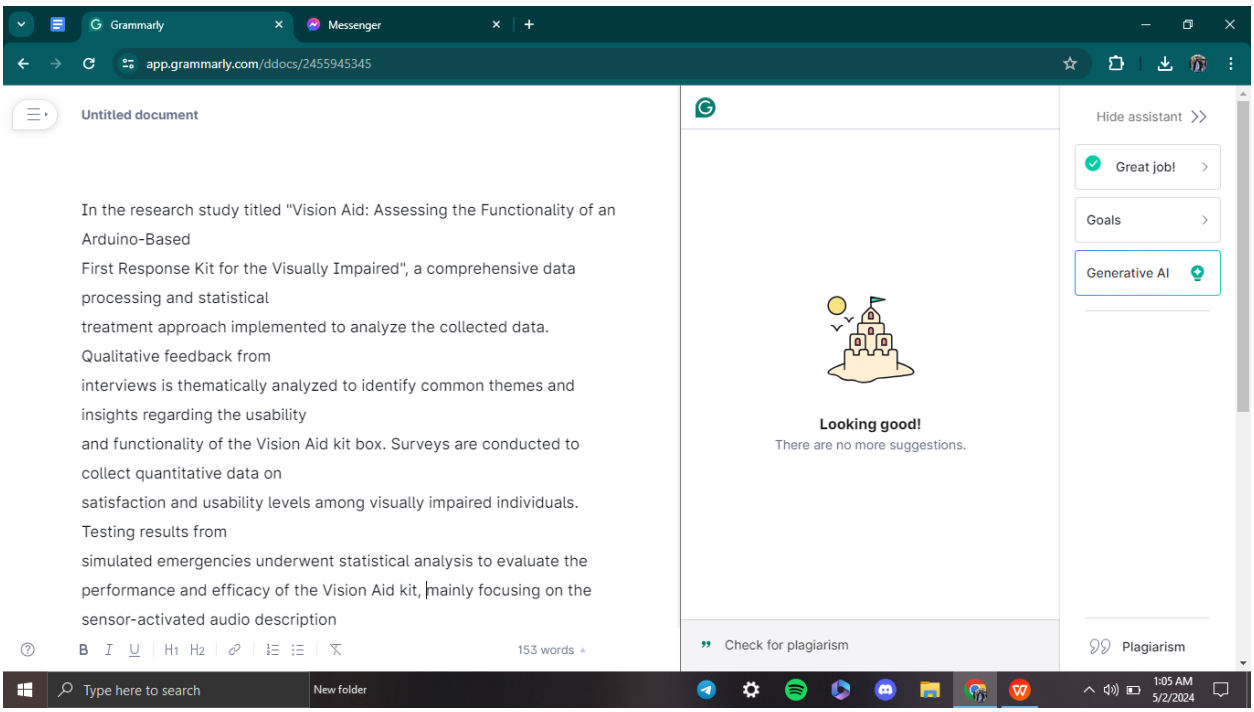
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In the research study titled "Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired", a comprehensive data processing and statistical treatment approach implemented to analyze the collected data. Qualitative feedback from interviews is thematically analyzed to identify common themes and insights regarding the usability and functionality of the Vision Aid kit box. Surveys are conducted to collect quantitative data on satisfaction and usability levels among visually impaired individuals. Testing results from simulated emergencies underwent statistical analysis to evaluate the performance and efficacy of the Vision Aid kit, mainly focusing on the sensor-activated audio description
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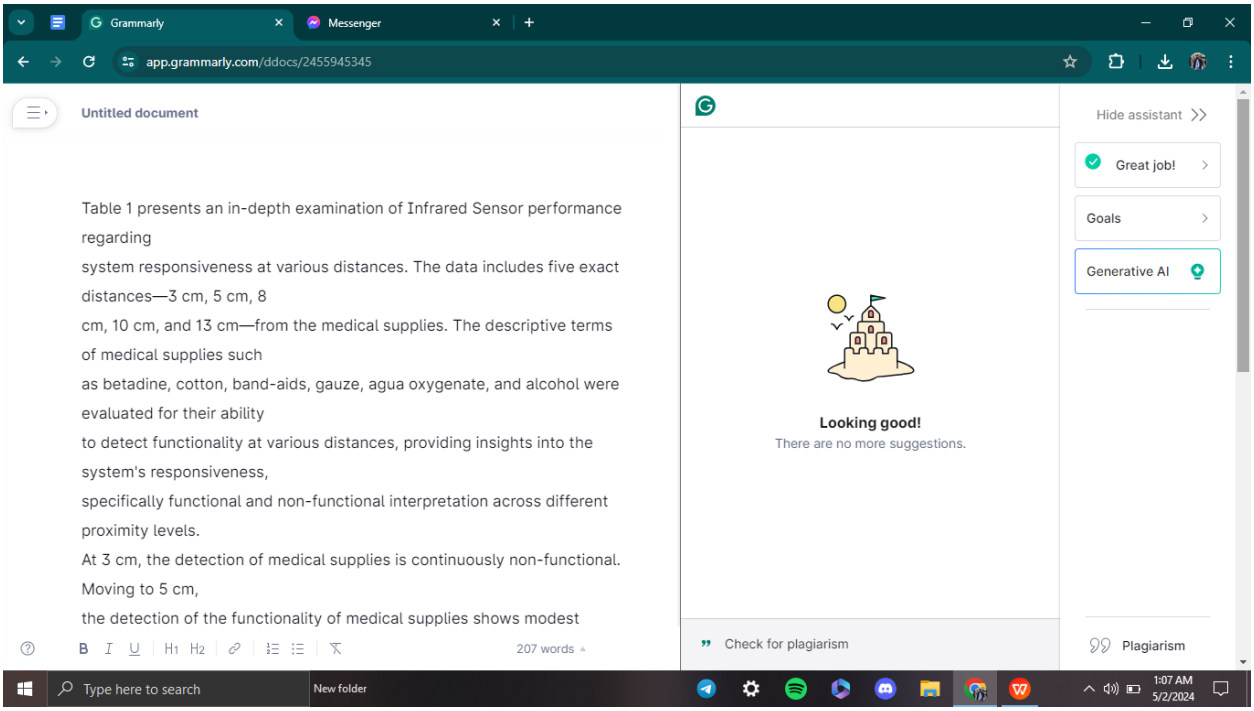
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Table 1 presents an in-depth examination of Infrared Sensor performance regarding
system responsiveness at various distances. The data includes five exact distances—3 cm, 5 cm, 8
cm, 10 cm, and 13 cm—from the medical supplies. The descriptive terms of medical supplies such
as betadine, cotton, band-aids, gauze, agua oxygenate, and alcohol were evaluated for their ability
to detect functionality at various distances, providing insights into the system's responsiveness,
specifically functional and non-functional interpretation across different proximity levels.
At 3 cm, the detection of medical supplies is continuously non-functional. Moving to 5 cm,
the detection of the functionality of medical supplies shows modest
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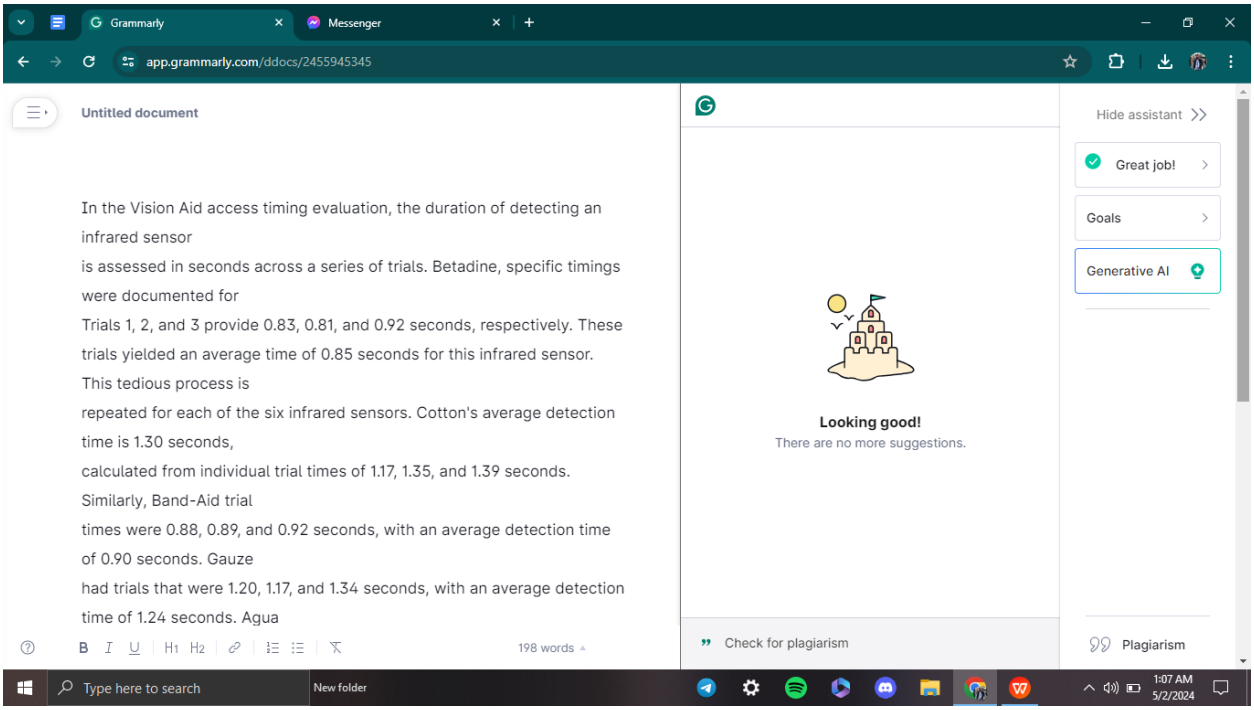
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In the Vision Aid access timing evaluation, the duration of detecting an infrared sensor
is assessed in seconds across a series of trials. Betadine, specific timings were documented for
Trials 1, 2, and 3 provide 0.83, 0.81, and 0.92 seconds, respectively. These trials yielded an average time of 0.85 seconds for this infrared sensor. This tedious process is
repeated for each of the six infrared sensors. Cotton's average detection time is 1.30 seconds,
calculated from individual trial times of 1.17, 1.35, and 1.39 seconds. Similarly, Band-Aid trial
times were 0.88, 0.89, and 0.92 seconds, with an average detection time of 0.90 seconds. Gauze
had trials that were 1.20, 1.17, and 1.34 seconds, with an average detection time of 1.24 seconds. Agua
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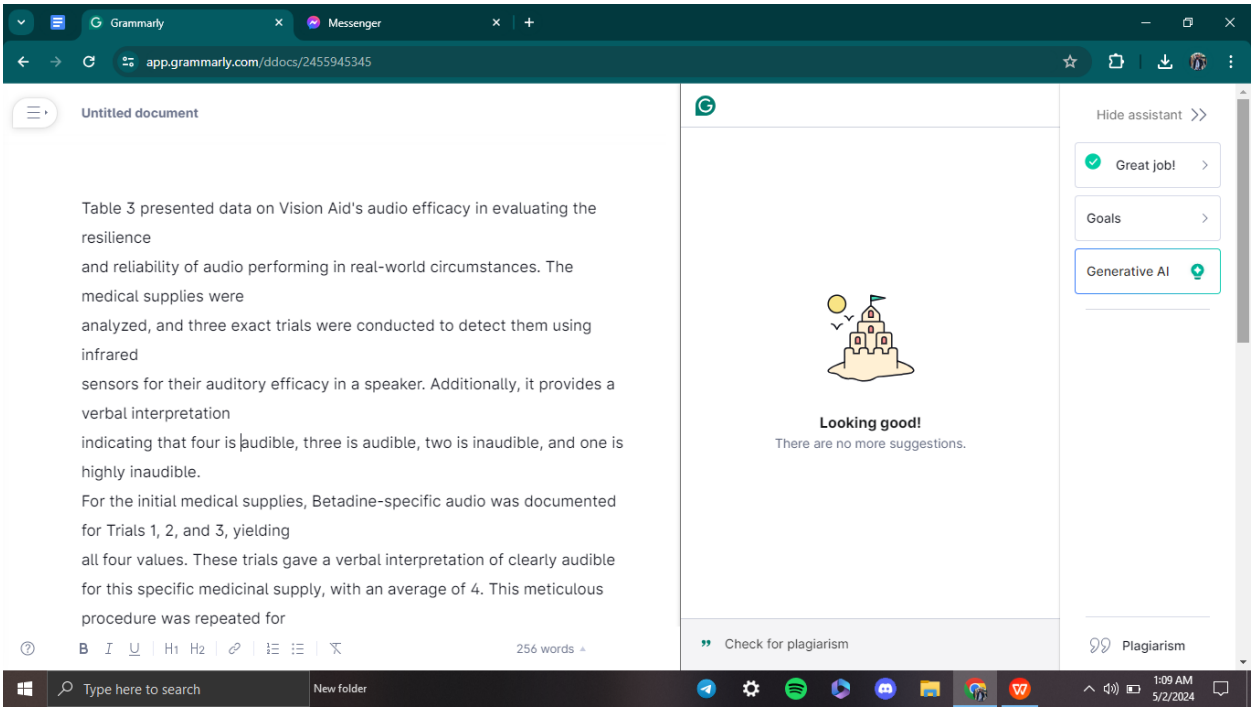
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Table 3 presented data on Vision Aid's audio efficacy in evaluating the resilience
and reliability of audio performing in real-world circumstances. The medical supplies were
analyzed, and three exact trials were conducted to detect them using infrared
sensors for their auditory efficacy in a speaker. Additionally, it provides a verbal interpretation
indicating that four is audible, three is audible, two is inaudible, and one is highly inaudible.
For the initial medical supplies, Betadine-specific audio was documented for Trials 1, 2, and 3, yielding
all four values. These trials gave a verbal interpretation of clearly audible for this specific medicinal supply, with an average of 4. This meticulous procedure was repeated for
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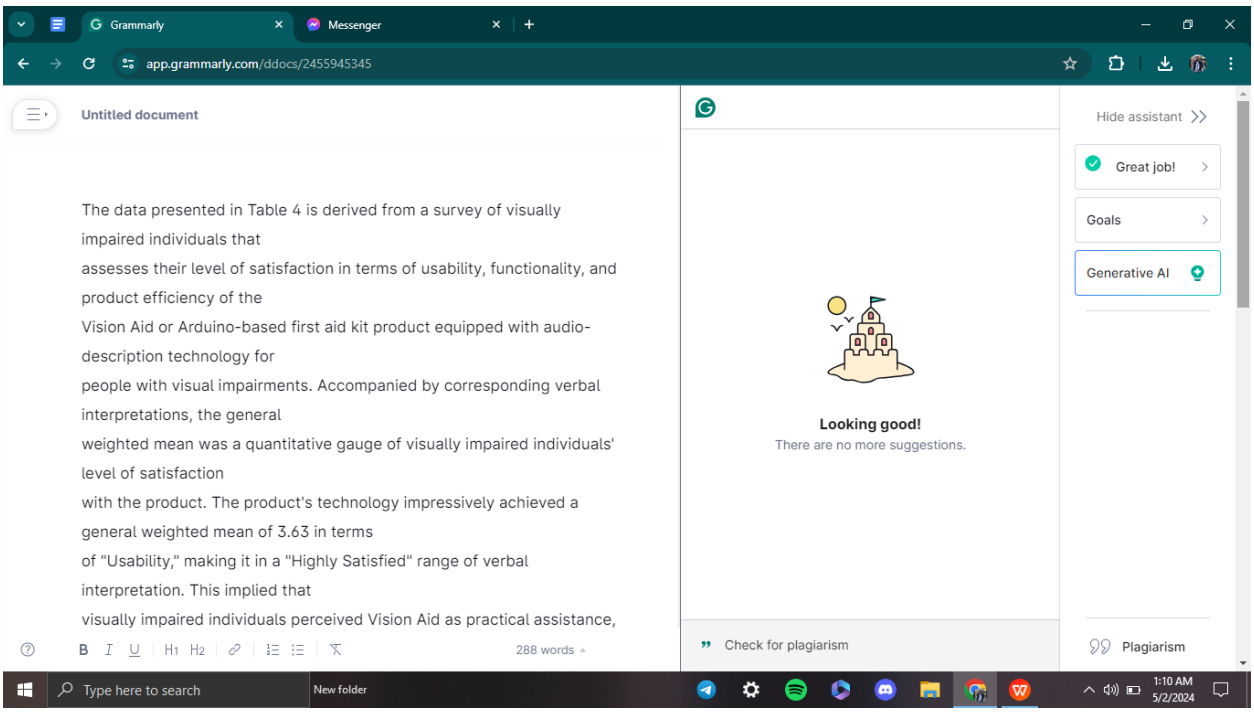
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The data presented in Table 4 is derived from a survey of visually impaired individuals that
assesses their level of satisfaction in terms of usability, functionality, and product efficiency of the
Vision Aid or Arduino-based first aid kit product equipped with audio-description technology for
people with visual impairments. Accompanied by corresponding verbal interpretations, the general
weighted mean was a quantitative gauge of visually impaired individuals' level of satisfaction
with the product. The product's technology impressively achieved a general weighted mean of 3.63 in terms
of "Usability," making it in a "Highly Satisfied" range of verbal interpretation. This implied that
visually impaired individuals perceived Vision Aid as practical assistance,
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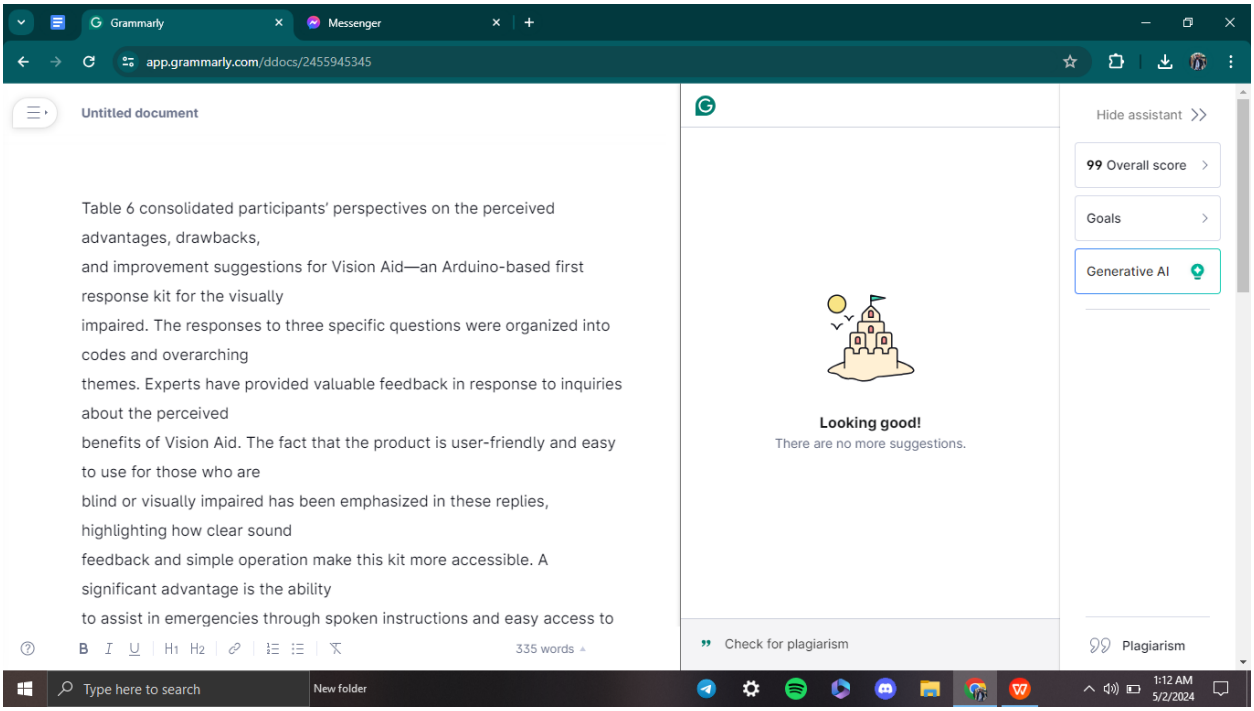
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Table 6 consolidated participants' perspectives on the perceived advantages, drawbacks, and improvement suggestions for Vision Aid—an Arduino-based first response kit for the visually impaired. The responses to three specific questions were organized into codes and overarching themes. Experts have provided valuable feedback in response to inquiries about the perceived benefits of Vision Aid. The fact that the product is user-friendly and easy to use for those who are blind or visually impaired has been emphasized in these replies, highlighting how clear sound feedback and simple operation make this kit more accessible. A significant advantage is the ability to assist in emergencies through spoken instructions and easy access to
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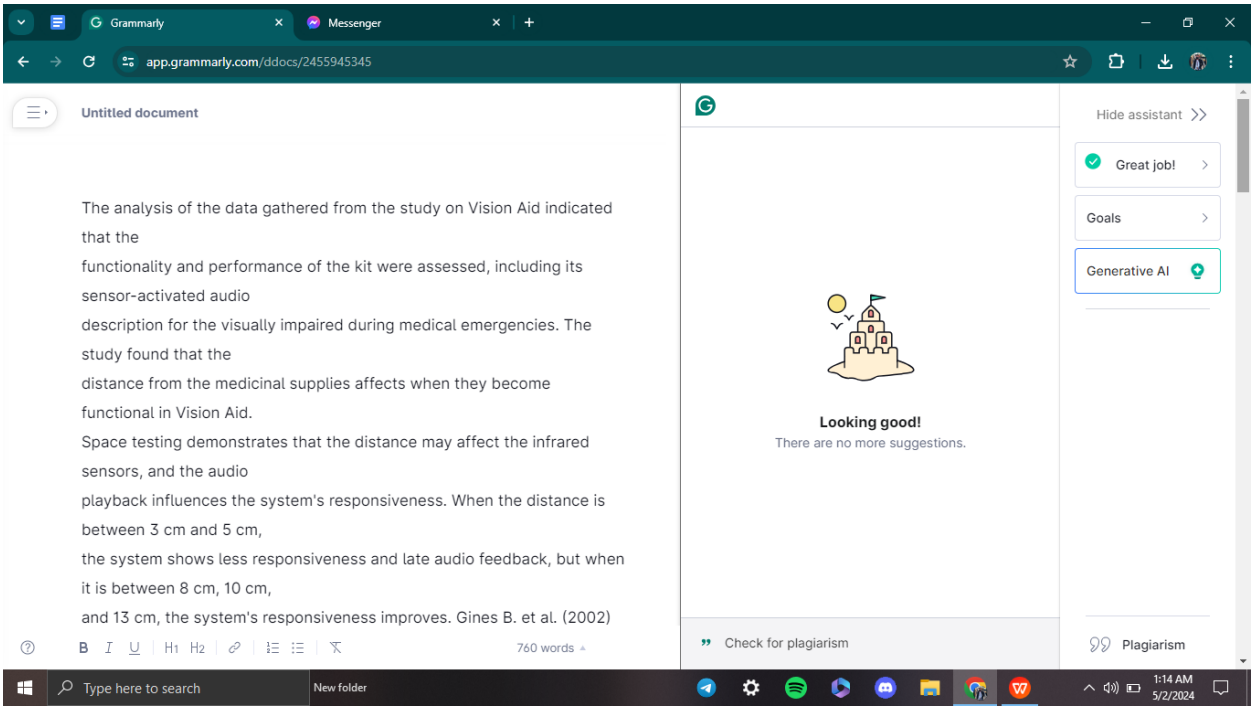
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The analysis of the data gathered from the study on Vision Aid indicated that the functionality and performance of the kit were assessed, including its sensor-activated audio description for the visually impaired during medical emergencies. The study found that the distance from the medicinal supplies affects when they become functional in Vision Aid. Space testing demonstrates that the distance may affect the infrared sensors, and the audio playback influences the system's responsiveness. When the distance is between 3 cm and 5 cm, the system shows less responsiveness and late audio feedback, but when it is between 8 cm, 10 cm, and 13 cm, the system's responsiveness improves. Gines B. et al. (2002)
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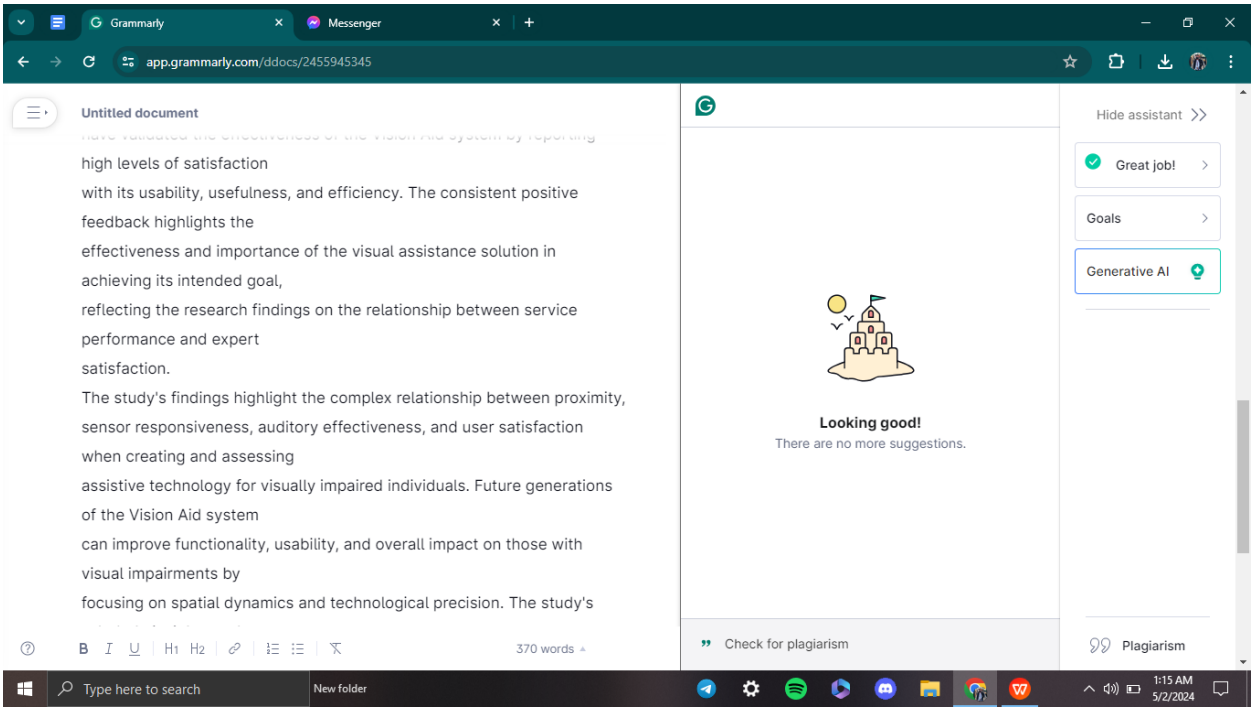
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high levels of satisfaction
with its usability, usefulness, and efficiency. The consistent positive feedback highlights the
effectiveness and importance of the visual assistance solution in achieving its intended goal,
reflecting the research findings on the relationship between service performance and expert
satisfaction.
The study's findings highlight the complex relationship between proximity, sensor responsiveness, auditory effectiveness, and user satisfaction when creating and assessing
assistive technology for visually impaired individuals. Future generations of the Vision Aid system
can improve functionality, usability, and overall impact on those with visual impairments by
focusing on spatial dynamics and technological precision. The study's
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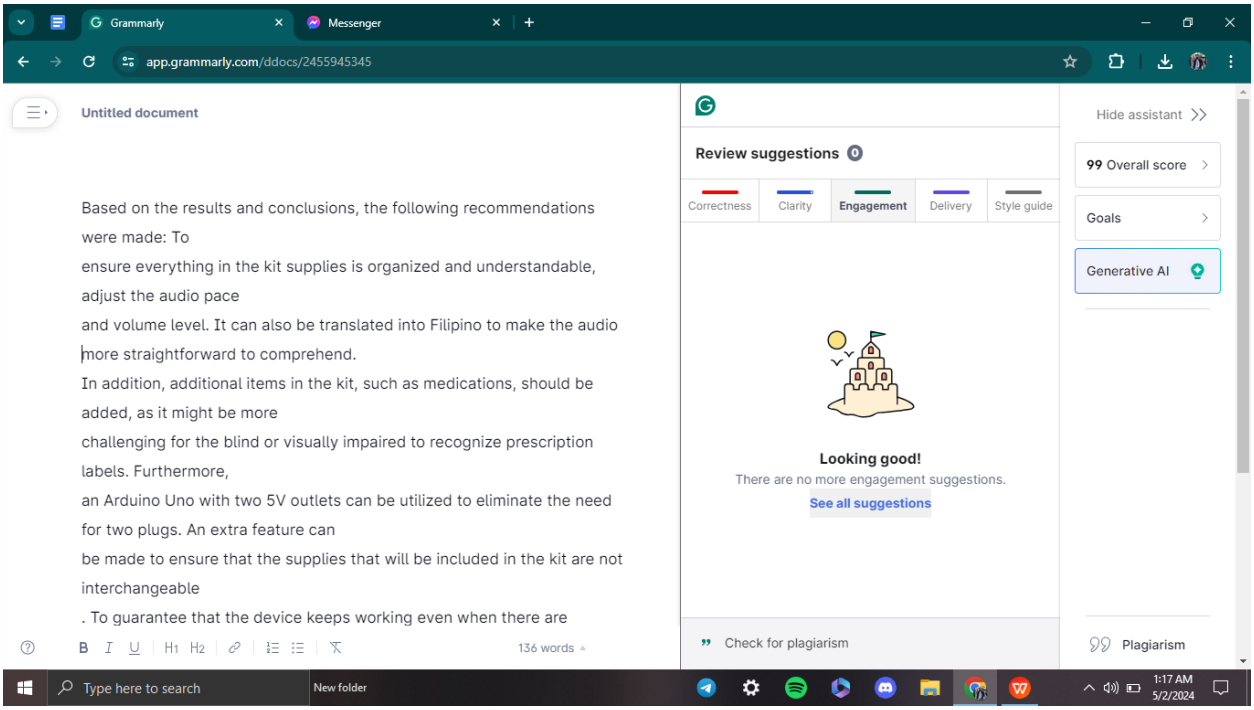
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Based on the results and conclusions, the following recommendations were made: To
ensure everything in the kit supplies is organized and understandable, adjust the audio pace
and volume level. It can also be translated into Filipino to make the audio more straightforward to comprehend.
In addition, additional items in the kit, such as medications, should be added, as it might be more
challenging for the blind or visually impaired to recognize prescription labels. Furthermore,
an Arduino Uno with two 5V outlets can be utilized to eliminate the need for two plugs. An extra feature can
be made to ensure that the supplies that will be included in the kit are not interchangeable
. To guarantee that the device keeps working even when there are
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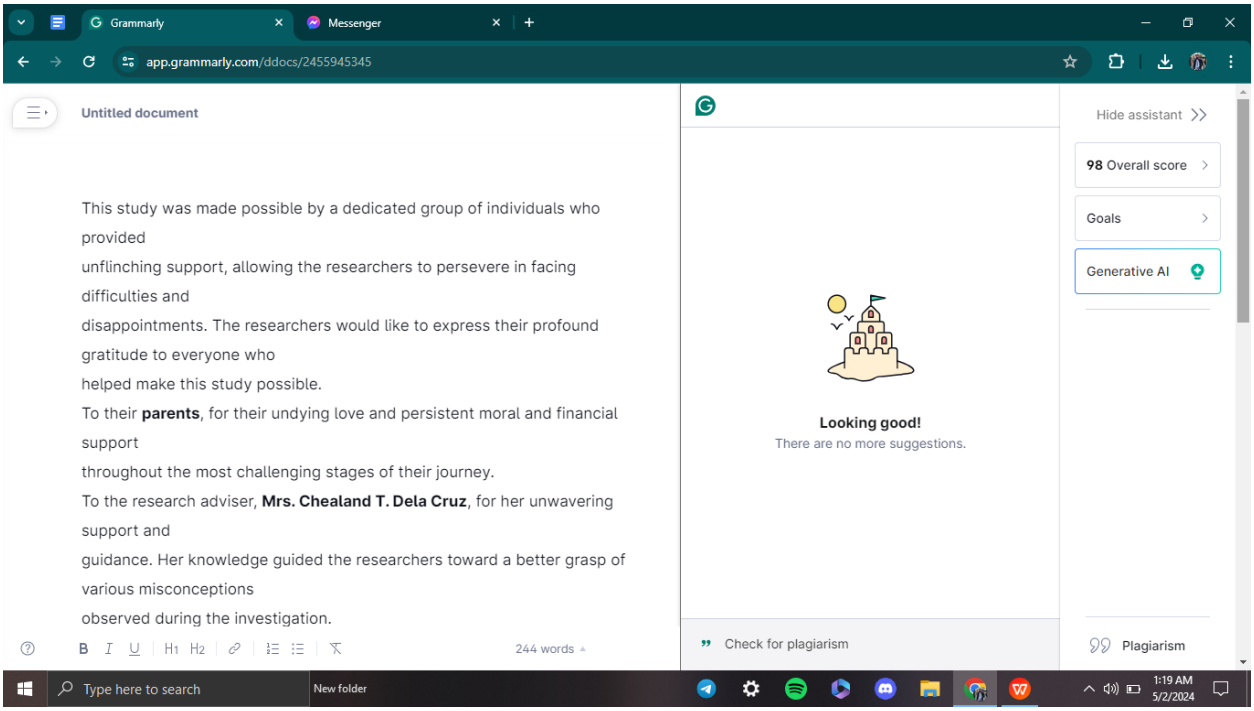
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This study was made possible by a dedicated group of individuals who provided
unflinching support, allowing the researchers to persevere in facing difficulties and
disappointments. The researchers would like to express their profound gratitude to everyone who
helped make this study possible.
To their parents, for their undying love and persistent moral and financial support
throughout the most challenging stages of their journey.
To the research adviser, Mrs. Chealand T. Dela Cruz, for her unwavering support and
guidance. Her knowledge guided the researchers toward a better grasp of various misconceptions
observed during the investigation.
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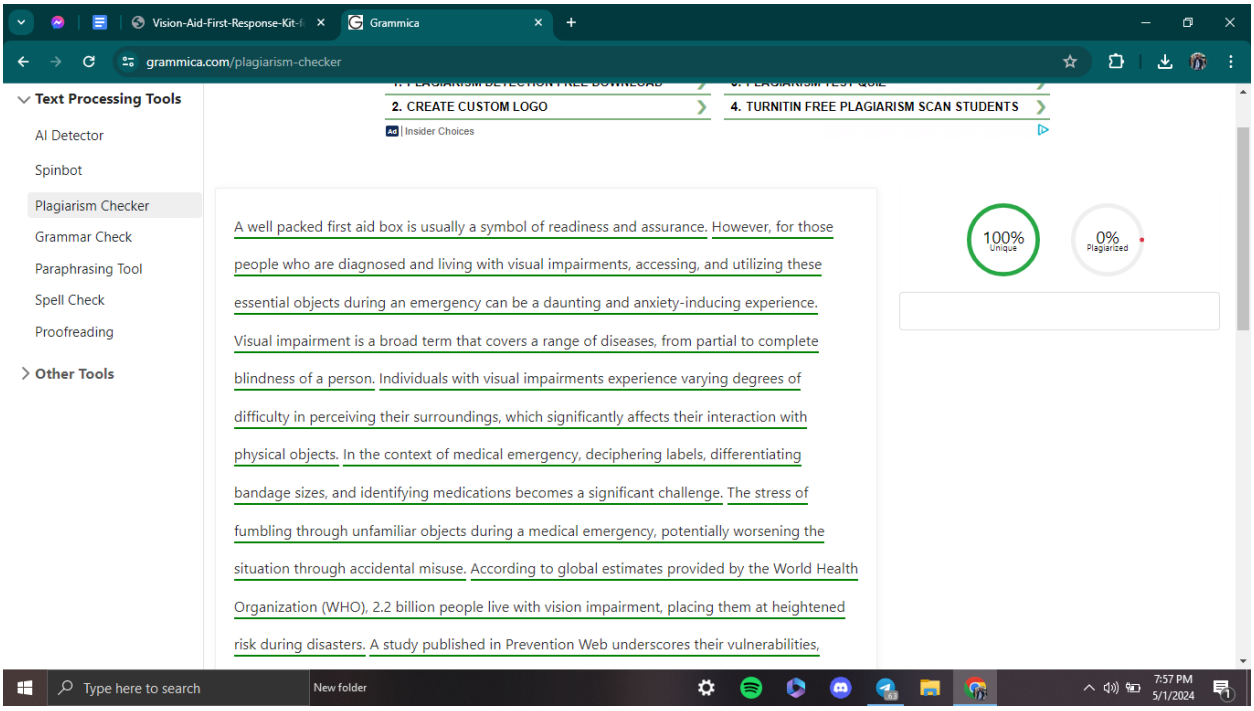
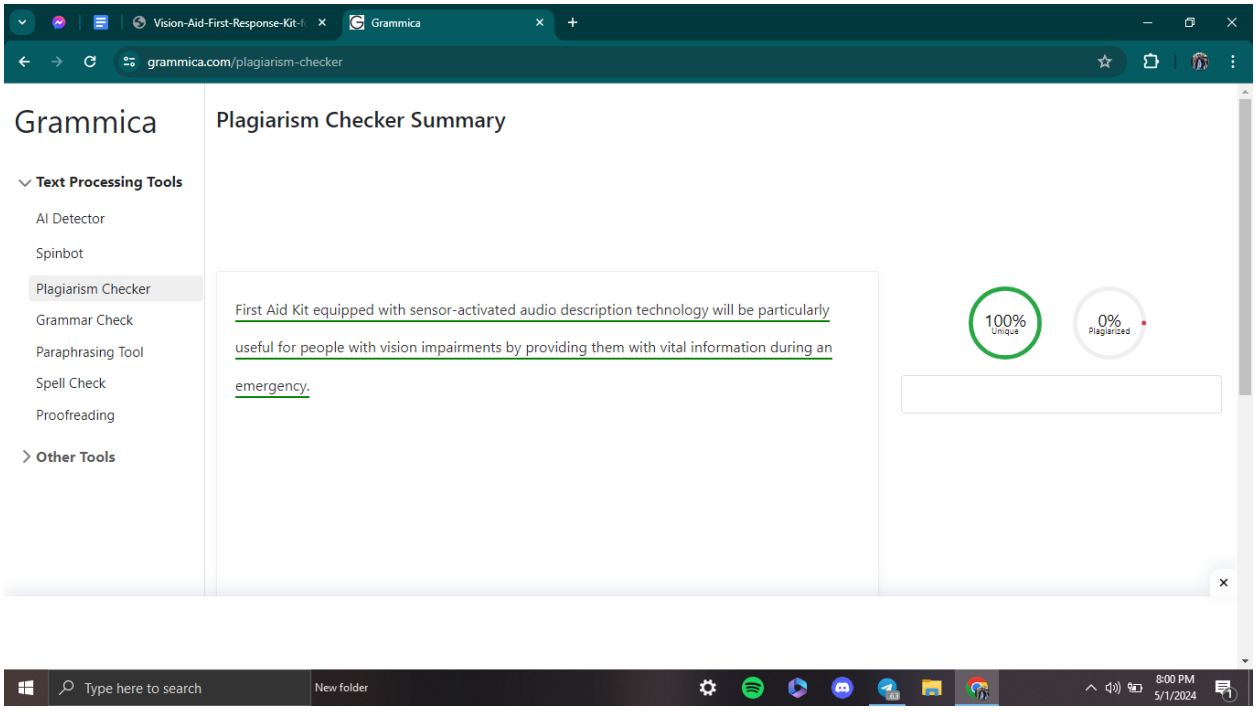
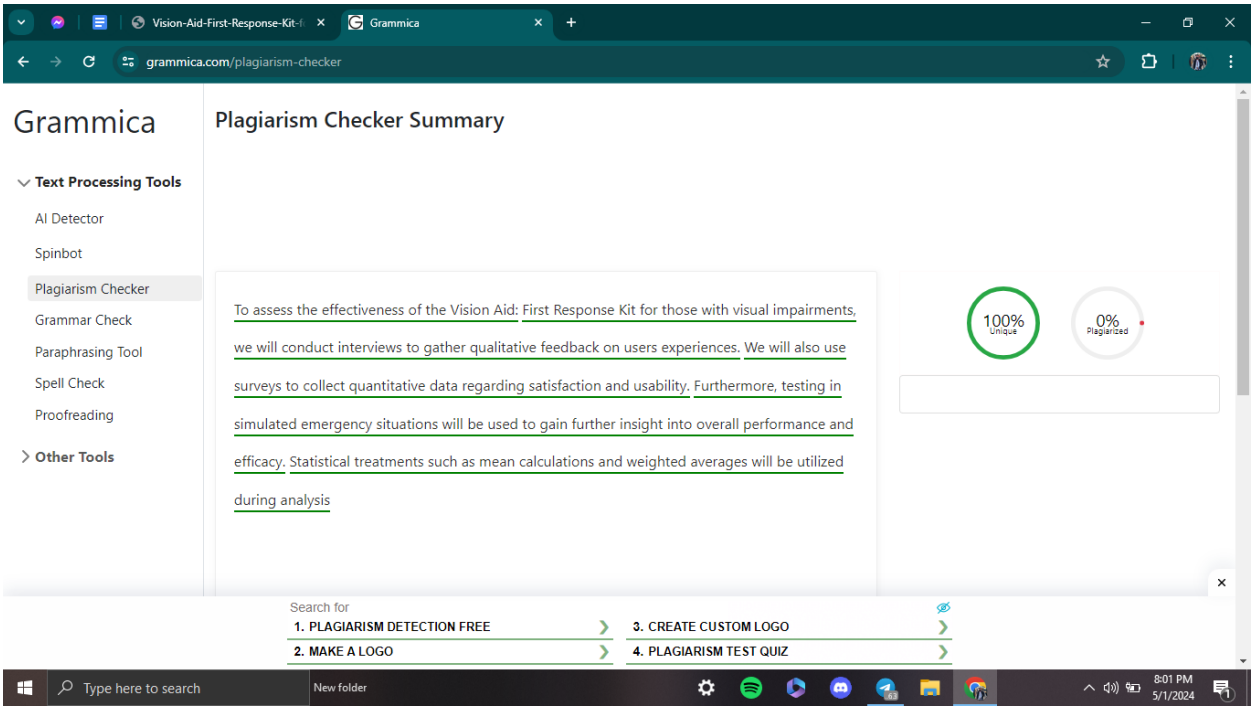
APPENDIX L

RESULTS OF PLAGIARISM TEST



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Proofreading
Other Tools
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To assess the effectiveness of the Vision Aid: First Response Kit for those with visual impairments, we will conduct interviews to gather qualitative feedback on users experiences. We will also use surveys to collect quantitative data regarding satisfaction and usability. Furthermore, testing in simulated emergency situations will be used to gain further insight into overall performance and efficacy. Statistical treatments such as mean calculations and weighted averages will be utilized during analysis
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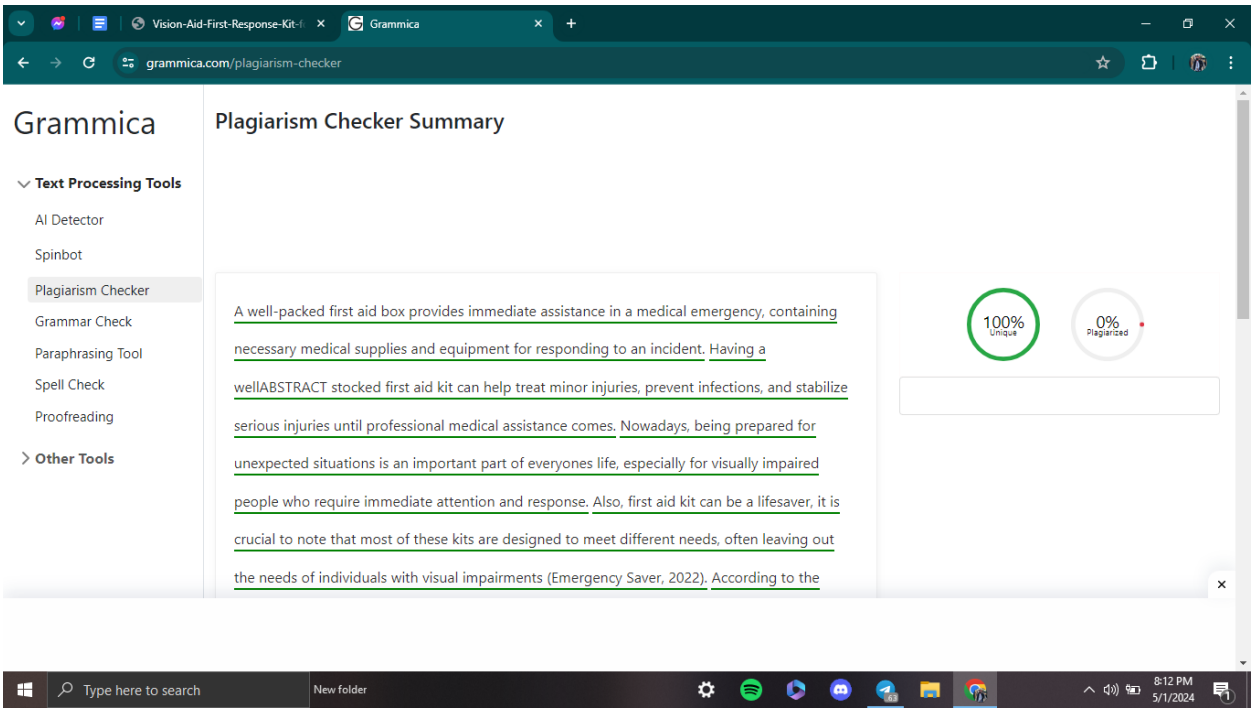
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A well-packed first aid box provides immediate assistance in a medical emergency, containing necessary medical supplies and equipment for responding to an incident. Having a wellABSTRACT stocked first aid kit can help treat minor injuries, prevent infections, and stabilize serious injuries until professional medical assistance comes. Nowadays, being prepared for unexpected situations is an important part of everyones life, especially for visually impaired people who require immediate attention and response. Also, first aid kit can be a lifesaver, it is crucial to note that most of these kits are designed to meet different needs, often leaving out the needs of individuals with visual impairments (Emergency Saver, 2022). According to the
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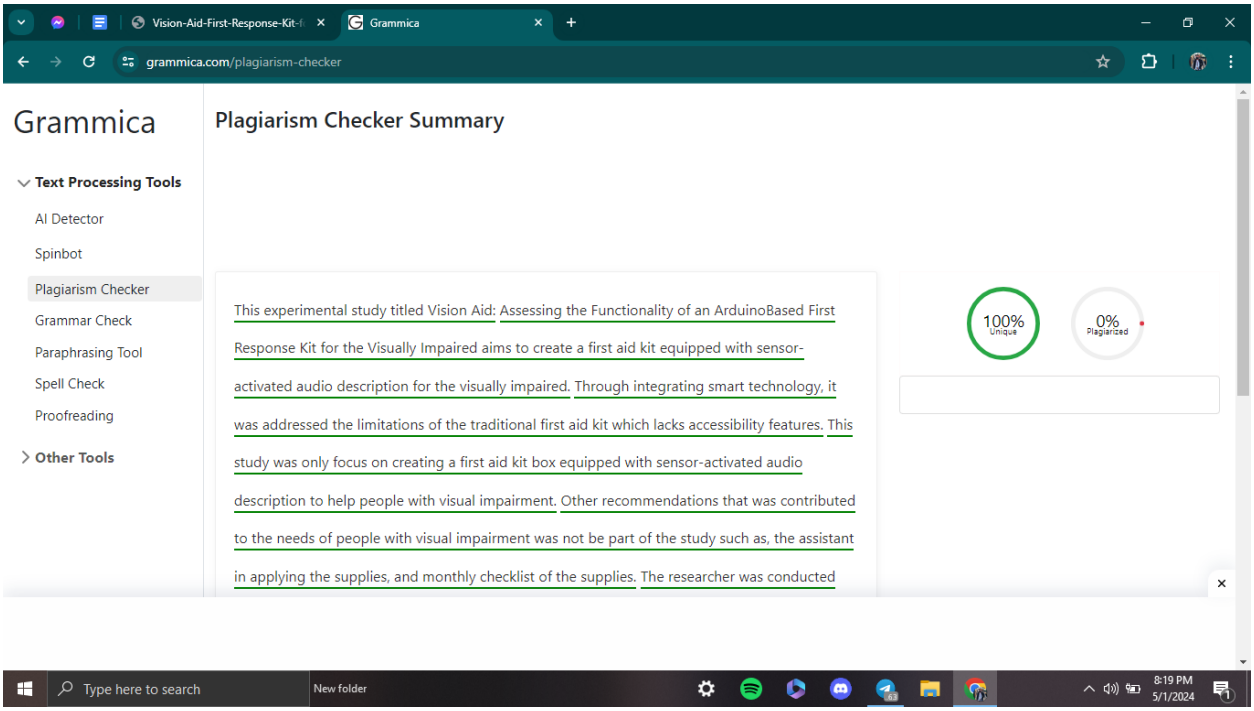
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This experimental study titled Vision Aid: Assessing the Functionality of an ArduinoBased First Response Kit for the Visually Impaired aims to create a first aid kit equipped with sensor-activated audio description for the visually impaired. Through integrating smart technology, it was addressed the limitations of the traditional first aid kit which lacks accessibility features. This study was only focus on creating a first aid kit box equipped with sensor-activated audio description to help people with visual impairment. Other recommendations that was contributed to the needs of people with visual impairment was not be part of the study such as, the assistant in applying the supplies, and monthly checklist of the supplies. The researcher was conducted
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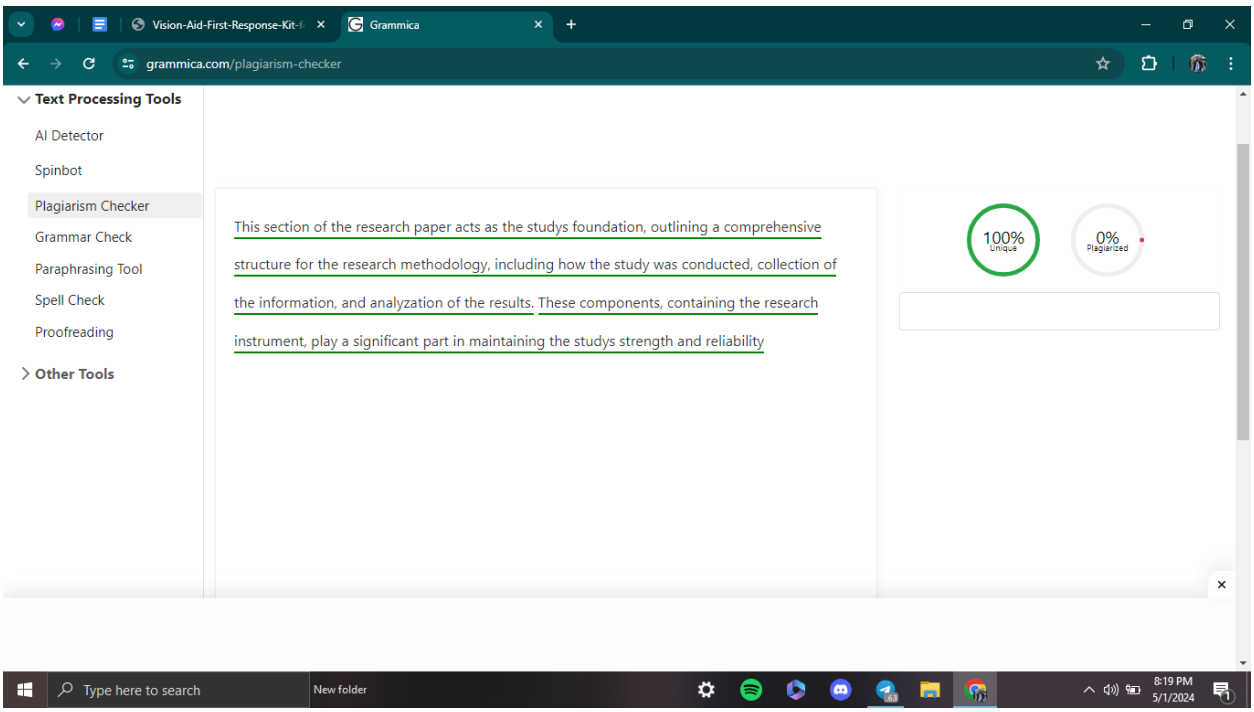
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This section of the research paper acts as the studys foundation, outlining a comprehensive structure for the research methodology, including how the study was conducted, collection of the information, and analyzation of the results. These components, containing the research instrument, play a significant part in maintaining the studys strength and reliability
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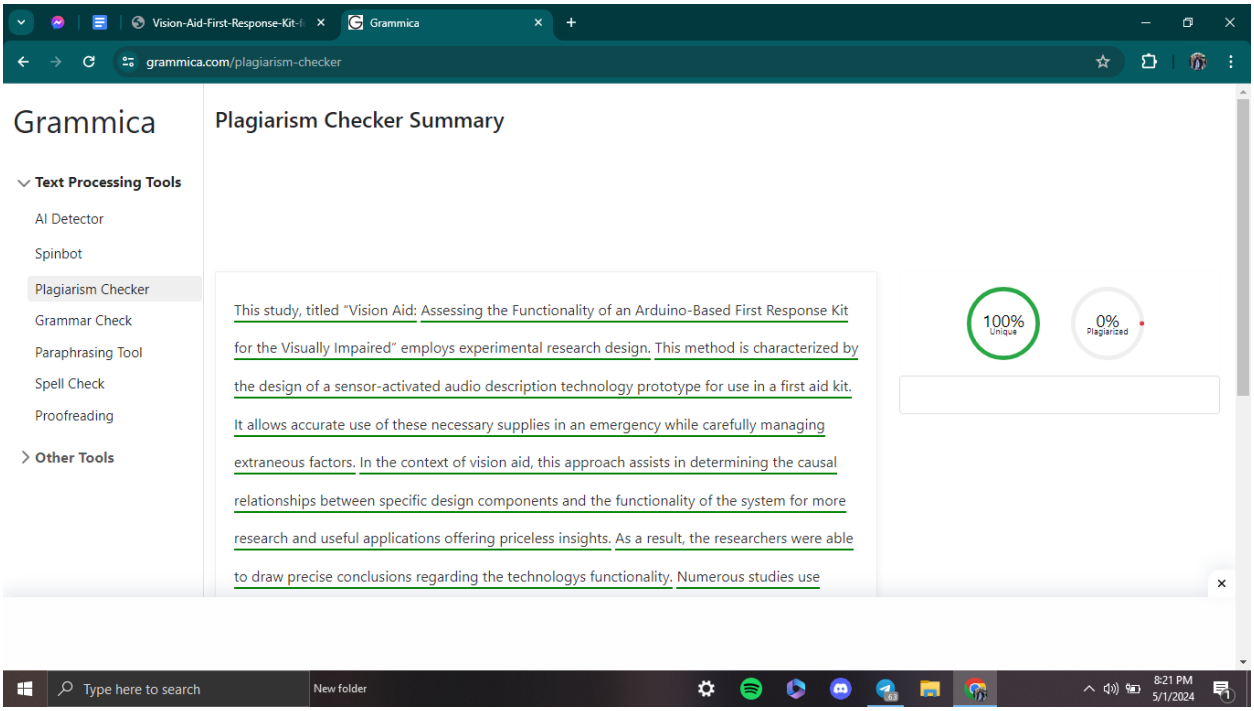
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This study, titled "Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired" employs experimental research design. This method is characterized by the design of a sensor-activated audio description technology prototype for use in a first aid kit. It allows accurate use of these necessary supplies in an emergency while carefully managing extraneous factors. In the context of vision aid, this approach assists in determining the causal relationships between specific design components and the functionality of the system for more research and useful applications offering priceless insights. As a result, the researchers were able to draw precise conclusions regarding the technologys functionality. Numerous studies use
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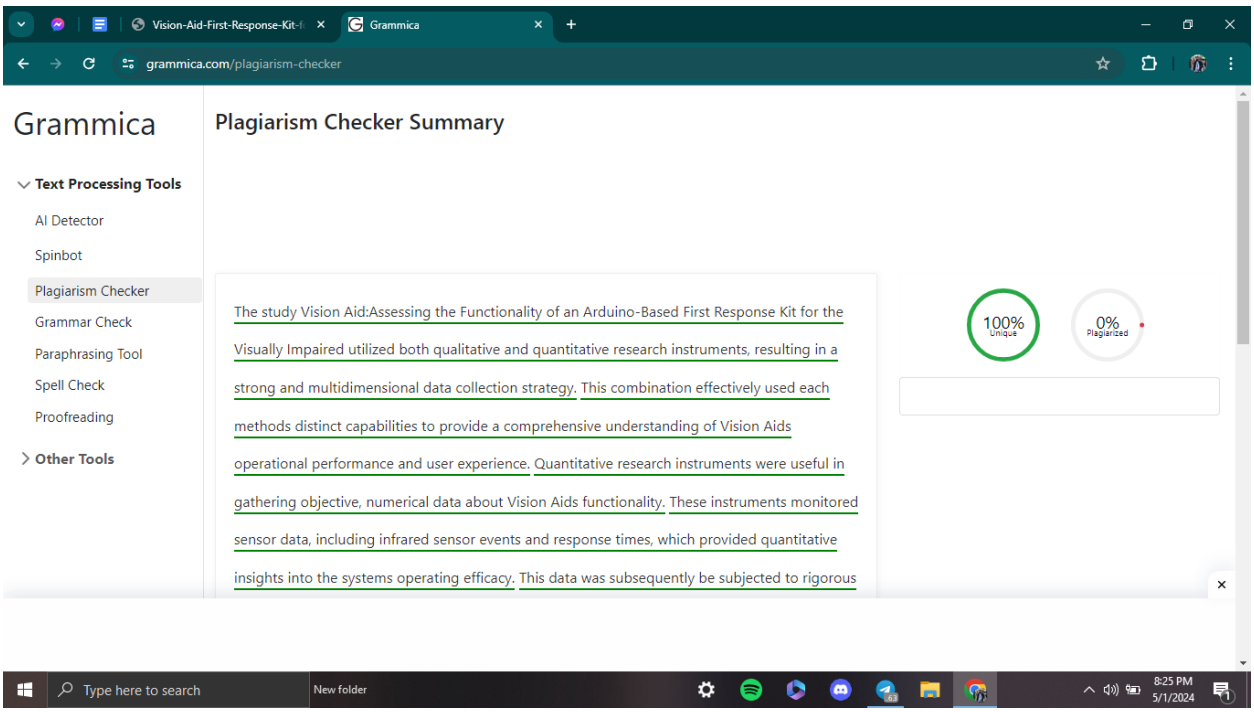
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The study Vision Aid:Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired utilized both qualitative and quantitative research instruments, resulting in a strong and multidimensional data collection strategy. This combination effectively used each methods distinct capabilities to provide a comprehensive understanding of Vision Aids operational performance and user experience. Quantitative research instruments were useful in gathering objective, numerical data about Vision Aids functionality. These instruments monitored sensor data, including infrared sensor events and response times, which provided quantitative insights into the systems operating efficacy. This data was subsequently be subjected to rigorous
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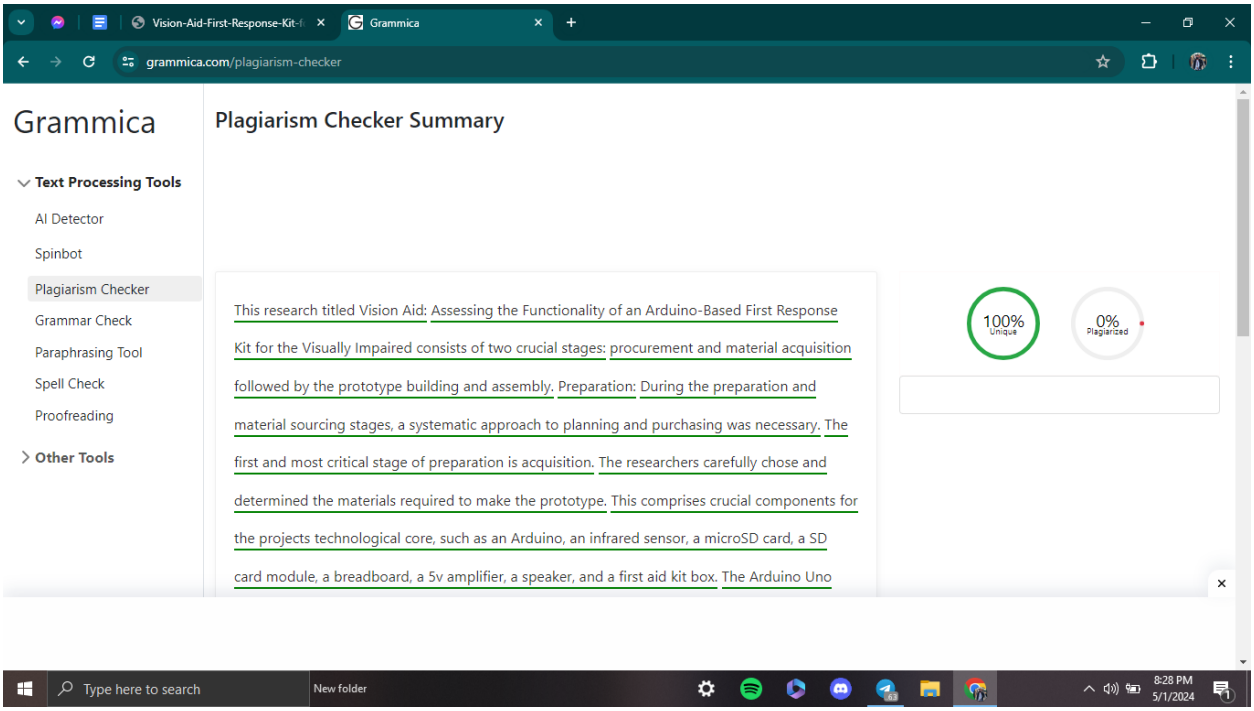
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This research titled Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired consists of two crucial stages: procurement and material acquisition followed by the prototype building and assembly. Preparation: During the preparation and material sourcing stages, a systematic approach to planning and purchasing was necessary. The first and most critical stage of preparation is acquisition. The researchers carefully chose and determined the materials required to make the prototype. This comprises crucial components for the projects technological core, such as an Arduino, an infrared sensor, a microSD card, a SD card module, a breadboard, a 5v amplifier, a speaker, and a first aid kit box. The Arduino Uno
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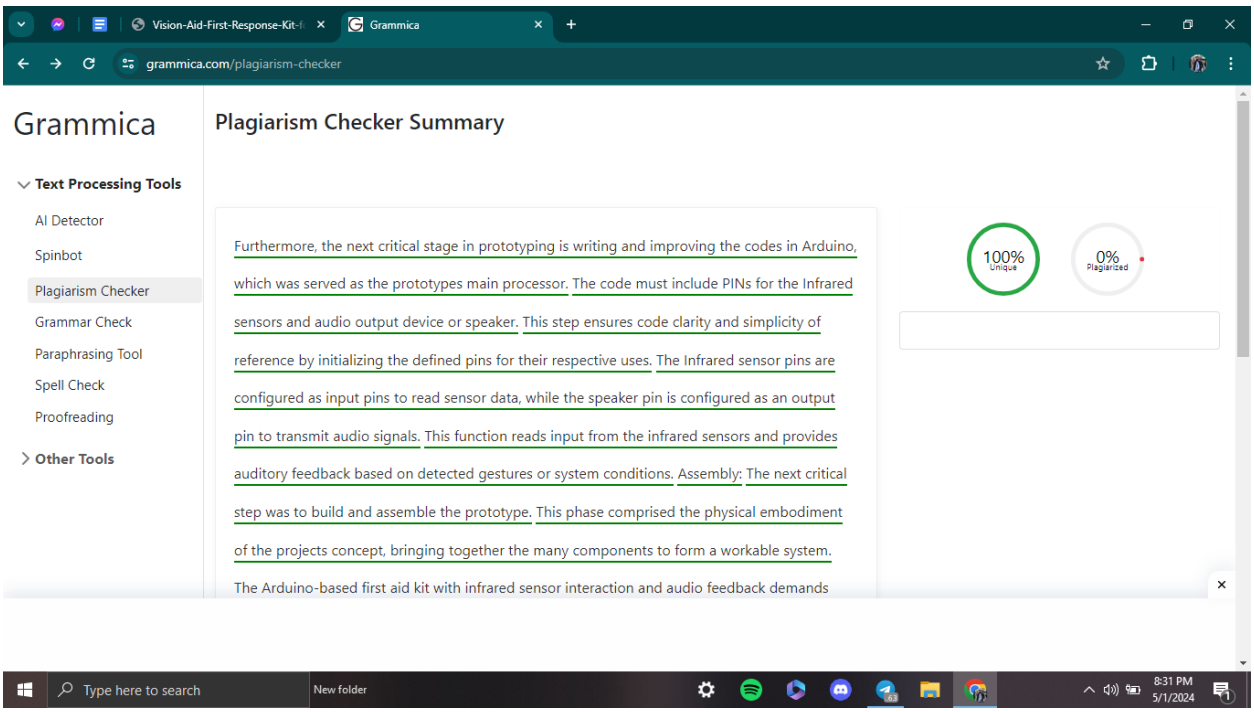
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Furthermore, the next critical stage in prototyping is writing and improving the codes in Arduino, which was served as the prototypes main processor. The code must include PINs for the Infrared sensors and audio output device or speaker. This step ensures code clarity and simplicity of reference by initializing the defined pins for their respective uses. The Infrared sensor pins are configured as input pins to read sensor data, while the speaker pin is configured as an output pin to transmit audio signals. This function reads input from the infrared sensors and provides auditory feedback based on detected gestures or system conditions. Assembly: The next critical step was to build and assemble the prototype. This phase comprised the physical embodiment of the projects concept, bringing together the many components to form a workable system. The Arduino-based first aid kit with infrared sensor interaction and audio feedback demands
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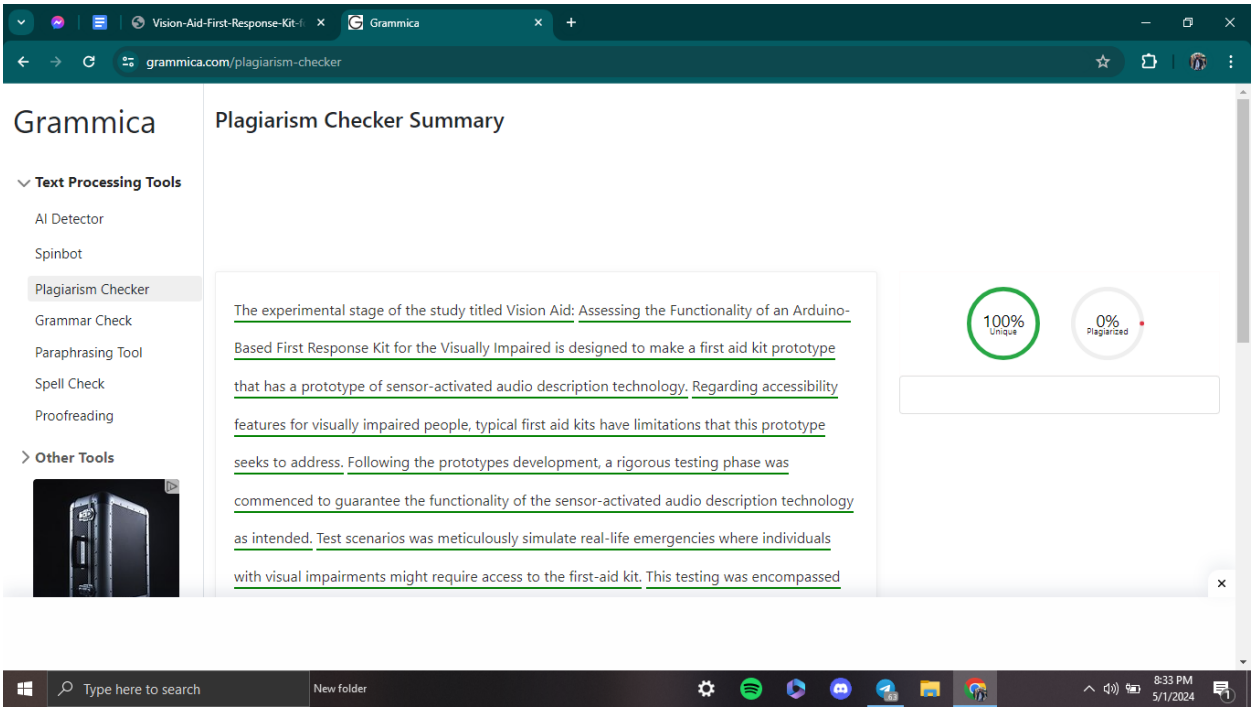
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The experimental stage of the study titled Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired is designed to make a first aid kit prototype that has a prototype of sensor-activated audio description technology. Regarding accessibility features for visually impaired people, typical first aid kits have limitations that this prototype seeks to address. Following the prototypes development, a rigorous testing phase was commenced to guarantee the functionality of the sensor-activated audio description technology as intended. Test scenarios was meticulously simulate real-life emergencies where individuals with visual impairments might require access to the first-aid kit. This testing was encompassed
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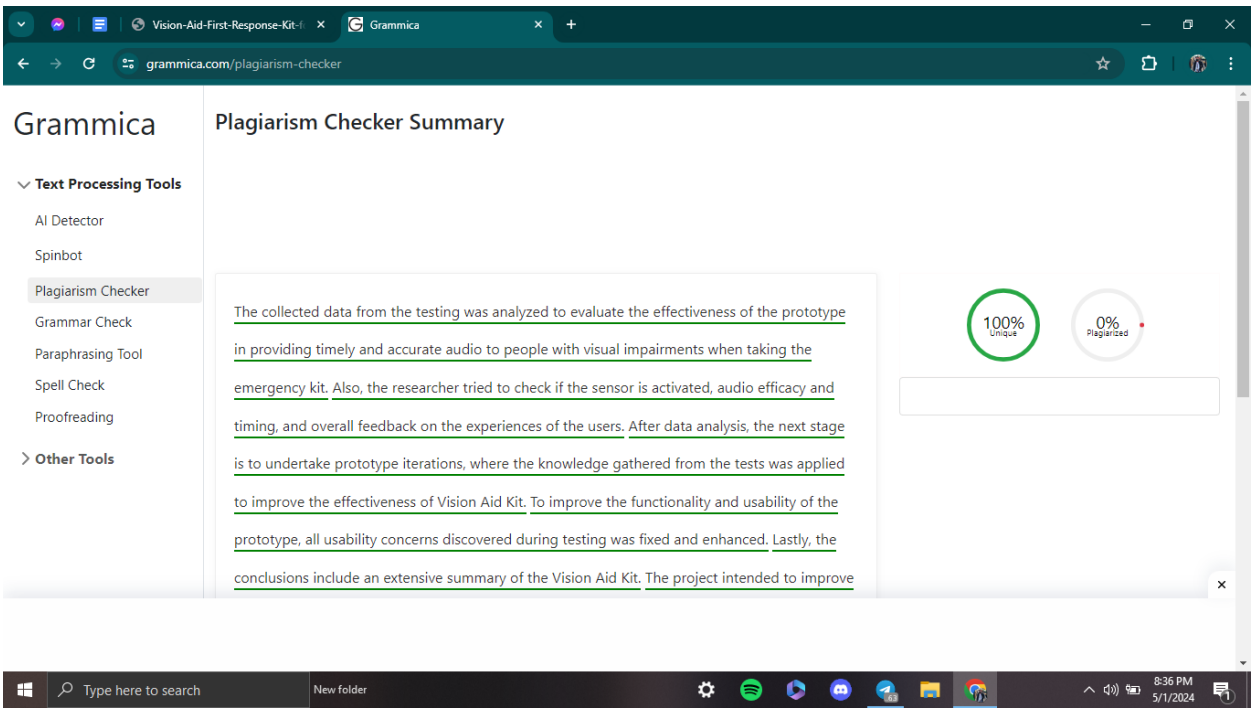
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The collected data from the testing was analyzed to evaluate the effectiveness of the prototype in providing timely and accurate audio to people with visual impairments when taking the emergency kit. Also, the researcher tried to check if the sensor is activated, audio efficacy and timing, and overall feedback on the experiences of the users. After data analysis, the next stage is to undertake prototype iterations, where the knowledge gathered from the tests was applied to improve the effectiveness of Vision Aid Kit. To improve the functionality and usability of the prototype, all usability concerns discovered during testing was fixed and enhanced. Lastly, the conclusions include an extensive summary of the Vision Aid Kit. The project intended to improve
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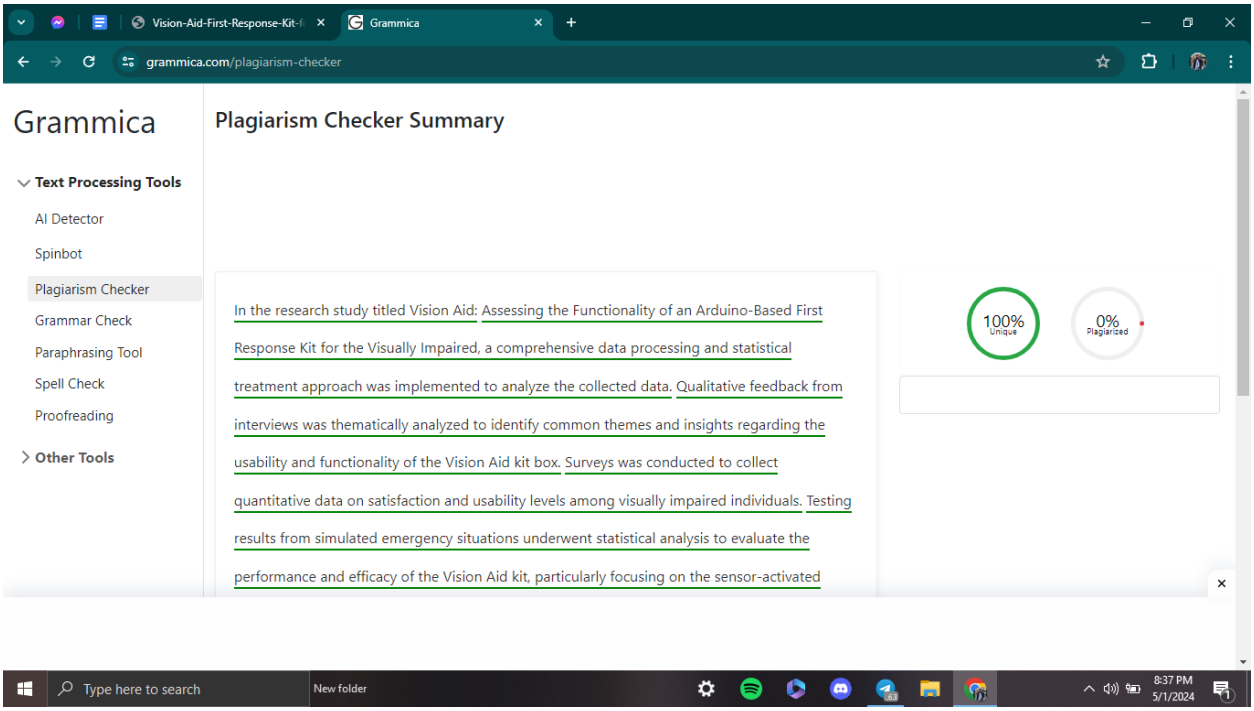
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In the research study titled Vision Aid: Assessing the Functionality of an Arduino-Based First Response Kit for the Visually Impaired, a comprehensive data processing and statistical treatment approach was implemented to analyze the collected data. Qualitative feedback from interviews was thematically analyzed to identify common themes and insights regarding the usability and functionality of the Vision Aid kit box. Surveys was conducted to collect quantitative data on satisfaction and usability levels among visually impaired individuals. Testing results from simulated emergency situations underwent statistical analysis to evaluate the performance and efficacy of the Vision Aid kit, particularly focusing on the sensor-activated
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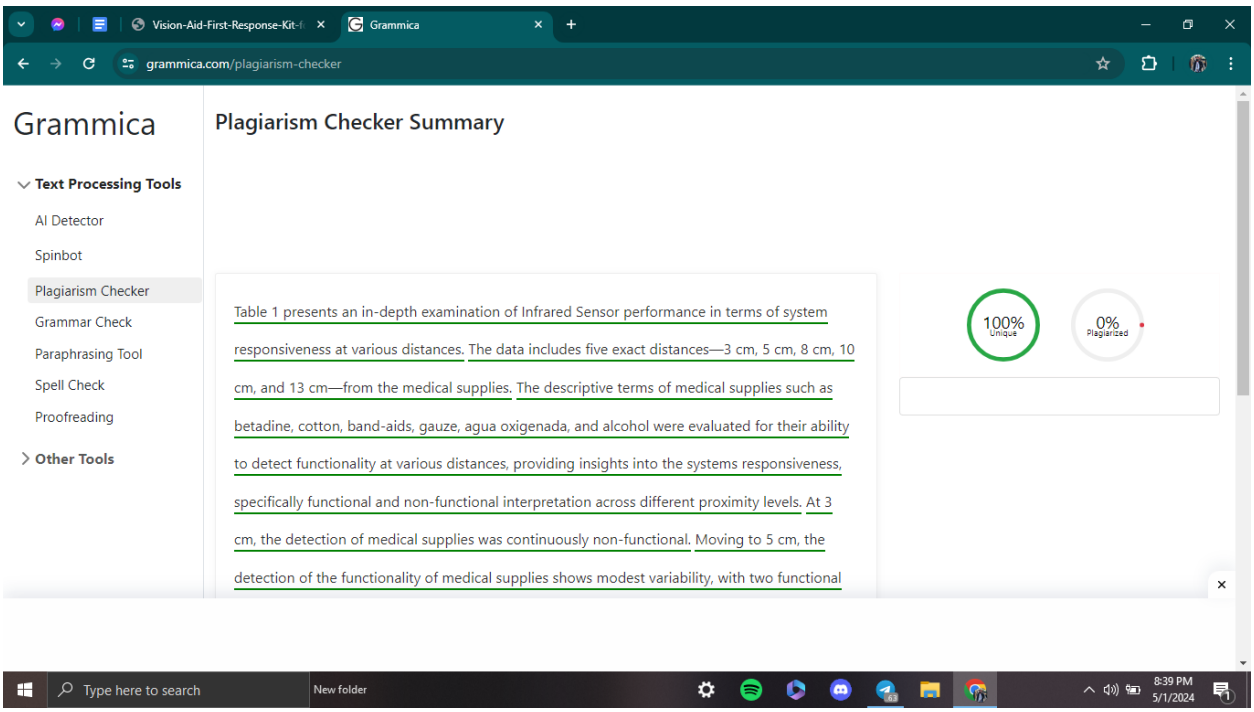
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Table 1 presents an in-depth examination of Infrared Sensor performance in terms of system responsiveness at various distances. The data includes five exact distances—3 cm, 5 cm, 8 cm, 10 cm, and 13 cm—from the medical supplies. The descriptive terms of medical supplies such as betadine, cotton, band-aids, gauze, agua oxigenada, and alcohol were evaluated for their ability to detect functionality at various distances, providing insights into the systems responsiveness, specifically functional and non-functional interpretation across different proximity levels. At 3 cm, the detection of medical supplies was continuously non-functional. Moving to 5 cm, the detection of the functionality of medical supplies shows modest variability, with two functional
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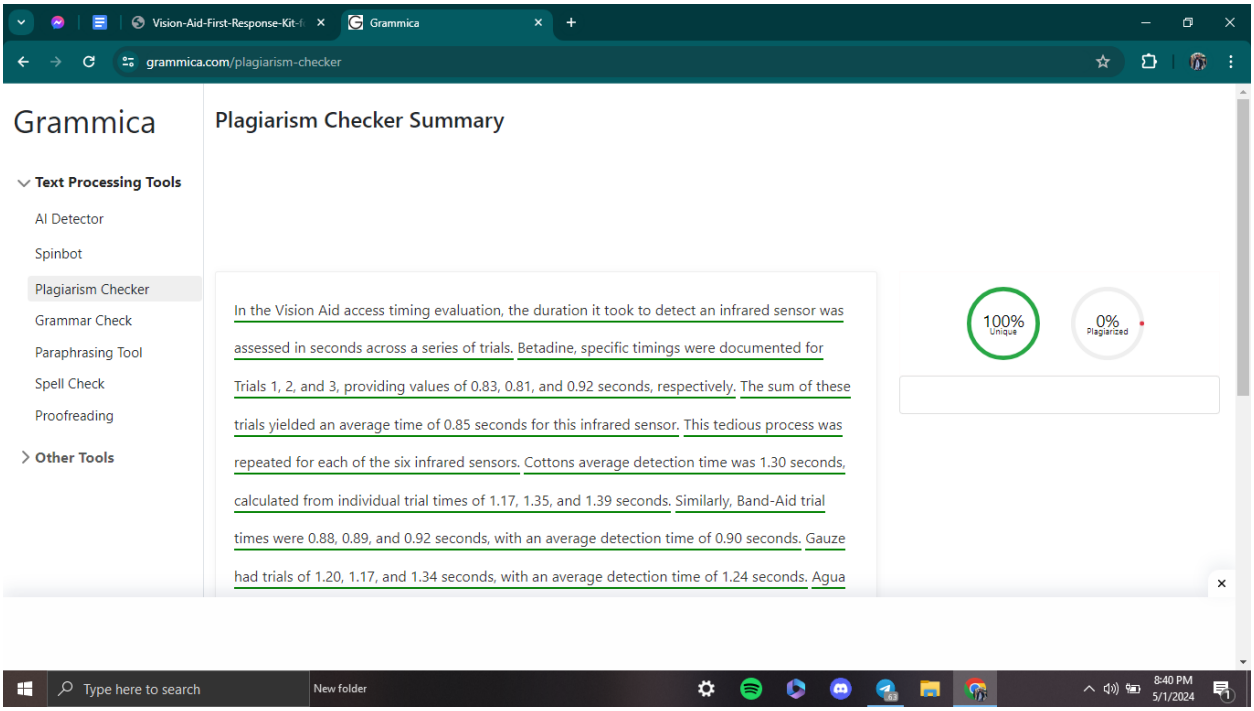
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In the Vision Aid access timing evaluation, the duration it took to detect an infrared sensor was assessed in seconds across a series of trials. Betadine, specific timings were documented for Trials 1, 2, and 3, providing values of 0.83, 0.81, and 0.92 seconds, respectively. The sum of these trials yielded an average time of 0.85 seconds for this infrared sensor. This tedious process was repeated for each of the six infrared sensors. Cottons average detection time was 1.30 seconds, calculated from individual trial times of 1.17, 1.35, and 1.39 seconds. Similarly, Band-Aid trial times were 0.88, 0.89, and 0.92 seconds, with an average detection time of 0.90 seconds. Gauze had trials of 1.20, 1.17, and 1.34 seconds, with an average detection time of 1.24 seconds. Agua
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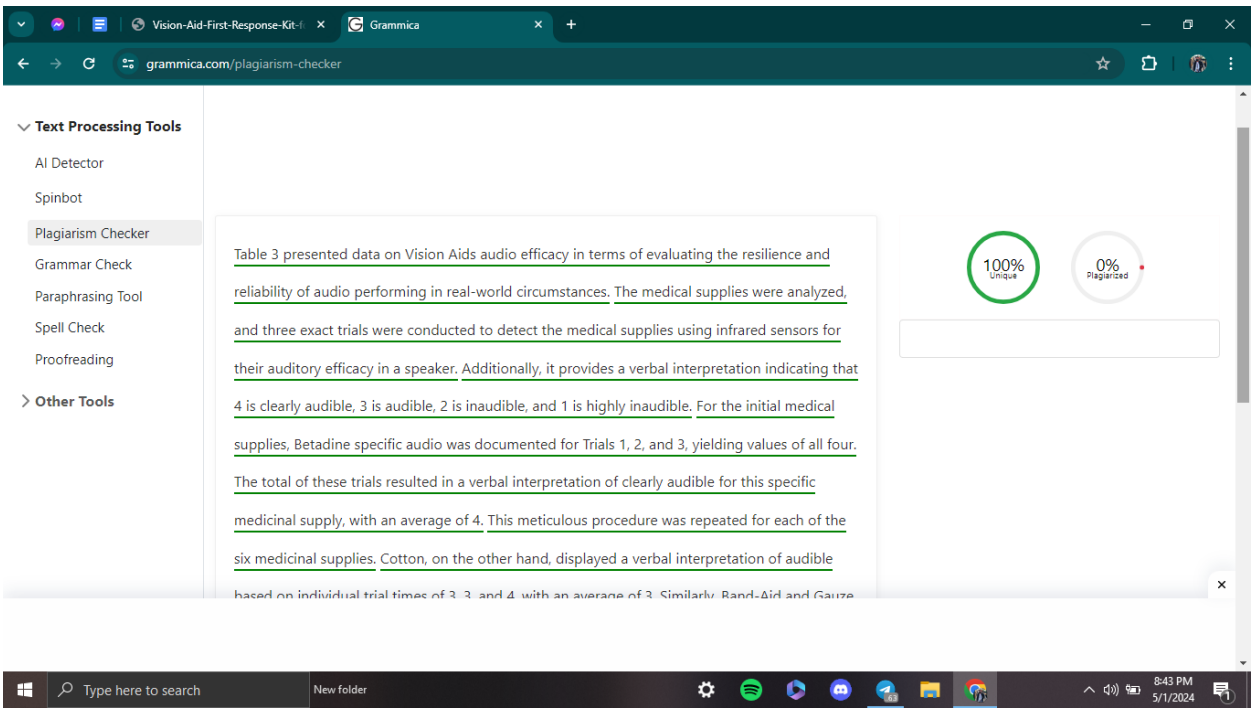
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Table 3 presented data on Vision Aids audio efficacy in terms of evaluating the resilience and reliability of audio performing in real-world circumstances. The medical supplies were analyzed, and three exact trials were conducted to detect the medical supplies using infrared sensors for their auditory efficacy in a speaker. Additionally, it provides a verbal interpretation indicating that 4 is clearly audible, 3 is audible, 2 is inaudible, and 1 is highly inaudible. For the initial medical supplies, Betadine specific audio was documented for Trials 1, 2, and 3, yielding values of all four. The total of these trials resulted in a verbal interpretation of clearly audible for this specific medicinal supply, with an average of 4. This meticulous procedure was repeated for each of the six medicinal supplies. Cotton, on the other hand, displayed a verbal interpretation of audible
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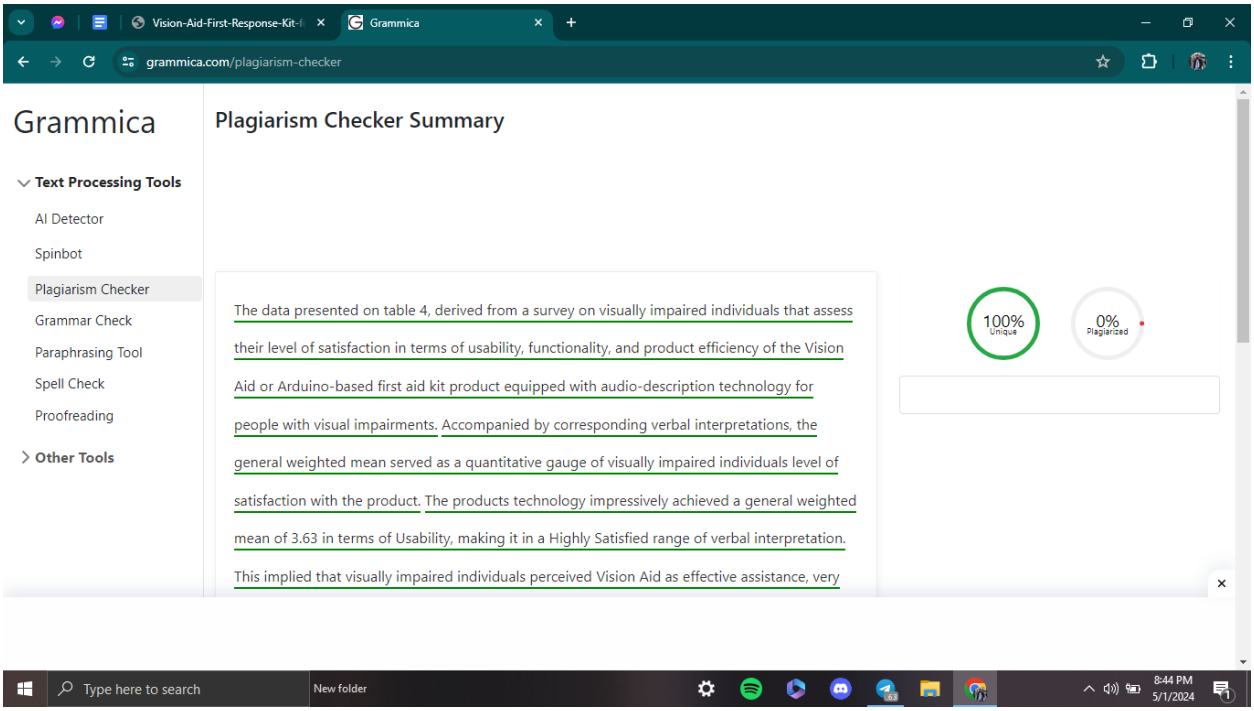
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The data presented on table 4, derived from a survey on visually impaired individuals that assess their level of satisfaction in terms of usability, functionality, and product efficiency of the Vision Aid or Arduino-based first aid kit product equipped with audio-description technology for people with visual impairments. Accompanied by corresponding verbal interpretations, the general weighted mean served as a quantitative gauge of visually impaired individuals level of satisfaction with the product. The products technology impressively achieved a general weighted mean of 3.63 in terms of Usability, making it in a Highly Satisfied range of verbal interpretation. This implied that visually impaired individuals perceived Vision Aid as effective assistance, very
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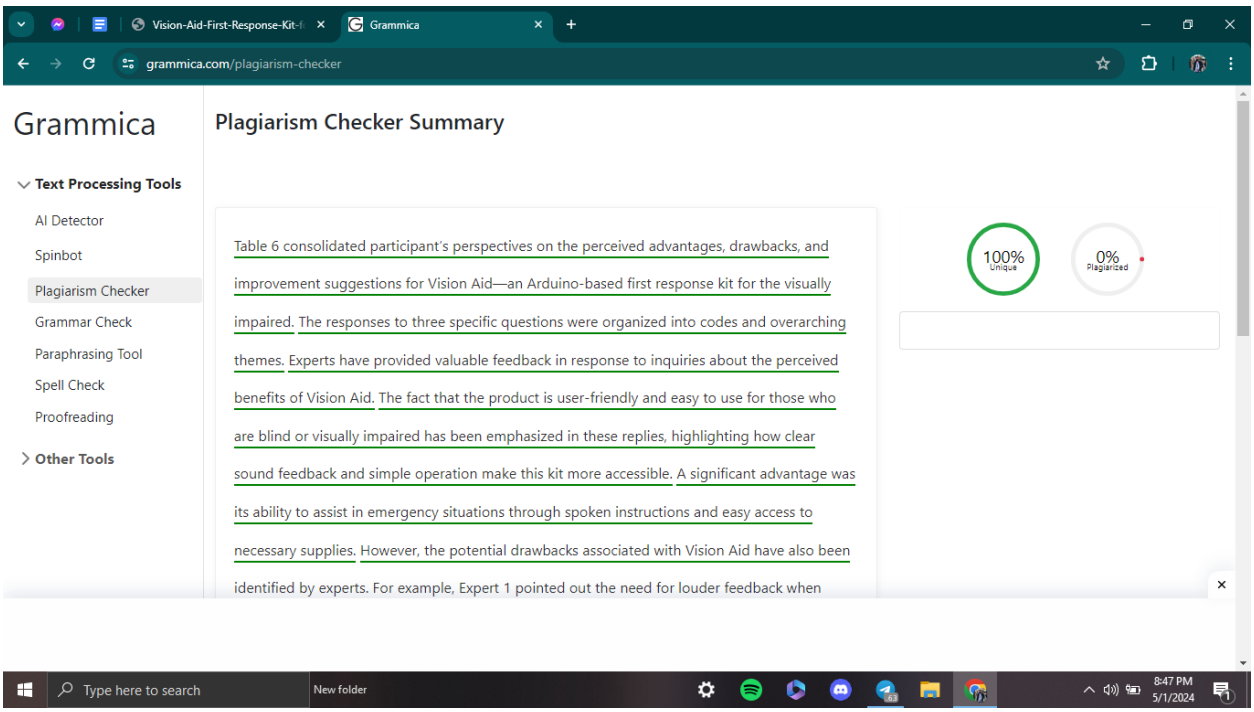
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Table 6 consolidated participant's perspectives on the perceived advantages, drawbacks, and improvement suggestions for Vision Aid—an Arduino-based first response kit for the visually impaired. The responses to three specific questions were organized into codes and overarching themes. Experts have provided valuable feedback in response to inquiries about the perceived benefits of Vision Aid. The fact that the product is user-friendly and easy to use for those who are blind or visually impaired has been emphasized in these replies, highlighting how clear sound feedback and simple operation make this kit more accessible. A significant advantage was its ability to assist in emergency situations through spoken instructions and easy access to necessary supplies. However, the potential drawbacks associated with Vision Aid have also been identified by experts. For example, Expert 1 pointed out the need for louder feedback when
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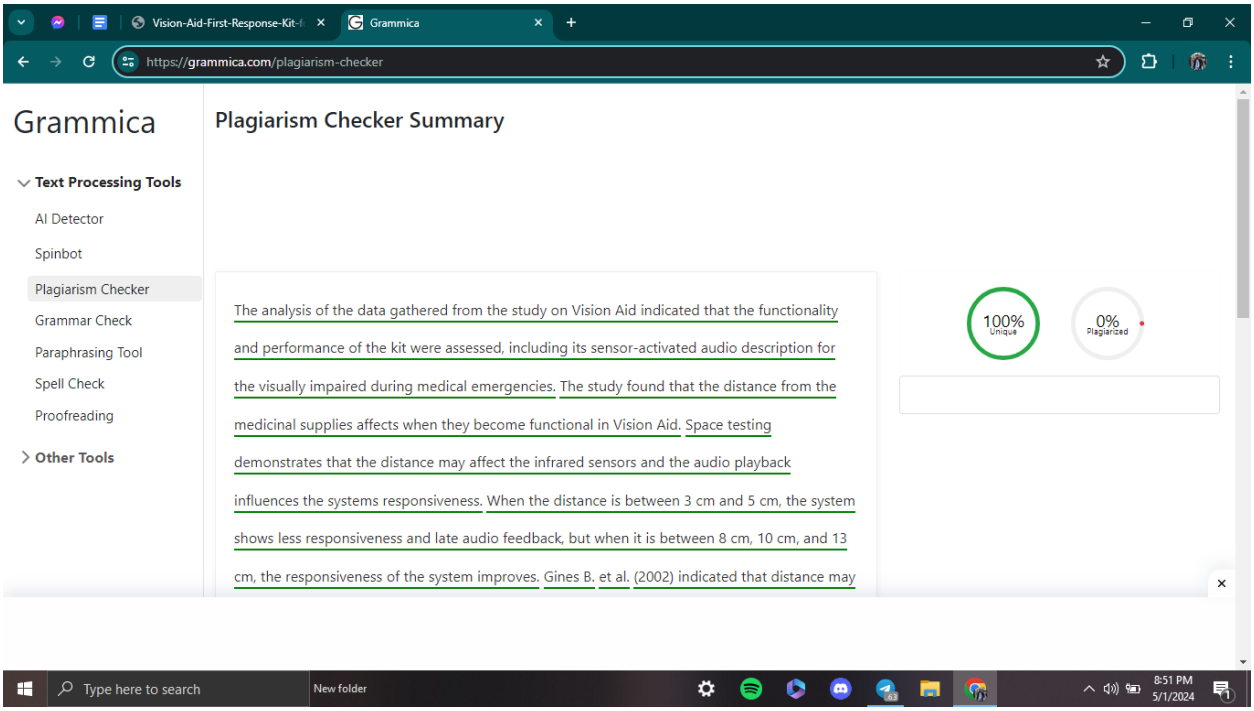
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The analysis of the data gathered from the study on Vision Aid indicated that the functionality and performance of the kit were assessed, including its sensor-activated audio description for the visually impaired during medical emergencies. The study found that the distance from the medicinal supplies affects when they become functional in Vision Aid. Space testing demonstrates that the distance may affect the infrared sensors and the audio playback influences the systems responsiveness. When the distance is between 3 cm and 5 cm, the system shows less responsiveness and late audio feedback, but when it is between 8 cm, 10 cm, and 13 cm, the responsiveness of the system improves. Gines B. et al. (2002) indicated that distance may
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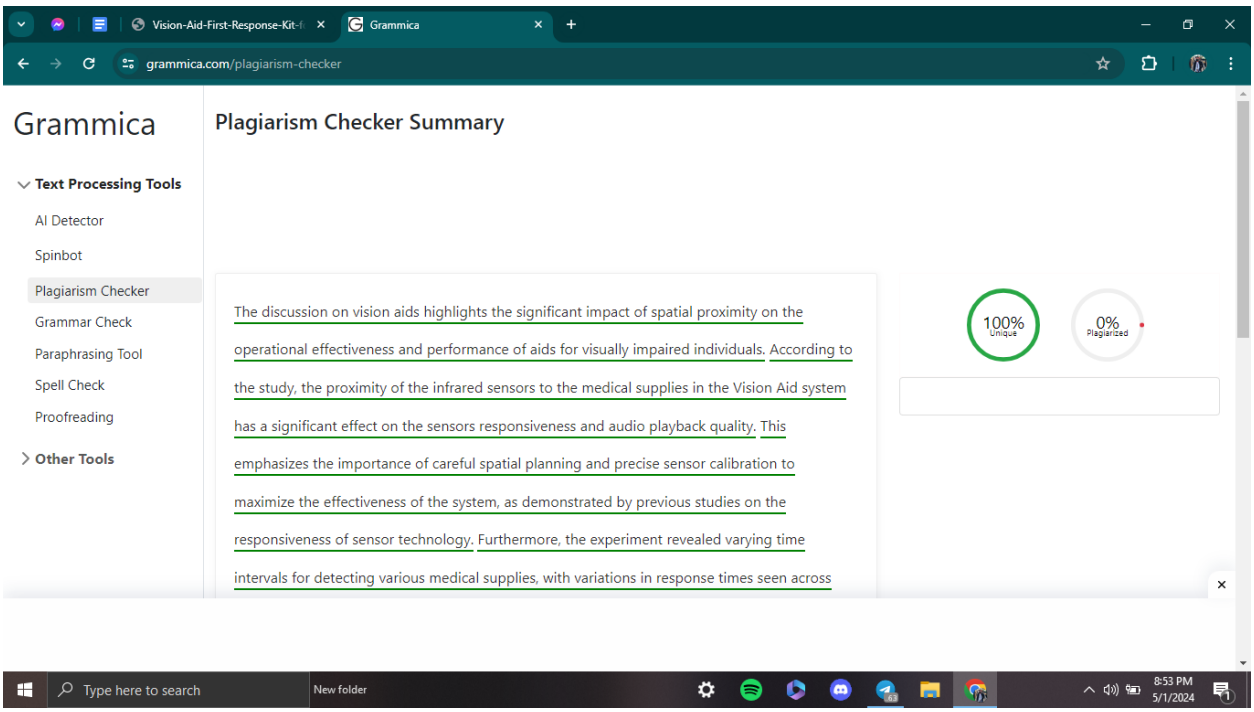
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The discussion on vision aids highlights the significant impact of spatial proximity on the operational effectiveness and performance of aids for visually impaired individuals. According to the study, the proximity of the infrared sensors to the medical supplies in the Vision Aid system has a significant effect on the sensors responsiveness and audio playback quality. This emphasizes the importance of careful spatial planning and precise sensor calibration to maximize the effectiveness of the system, as demonstrated by previous studies on the responsiveness of sensor technology. Furthermore, the experiment revealed varying time intervals for detecting various medical supplies, with variations in response times seen across
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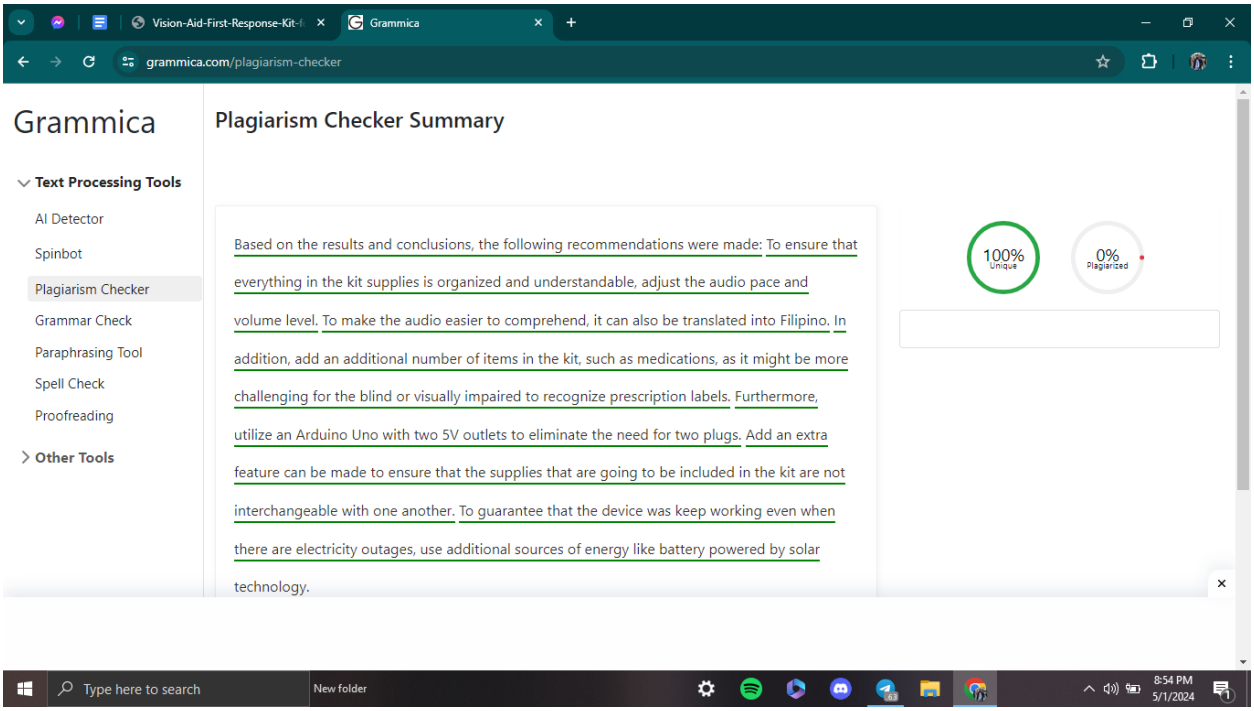
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Based on the results and conclusions, the following recommendations were made: To ensure that everything in the kit supplies is organized and understandable, adjust the audio pace and volume level. To make the audio easier to comprehend, it can also be translated into Filipino. In addition, add an additional number of items in the kit, such as medications, as it might be more challenging for the blind or visually impaired to recognize prescription labels. Furthermore, utilize an Arduino Uno with two 5V outlets to eliminate the need for two plugs. Add an extra feature can be made to ensure that the supplies that are going to be included in the kit are not interchangeable with one another. To guarantee that the device was keep working even when there are electricity outages, use additional sources of energy like battery powered by solar technology.
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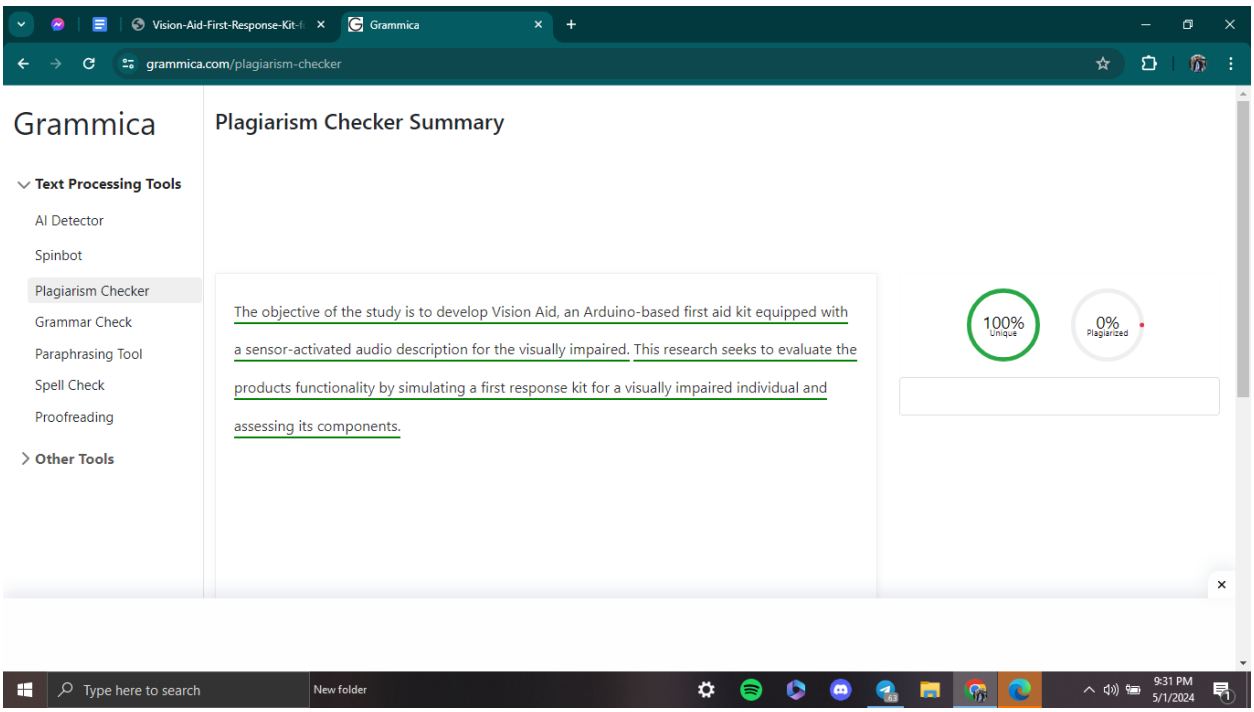
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The objective of the study is to develop Vision Aid, an Arduino-based first aid kit equipped with a sensor-activated audio description for the visually impaired. This research seeks to evaluate the products functionality by simulating a first response kit for a visually impaired individual and assessing its components.
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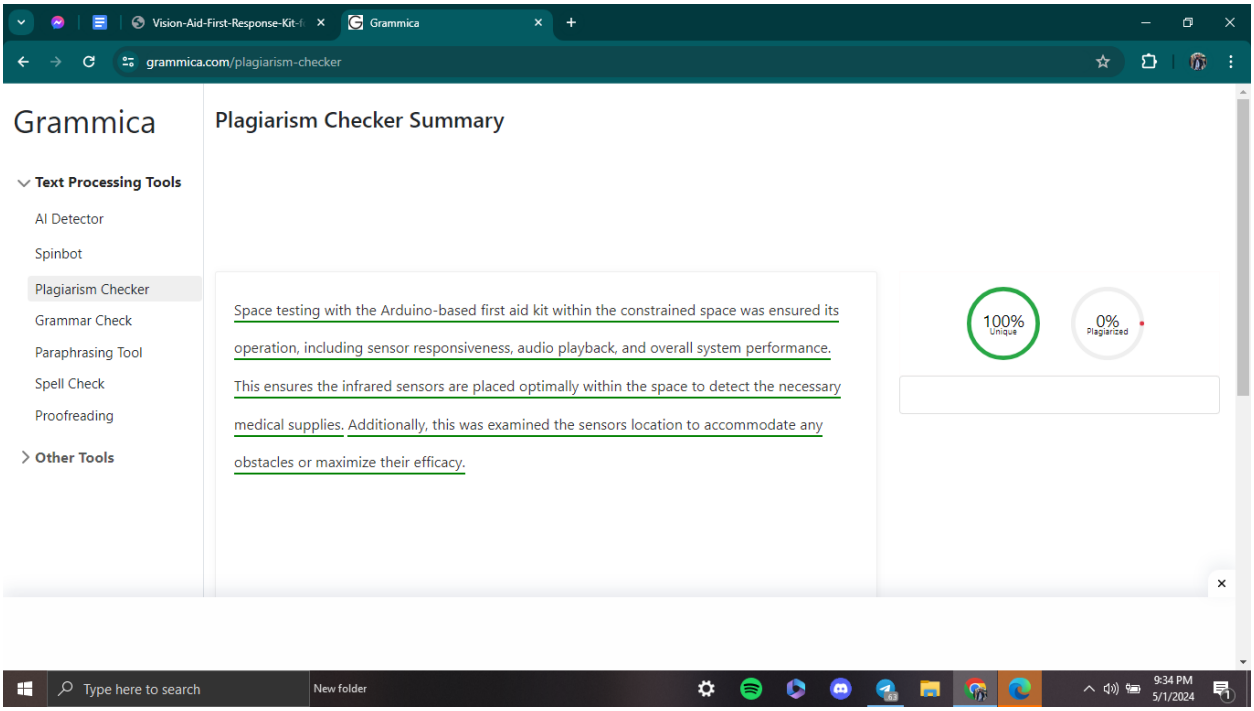
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Space testing with the Arduino-based first aid kit within the constrained space was ensured its operation, including sensor responsiveness, audio playback, and overall system performance. This ensures the infrared sensors are placed optimally within the space to detect the necessary medical supplies. Additionally, this was examined the sensors location to accommodate any obstacles or maximize their efficacy.
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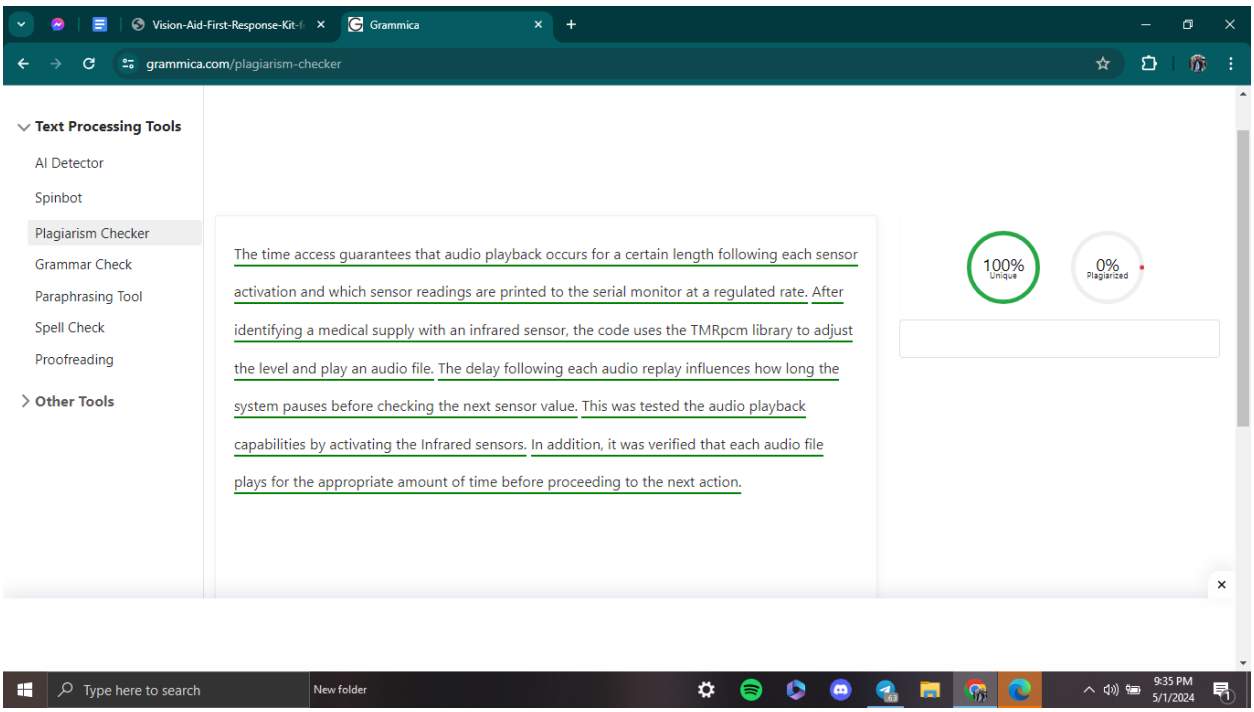
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The time access guarantees that audio playback occurs for a certain length following each sensor activation and which sensor readings are printed to the serial monitor at a regulated rate. After identifying a medical supply with an infrared sensor, the code uses the TMRpcm library to adjust the level and play an audio file. The delay following each audio replay influences how long the system pauses before checking the next sensor value. This was tested the audio playback capabilities by activating the Infrared sensors. In addition, it was verified that each audio file plays for the appropriate amount of time before proceeding to the next action.
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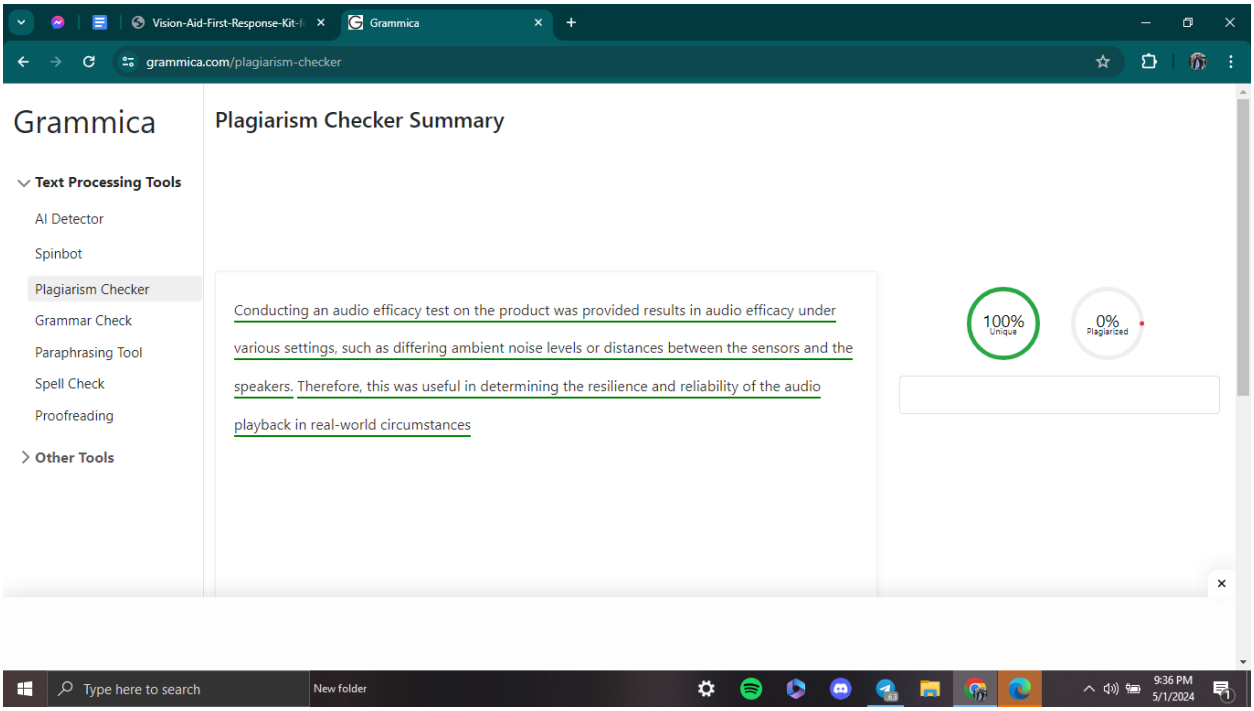
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Grammica
grammica.com/plagiarism-checker
Grammica
Plagiarism Checker Summary
Text Processing Tools
AI Detector
Spinbot
Plagiarism Checker
Grammar Check
Paraphrasing Tool
Spell Check
Proofreading
Other Tools
Conducting an audio efficacy test on the product was provided results in audio efficacy under various settings, such as differing ambient noise levels or distances between the sensors and the speakers. Therefore, this was useful in determining the resilience and reliability of the audio playback in real-world circumstances
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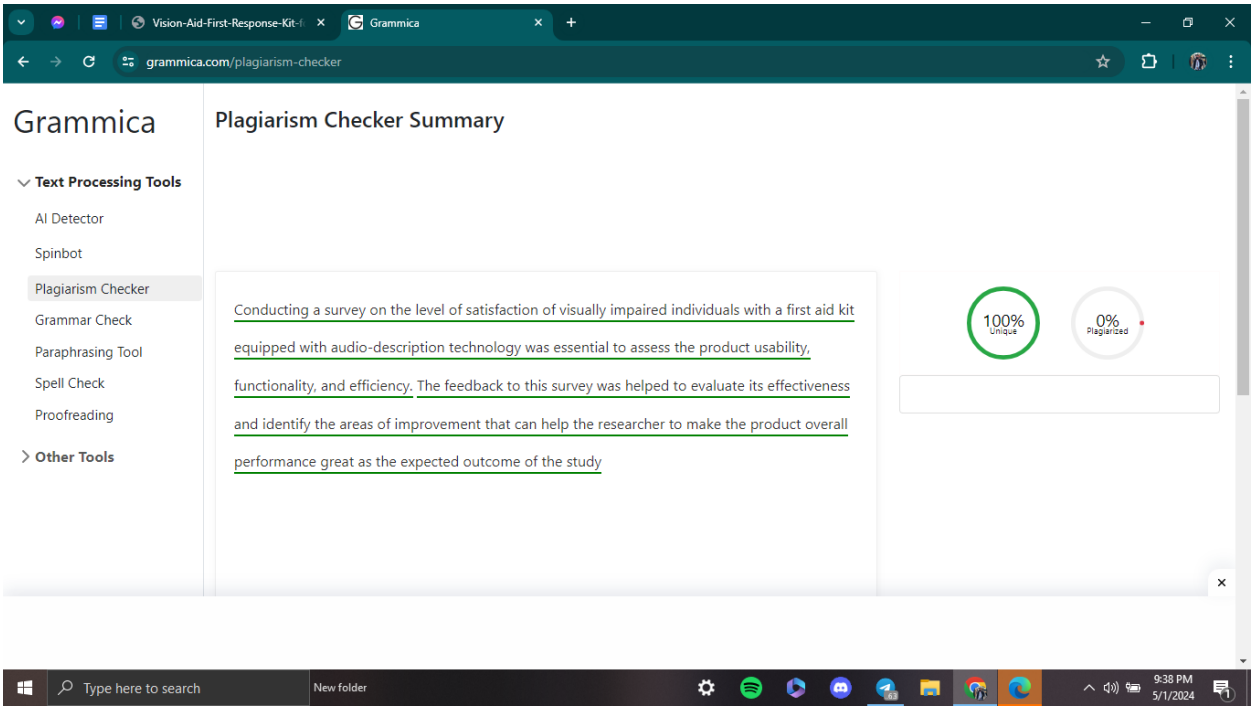
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Grammar Check
Paraphrasing Tool
Spell Check
Proofreading
Other Tools
Conducting a survey on the level of satisfaction of visually impaired individuals with a first aid kit equipped with audio-description technology was essential to assess the product usability, functionality, and efficiency. The feedback to this survey was helped to evaluate its effectiveness and identify the areas of improvement that can help the researcher to make the product overall performance great as the expected outcome of the study
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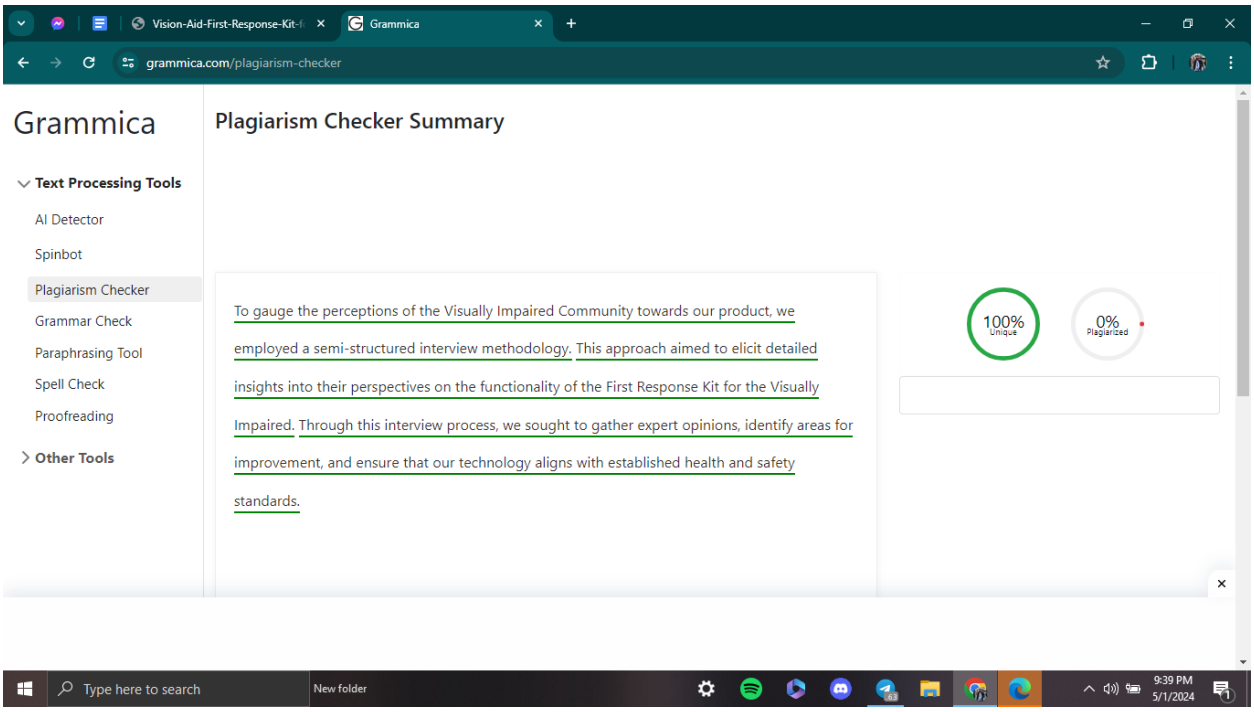
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Grammica
Text Processing Tools
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Plagiarism Checker
Grammar Check
Paraphrasing Tool
Spell Check
Proofreading
Other Tools
Plagiarism Checker Summary
To gauge the perceptions of the Visually Impaired Community towards our product, we employed a semi-structured interview methodology. This approach aimed to elicit detailed insights into their perspectives on the functionality of the First Response Kit for the Visually Impaired. Through this interview process, we sought to gather expert opinions, identify areas for improvement, and ensure that our technology aligns with established health and safety standards.
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5/1/2024

capstone

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Member:	Vecina, John Francis V.
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Data Logbook Entry # 1

February 12, 2024

During the first meeting, the paper was rotated so that each of us could vote for the idea that would be proposed to Ma'am Cruzland.

Christopher John H. Cruz
TEAM LEADER

Cheslanie T Dela Cruz
RESEARCH ADVICER

Data logbook entry # 2

February 16, 2024

We had a meeting to find a programmer because Ma'am Chrelan d accept me problem number 3 which required programming.

Christopher John H. Cruz
TEAM LEADER

Checlarie T. Dela Cruz
RESEARCH ADVISER

Data Logbook Entry # 5

February 18, 2024

We set a meeting with the programmer we talk through the Google Meet App, discribed the product and discussed the sort of programming as well as what materials we might need.

Christopher John H. Cruz
TEAM LEADER

Chealant T. Dela Cruz
RECEARCH ADVISER

Data Logbook Entry # 4

February 19, 2024

We assigned all the research plan into parts and talked about the deadline for submission. We also discussed when we'd meet the programmer we got.

Christopher John H. Cruz
TEAM LEADER

Chealant T. Dela Cruz
RESEARCH ADVISER

Data Logbook Entry # 5

February 28, 2024

Our research plan was transferred to the word after that we donated funds to pay for the cost of the materials and for the programmer's fee. Also we discussed wether to continue working with the previous programmer because we had a bad experience. And in the afternoon we got a medical kit box that we can use for our product. We also met with the new programmer and talked again about the product to be made and materials and also talked about the possible price and signed the contract.

Christopher John H. Cruz
TEAM LEADER

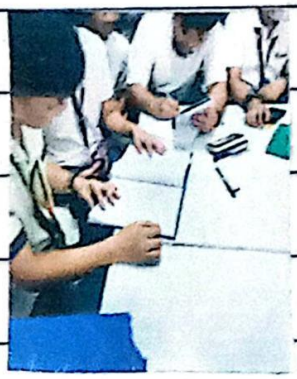
Chealane T. DelaCruz
RESEARCH ADVISER

Data logbook entry # 6

March 07, 2024

We visited scilab today to check over and view the work of the last batchty that are similar to our productr. we also obsrtved how mey completed their paperr so that we would also obtain guidance ffom them.

Chrisfopher John H. Cruz
TEAM LEADER



Chealand T. Dela Cruz
RECEARCH ADVISER

Data Logbook Entry # 7

March 8, 2024

We had a meeting today to discuss the final payment we will prepare to the programmer and the total amount required. Besides that, we were seperated into groups and given the introduction's parts along with the deadline for each.

Chrishopher John H. Cruz
TEAM LEADER

Cheaiann T. Dela Cruz
RESEARCH ADVICER

Data logbook . Data # 8

March 11, 2024

We devided up the materials and . method into parts today, because we have finished the IMRADC introduction. Next, we will work on the method and materials. We also talked about when the submission deadline is, as well as whether the programmer's final payment was agreed upon.

Christopher John H. Cruz
TEAM LEADER

Checlanie T. Dela Cruz
RESEARCH ADVISER

Data Logbook Entry # 9

March 14, 2024

We paid the remaining balance from the agreement cost, also we went back to the programmer to pick up the product and pay the complete amount had to pay, they went also to DEECO to purchase matenair such as transistors and power adapters.

Christopher John H. Cruz
TEAM LEADER

Chealand T. Dela Cruz
RECEARCH ADVICER

Data logbook entry # 10

March 15, 2024

today, our two group member went to the RAON to reserch of a speaker that runs on just 3 volts, as instructed by the programm they questioned locals when they could purchase speakers until they found upon a seller.

Christopher John H. Cruz
TEAM LEADER

Chealane T. Dela Cruz
RESEARCH ADVISER

Data logbook entry # 11

March 16, 2024

Today, we come together at one of the group's house to work on fixing the First Aid kit. After assembling it and trying to turn it on, we discovered that it is really working but the speaker isn't making sounds that correspond with the medications in the kit.

Christopher John H. Cruz
TEAM LEADER

Cheolanie T. Dela Cruz
RESEARCH ADVICER

Data logbook Entry # 12

March 22, 2024

Today, the product was brought back to the programmer due to coding issue. Because by next week, we must demonstrate that our product has been fixed and is capable of producing sounds related to the materials provided inside the kit.

Christopher John H. Cruz
TEAM LEADER

Chealant T. Dela Cruz
RESEARCH ADVICER

DATE _____

Data Logbook Entry # 13

March 25, 2024

Today, we went to our leader's house to test our product, which arrived from our programmer. Although the sound are a little better now, we still need an amplifier. We tried replacing the small speaker with a larger one, but the sound quality remains low.

Christopher John H. Cruz
TEAM LEADER

Checkard T. Dela Cruz
RESEARCH ADVISER

Data Logbook Entry # 14

March 31, 2024

Today, we go back to our leader's house because as unfortunately, the original Arduino explode and was unable to power on. As a result, we purchased a replacement Arduino and attempted to upload the code to it using different laptops, but the attempt was unsuccessful, sure, we made the decision to test it tomorrow on our other member's laptop. Also today the schematic diagram is starting to work on.

Christopher John H. Cruz
TEAM LEADER

Cherelane T. Dela Cruz
RESEARCH ADVISER

Data logbook Entry # 15

April 01, 2024

Today, we try to upload the code on our other's member's laptop, we were able to successfully upload the code to the new Arduino. Adaitionally, the wiring for the Arduino was organized after the code was uploaded.

Christopher John H. Cruz
TEAM LEADER

Chealanz T. Dela Cruz
RESEARCH ADVISER

DATE _____

Data logbook entry # 16

April 02, 2024

We tried it today to see if it would work, but the result is the same as it was the last time - only the brat is heard, and the material found inside the kit doesn't make any sound. Also, we were divided on the same day into two groups: one for the prototype and the other for the paperwork.

Christopher John H. Cruz
TEAM LEADER

Chealand T. Dela Cruz
RESEARCH ADVISER

Data logbook entry # 17

April 03, 2024

Today, we tested the product again at our leader's house. It worked, but the sounds were weak again. For this cause, we purchased an amplifier. After buying and applied it to our product, our product is fully functiona and produces clear sounds.

Christopher John H. Cruz
TEAM LEADER

Chealard T. Dela Cruz
RESEARCH ADVISER

Data logbook entry # 18

April 08, 2024

Today, we went to our leaders house to work on the product, paint it, and add styrofoam inside the kit so that those who are blind or visually impaired can return the items in the kit. We also had a conversation about the schedule for when we can start the implementations. The consent paperwork required for the implementation has also been finalized.

Christopher John H. Cruz
TEAM LEADER

Cheakane T. Dela Cruz
RESEARCH ADVISER

Data logbook Entry # 19

April 11, 2024

Today, it was our first implementation. We went to the south first, where there were a lot of experts that we could get for our study. However, the watchman told us to come back tomorrow because they were busy. Next, we went to SM Valenzuela Vibes Massage service, where we got three experts. They test our product and then we asked them some survey questions. Once we were done, they suggested to visit their other branch in the monumento. However, when we we went there and talked with the head, they said they were busy, so we went to the Pure Gold Paso de Blas D.O.C Wellness Massage corner, where we at least had four experts. Following that, we decide to finish the implementation again in the following day.

Christopher John H. Cruz
TEAM LEADER

Chesklaine T. Dela Cruz
RESEARCH ADVISER

Data Logbook Entry # 20

April 12 2024

Today represents day two of our implementation. We started by going back to the south, but there was only one expert available for the study. Since there aren't many experts available, our group decided to go to SM. Marilao because there is another branch of Vibra Massage. Fortunately, four experts agreed, and everything went easily. They also provided some recommendations regarding the product and accurately answered survey questions about their experience using the virtually impaired kit.

Christopher John H. Cruz

TEAM LEADER

RESEARCH ADVISER:

Cheelaine T. Dela Cruz

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CHRISTOPHER JOHN H. CRUZ

PERSONAL DATA

Date of Birth	December 27, 2005
Place of Birth	Acuñia Maternity Clinic, Valenzuela City
Civil Status	Single
Citizenship	Filipino
Religion	Catholic
Height	163 cm
Weight	64 kg
Father's Name	Rolando G. Cruz
Mother's Name	Crisanta H. Cruz

EDUCATIONAL BACKGROUND

Primary — **Dalandanan Elementary School**

124 Villanueva St. Sumilang Subd, Dalandanan Valenzuela City

2012-2018

Secondary — **Malinta National High School - Junior High**

A Pablo St., Malinta Valenzuela City

2018-2022

Malinta National High School - Senior High

St. Jude Subdivision, Malinta, Valenzuela City

2022-Present

Ph.2 Blk.8 L.2 Assumption Ville, Lingunan Valenzuela City

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MIRKO R. BARCELLANO

PERSONAL DATA

Date of Birth	January 9, 2006
Place of Birth	Valenzuela City
Civil Status	Single
Citizenship	Filipino
Religion	Iglesia Ni Cristo
Height	5’5 ft
Weight	60 kg
Father's Name	Marlon Barcellano
Mother's Name	Evelyn Barcellano

EDUCATIONAL BACKGROUND

Primary — **Canumay West Elementary School**

L Sandiego, Canumay West Valenzuela City

2012-2018

Secondary — **Canumay West National High School**

5 Kaypandan Street, Canumay West Valenzuela City

2018-2022

Malinta National High School - Senior High

St. Jude Subdivision, Malinta, Valenzuela City

2022-Present

B5 L10 Angeles Village Malinta Valenzuela City

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ARGIE G. JALDO

PERSONAL DATA

Date of Birth	December 27, 2001
Place of Birth	Metro Manila
Civil Status	Single
Citizenship	Filipino
Religion	Roman Catholic
Height	5’8 ft
Weight	99 kg
Father's Name	Teodoro R. Jaldo
Mother's Name	Rosie G. Jaldo

EDUCATIONAL BACKGROUND

Primary — **Malinta Elementary School**

A Pablo St., Malinta Valenzuela City

Secondary — **Malinta National High School - Junior High**

A Pablo St., Malinta Valenzuela City

Malinta National High School - Senior High

St. Jude Subdivision, Malinta, Valenzuela City

2022-Present

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CLARENZ L. DELA CRUZ

PERSONAL DATA

Date of Birth	June 1, 2003
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Citizenship	Filipino
Religion	Catholic
Height	n/a
Weight	n/a
Father's Name	Chito Dela Cruz
Mother's Name	Fe Dela Cruz

EDUCATIONAL BACKGROUND

Primary — **Malinta Elementary School**

A Pablo St., Malinta Valenzuela City

Secondary — **Malinta National High School - Junior High**

A Pablo St., Malinta Valenzuela City

Malinta National High School - Senior High

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2022-Present

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CRISALYN D. ESPINAR

PERSONAL DATA

Date of Birth	January 15, 2006
Place of Birth	Silvino Lubos N.Samar
Civil Status	Single
Citizenship	Filipino
Religion	Catholic
Height	n/a
Weight	49 kg
Father's Name	Wilfredo Espinar
Mother's Name	Asteria Espinar

EDUCATIONAL BACKGROUND

Primary — **Cawayan Elementary School**

Cawayan Catarman N. Samar

2017-2021

Secondary — **Lingunan National High School**

Lingunan, Valenzuela City

2021-2022

Malinta National High School - Senior High

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2022-Present

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JAMES I. SERILLA

PERSONAL DATA

Date of Birth	October 22, 2002
Place of Birth	Valenzuela City
Civil Status	Single
Citizenship	Filipino
Religion	Roman Catholic
Height	163 cm
Weight	68 kg
Father's Name	Mariano Serilla
Mother's Name	Teodora Serilla

EDUCATIONAL BACKGROUND

Primary — **Canumay West Elementary School**

L Sandiego, Canumay West Valenzuela City

2012-2018

Secondary — **Canumay West National High School**

5 Kaypandan Street, Canumay West Valenzuela City

2018-2022

Malinta National High School - Senior High

St. Jude Subdivision, Malinta, Valenzuela City

2022-Present

066 I. Marcelo St. Malanday, Valenzuela City

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JOHN FRANCIS V. VECINA

PERSONAL DATA

Date of Birth	June 24, 2006
Place of Birth	Karuhatan, Valenzuela City
Civil Status	Single
Citizenship	Filipino
Religion	Roman Catholic
Height	161 cm
Weight	58 kg
Father's Name	Noli Vecina
Mother's Name	Maria Vecina

EDUCATIONAL BACKGROUND

Primary — **Andres Fernando Elementary School**

309 Marcelo H. Del Pilar St, Valenzuela City

2012-2018

Secondary — **Dalandanan National High School**

258 G. Lazaro Rd, Valenzuela City

2018-2022

Malinta National High School - Senior High

St. Jude Subdivision, Malinta, Valenzuela City

2022-Present

7 Karamay St. Balangkas, Valenzuela City

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JULIE MAE G. SATIMBRE

PERSONAL DATA

Date of Birth	March 19, 2006
Place of Birth	Valenzuela City
Civil Status	Single
Citizenship	Filipino
Religion	Roman Catholic
Height	5 ft
Weight	49 kg
Father's Name	Julio P. Satimbre
Mother's Name	Rechelle G. Satimbre

EDUCATIONAL BACKGROUND

Primary **Arcadio F. Deato Elementary School**

2 Sampaguita, Valenzuela, Metro Manila

2012-2018

Secondary **Polo National High School**

PVES Compound, Marcelo H. Del Pilar St, Valenzuela

2018-2022

Malinta National High School - Senior High

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KYAN ACE C. LUNGSOD

PERSONAL DATA

Date of Birth	January 04, 2006
Place of Birth	Malajog, Samar
Civil Status	Single
Citizenship	Filipino
Religion	Catholic
Height	5’8 ft
Weight	45 kg
Father's Name	Nelbert B. Lungsod
Mother's Name	Liezel P. Cajandab

EDUCATIONAL BACKGROUND

Primary **Caybiga Elementary School**

General Luis, Caybiga, 166, Caloocan, 1420 Metro Manila

2013-2018

Secondary **Bagbaguin National High School**

P253+6C5, Bragados, Valenzuela, 1440 Metro Manila

2018-2022

Malinta National High School - Senior High

St. Jude Subdivision, Malinta, Valenzuela City

2022-Present

#26 Pablo Street Victoria Village Canumay East Valenzuela City

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PHILIP JAMES L. CARICUNGAN

PERSONAL DATA

Date of Birth	February 17, 2005
Place of Birth	Valenzuela City
Civil Status	Single
Citizenship	Filipino
Religion	Catholic
Height	171 cm
Weight	50 kg
Father's Name	Freddie Caricungan
Mother's Name	Dolores Caricungan

EDUCATIONAL BACKGROUND

Primary — **Andres Mariano Elementary School**

124 Villanueva St. Sumilang Subd, Dalandanan Valenzuela City

2012-2018

Secondary — **Paso De Blas National High School**

170 Paso De Blas Road Valenzuela City

2018-2022

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2022-Present

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SHAFALIE G. DELA TORRE

PERSONAL DATA

Date of Birth	January 19, 2006
Place of Birth	Maysan, Valenzuela City
Civil Status	Single
Citizenship	Filipino
Religion	Roman Catholic
Height	158 cm
Weight	48.8 kg
Father's Name	Pepito M. Dela Torre
Mother's Name	Melchora G. Dela Torre

EDUCATIONAL BACKGROUND

Primary — **Maysan Elementary School**

102 Maysan Road, Valenzuela City

2012-2018

Secondary — **Maysan National High School**

104 Maysan Road, Valenzuela City

2018-2022

Malinta National High School - Senior High

St. Jude Subdivision, Malinta, Valenzuela City

2022-Present